
Section 0 — The Algebraic Foundation

0.0 Purpose of the Foundation

0.0.1 The Foundational Question

The purpose of this section is to specify the minimal mathematical structure required for the theory before any appeal is made to spacetime, geometry, matter fields, particles, observers, measurement, or external time.

The framework begins from the claim that physical admissibility is not primarily a matter of locating objects in a pre-existing background, but of satisfying a system of relational constraints. What is fundamental is therefore not a manifold, a metric, a field on spacetime, or a Hilbert-space state vector taken in isolation, but a constrained relational structure whose admissible configurations are those in which inconsistency vanishes.

0.0.2 The Central Slogan

The central foundational commitment is:

> **Physicality is constraint compatibility, not prior placement in spacetime.**

This statement must be understood carefully. It does not deny spacetime, fields, clocks, particles, or geometry. Rather, it denies that these are primitive. They are reconstructed later from the behaviour of physical states, zero-preserving observables, correlation structure, causal dependency, burden, and coarse-grained effective descriptions.

0.0.3 Why the Foundation Must Be Algebraic

If one begins by assuming a spacetime manifold, then causal order, distance, locality, field propagation, and temporal duration are already built into the background. In that case, the theory would no longer explain the emergence of those structures; it would merely redescribe them.

The foundation therefore deliberately avoids assuming:

- a spacetime manifold,
- a metric,
- a topology,
- a causal order,
- a Hilbert space,
- a Hamiltonian,
- a Lagrangian,
- a probability measure,
- a classical configuration space,
- a background time parameter.

Instead, the primitive structure is a **unital $(*)$ -algebra** $\mathcal{A}$, whose elements represent possible relational operations or quantities. Physical admissibility is then imposed by a **constraint ideal** $\mathcal{I}_C$ inside $\mathcal{A}$, and physical states are positive linear functionals that annihilate the constraints in the appropriate quadratic sense.

0.0.4 The Role of This Section

This section has five roles.

First, it prevents a category error. If one begins with a spacetime manifold, then causal order, distance, and locality are already assumed. The foundation establishes the admissible relational substrate from which those notions may later emerge.

Second, it clarifies the meaning of "zero". The zero condition is not a claim that reality is empty. It is a consistency condition. To say that the master constraint vanishes is to say that relevant constraint violations vanish. It is a statement about admissibility, not nothingness.

Third, it distinguishes the primitive algebraic layer from its later representations. A Hilbert space arises through a GNS representation. A master constraint operator is introduced as a representation-level packaging. A classical master functional appears as an effective description. None of these is taken as fundamental.

Fourth, it separates physical admissibility from dynamics. The master constraint is not an equation of motion. It does not describe evolution with respect to external time. It selects the physical sector. Temporal notions are introduced later through causal dependency, burden, and clock-rate suppression.

Fifth, it makes later emergence claims theorem-safe. Claims about emergent space, time, particles, gravity, and cosmology must not be smuggled into the starting assumptions. They must be built in layers.

0.0.5 The Foundational Hierarchy

The foundational construction proceeds in the following order:

1. The relational algebra $\mathcal{A}$ is defined.
2. The primitive constraints $\{\hat{C}_\alpha\}$ and their generated ideal $\mathcal{I}_C$ are introduced.
3. Physical states are defined as positive linear functionals satisfying the quadratic constraint-vanishing condition.
4. The GNS representation is used to show how Hilbert space and represented constraint operators arise from a physical state.
5. The master constraint operator is introduced only as a representation-level packaging of the primitive constraints.
6. The effective classical master functional is introduced as a derived, coarse-grained description.

7. The section records the guardrails needed to avoid overinterpreting the zero condition, the Hilbert-space representation, or the master constraint.

The foundational commitment can therefore be stated as:

> **A physical structure is an admissible relational structure whose constraint violations vanish in the physical state.**

0.1 The Relational Algebra

0.1.1 Purpose of the Relational Algebra

The first object required by the framework is a mathematical structure capable of carrying primitive relations before those relations are interpreted as spatial, temporal, material, causal, or observational.

Since the theory does not begin with a spacetime manifold, it cannot define its basic objects as fields on spacetime points. Since it does not begin with an external time parameter, it cannot define its basic objects as time-evolving configurations. Since it does not begin with a fixed Hilbert space, it cannot treat vectors in that Hilbert space as the primitive layer.

The starting point must therefore be more abstract. It must be able to describe relational quantities, allow those quantities to compose, allow a notion of reversal or adjoint, and support constraints. A unital $(*)$ -algebra provides precisely this minimal structure.

The relational algebra is not introduced as a space of already physical observables. At this stage, it is only the kinematic carrier of possible relational expressions. Some of its elements will later be ruled physical, some will be identified modulo constraints, and some will become observables only after they are shown to preserve the physical sector.

> **The relational algebra contains possible relational expressions before physical admissibility is imposed.**

0.1.2 Definition — Relational $(*)$ -Algebra

Definition 0.1. A relational algebra is a unital complex $(*)$ -algebra $(\mathcal{A}, \dagger, \mathbf{1})$, where:

1. $\mathcal{A}$ is a complex associative algebra over $\mathbb{C}$.
2. $\mathbf{1} \in \mathcal{A}$ is a multiplicative identity satisfying $\mathbf{1}A = A\mathbf{1} = A$ for all $A \in \mathcal{A}$.
3. $\dagger: \mathcal{A} \rightarrow \mathcal{A}$ is an involution satisfying, for all $A, B \in \mathcal{A}$ and all $\lambda, \mu \in \mathbb{C}$:
 - $(A^\dagger)^\dagger = A$,
 - $(AB)^\dagger = B^\dagger A^\dagger$,
 - $(\lambda A + \mu B)^\dagger = \bar{\lambda} A^\dagger + \bar{\mu} B^\dagger$.

0.1.3 Minimal Assumptions

****Assumption 0.1 (Non-trivial algebra).** $\mathcal{A} \neq \{0\}$.

****Assumption 0.2 (No additional structure).** At this stage, the framework does not assume:

- any norm,
- any topology,
- any completeness,
- any Hilbert-space representation,
- commutativity,
- a spacetime manifold,
- a metric,
- a causal order,
- a Hamiltonian,
- a Lagrangian,
- a probability measure,
- a classical configuration space,
- a background time parameter.

The only basic non-triviality condition is that the algebra is not the zero algebra.

0.1.4 Interpretation of Algebra Elements

An element $A \in \mathcal{A}$ should be understood as a ****possible relational expression****. It may encode a relation between primitive degrees of freedom, a composite relation built from simpler relations, or a candidate operation acting on a relational structure.

The product AB represents ****composition****: the relational expression B is followed, combined, or algebraically composed with the relational expression A . The ordering matters in general, since the algebra is not assumed to be commutative:

$$AB \neq BA \quad \text{in general.}$$

This non-commutativity is not yet a quantum postulate. It is a structural allowance: the order in which relational operations are composed may matter.

The adjoint $A^\dagger$ provides a formal ****reversal, dual, or compatibility operation****. It allows the theory to form positive expressions of the type $A^\dagger A$, which are essential for defining positive states and quadratic constraint-vanishing conditions.

The unit $\mathbf{1}$ represents the ****trivial relational operation****. It is the identity element of the algebraic language.

0.1.5 Why a $(*)$ -Algebra Is Used

The use of a unital $(*)$ -algebra is motivated by four requirements.

****First****, the framework needs composition. If two relational expressions are meaningful, then their composition should also be meaningful. This requires a multiplication operation.

****Second****, the framework needs linearity. Physical states will later be defined as linear functionals on the algebra. Linear combinations of relational expressions must therefore remain within the same formal structure.

****Third****, the framework needs positivity. Constraint satisfaction will be expressed through terms such as $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$. Such expressions require an adjoint operation and a notion of positive quadratic form.

****Fourth****, the framework needs representation flexibility. A $(*)$ -algebra can later be represented on a Hilbert space, but it does not require Hilbert space to be primitive.

0.1.6 Immediate Consequences

****Lemma 0.1 (Closure properties).**** For all $A, B \in \mathcal{A}$:

1. $AB \in \mathcal{A}$,
2. $A^\dagger \in \mathcal{A}$,
3. $A^\dagger A \in \mathcal{A}$.

Proof. Both statements follow directly from the definition of algebra multiplication and the involution map. $\square$

****Lemma 0.2 (Adjoint identities).**** For all $A, B \in \mathcal{A}$ and $\lambda \in \mathbb{C}$:

1. $(A^\dagger B^\dagger)^\dagger = BA$,
2. $(\mathbf{1})^\dagger = \mathbf{1}$.

Proof.

1. Using involution and anti-multiplicativity: $(A^\dagger B^\dagger)^\dagger = (B^\dagger)^\dagger (A^\dagger)^\dagger = BA$.
2. Since $\mathbf{1}A = A$, applying adjoint gives $(\mathbf{1}A)^\dagger = A^\dagger \rightarrow \mathbf{1}^\dagger A^\dagger = A^\dagger \rightarrow \mathbf{1}^\dagger = A^\dagger A = \mathbf{1}$. $\square$

****Lemma 0.3 (Positive expressions are algebraically available).**** For every $A \in \mathcal{A}$, the quadratic expression $A^\dagger A$ is a well-defined element of $\mathcal{A}$.

Proof. By Lemma 0.1, $A^\dagger \in \mathcal{A}$ and $A^\dagger A \in \mathcal{A}$. $\square$

This lemma is simple but foundational. It guarantees that the algebra can express quantities of the same form as $\hat{C}_\alpha^\dagger \hat{C}_\alpha$, which will be used to define physical states.

0.1.7 What Is Not Assumed

The relational algebra is deliberately minimal. At this stage:

- There is no metric, no space, no notion of locality.
- There is no dynamics, no time, no evolution law.
- There is no Hilbert space and no representation.
- The adjoint operation does not by itself define time reversal, measurement, Hermitian observables, or physical duality. It is an algebraic operation needed for positivity and constraint formulation.
- Membership in $\mathcal{A}$ does not imply physical admissibility.

0.1.8 Relation to Later Layers

This algebra will serve as the primitive carrier structure for the entire framework:

- Section 0.2: constraints will be defined as elements of $\mathcal{A}$.
- Section 0.3: states will be linear functionals on $\mathcal{A}$.
- Section 1: observables will be defined via constraints and commutation.
- Later sections: geometry, time, and physics will be derived from states and correlations on $\mathcal{A}$.

0.2 Constraint Operators and the Constraint Ideal

0.2.1 Purpose of the Constraint Layer

The relational algebra $\mathcal{A}$ represents the full set of possible relational observables. However, not all algebraic configurations correspond to physically admissible structures. We therefore introduce constraints to encode:

- consistency conditions,
- admissibility requirements,
- the structural rules governing physical configurations.

These constraints are internal to the algebra and do not refer to external structures.

0.2.2 Definition — Constraint Operators

Definition 0.2. A constraint operator is an element $\hat{C}_\alpha \in \mathcal{A}$ labelled by an index $\alpha \in I$, where I is an index set. The full family of primitive constraints is written as $\{\hat{C}_\alpha\}_{\alpha \in I} \subset \mathcal{A}$.

Each $\hat{C}_\alpha$ represents a primitive algebraic condition that must vanish, in the appropriate physical sense, for a relational structure to be admissible.

Notation. The hat notation is used to remind us that these are algebraic constraint elements which may later be represented as operators. Fundamentally, $\hat{C}_\alpha \in \mathcal{A}$, not necessarily $\hat{C}_\alpha \in \mathcal{B}(\mathcal{H})$ for some pre-existing Hilbert space.

0.2.3 Why Constraints Live Inside the Algebra

The constraints must be elements of $\mathcal{A}$ because the framework requires consistency to be expressed internally. If constraints were imposed from outside the algebra, then physical admissibility would depend on an additional background structure not contained in the relational system itself.

By taking $\hat{C}_\alpha \in \mathcal{A}$, the theory ensures that constraints can be composed with other relational expressions, adjointed, inserted into state evaluations, and represented through the same machinery as all other algebraic elements.

This allows the following expressions to be meaningful within the theory:

$A\hat{C}_\alpha, \quad \hat{C}_\alpha B, \quad A\hat{C}_\alpha B, \quad \hat{C}_\alpha^\dagger, \quad \hat{C}_\alpha^\dagger \hat{C}_\alpha$,
for arbitrary $A, B \in \mathcal{A}$.

The final expression, $\hat{C}_\alpha^\dagger \hat{C}_\alpha$, will become especially important because physical states will be required to annihilate this positive quadratic form.

0.2.4 Primitive Constraints Versus Their Consequences

It is not enough to specify only the primitive list $\{\hat{C}_\alpha\}_{\alpha \in I}$. If a constraint $\hat{C}_\alpha$ represents an inadmissible direction or inconsistency, then any algebraic expression containing that constraint as a factor should also be regarded as belonging to the inconsistency sector. For example, if $\hat{C}_\alpha$ is constrained, then expressions such as $A\hat{C}_\alpha B$ should also be constrained for arbitrary relational expressions $A, B \in \mathcal{A}$.

This is the reason for introducing an ideal rather than merely a set of generators. The primitive constraints are the starting points. The ideal generated by them is the algebraic closure of all their consequences under addition, multiplication by arbitrary algebra elements, and adjoint.

> **Constraint operators are primitive generators; the constraint ideal is their full algebraic consequence class.**

0.2.5 Definition — Constraint Ideal

Definition 0.3. The constraint ideal generated by $\{\hat{C}_\alpha\}_{\alpha \in I}$ is the smallest two-sided $(*)$ -ideal of $\mathcal{A}$ containing all $\hat{C}_\alpha$. It is denoted:
 $\mathcal{I}_C = \langle \hat{C}_\alpha : \alpha \in I \rangle$.

Equivalently, $\mathcal{I}_C$ is the smallest subset of $\mathcal{A}$ satisfying:

1. **Contains all primitive constraints:** $\hat{C}_\alpha \in \mathcal{I}_C$ for all $\alpha \in I$.
2. **Closed under finite linear combinations:** $X, Y \in \mathcal{I}_C, \lambda, \mu \in \mathbb{C} \Rightarrow \lambda X + \mu Y \in \mathcal{I}_C$.
3. **Closed under left multiplication:** $X \in \mathcal{I}_C, A \in \mathcal{A} \Rightarrow AX \in \mathcal{I}_C$.

4. **Closed under right multiplication:** $X \in \mathcal{I}_C, A \in \mathcal{A} \Rightarrow XA \in \mathcal{I}_C$.
5. **Closed under adjoint:** $X \in \mathcal{I}_C \Rightarrow X^\dagger \in \mathcal{I}_C$.

Because it is both left and right closed, $\mathcal{I}_C$ is a two-sided ideal. Because it is also closed under the involution † , it is a $(*)$ -ideal.

0.2.6 Explicit Generated Form

Elements of the constraint ideal may be written schematically as finite sums of the form:

$$X = \sum_{k=1}^n A_k \hat{C}_{\alpha_k} B_k,$$

where $A_k, B_k \in \mathcal{A}$ and $\alpha_k \in I$.

Because the ideal is required to be $(*)$ -closed, it also contains adjoints of such expressions.

If $X = \sum_k A_k \hat{C}_{\alpha_k} B_k$, then:

$$X^\dagger = \sum_{k=1}^n B_k^\dagger \hat{C}_{\alpha_k}^\dagger A_k^\dagger$$

also belongs to $\mathcal{I}_C$.

More explicitly, one may write:

$$\mathcal{I}_C = \left\{ \sum_{k=1}^n A_k \hat{C}_{\alpha_k} B_k + \sum_{\ell=1}^m D_\ell \hat{C}_{\beta_\ell}^\dagger E_\ell \mid A_k, B_k, D_\ell, E_\ell \in \mathcal{A} \right\},$$

with the understanding that this represents the finite algebraic span of the primitive constraints and their adjoints under arbitrary left and right multiplication.

0.2.7 Structural Properties

Lemma 0.3 (Ideal properties). $\mathcal{I}_C$ satisfies:

1. **Linear subspace:** $X, Y \in \mathcal{I}_C \Rightarrow \lambda X + \mu Y \in \mathcal{I}_C$.
2. **Two-sided closure:** $AX, XA \in \mathcal{I}_C$ for all $A \in \mathcal{A}$.
3. **$(*)$ -closure:** $X \in \mathcal{I}_C \Rightarrow X^\dagger \in \mathcal{I}_C$.

Proof. All properties follow directly from construction: linearity from finite sums, two-sided closure from definition, and $(*)$ -closure imposed explicitly. $\square$

Lemma 0.4 (Minimal generating property). $\mathcal{I}_C$ is the smallest two-sided $(*)$ -ideal in $\mathcal{A}$ containing all $\hat{C}_\alpha$.

Proof. Let $\mathcal{J} \subset \mathcal{A}$ be any two-sided $(*)$ -ideal with $\hat{C}_\alpha \in \mathcal{J}$ for all α . Then $A \hat{C}_\alpha B \in \mathcal{J}$ for all $A, B \in \mathcal{A}$, any finite sum of such elements lies in $\mathcal{J}$, and adjoints also lie in $\mathcal{J}$. Hence $\mathcal{I}_C \subset \mathcal{J}$. Therefore $\mathcal{I}_C$ is minimal. $\square$

Lemma 0.5 (Adjoint consistency). If $\hat{C}_\alpha \in \mathcal{I}_C$, then $\hat{C}_\alpha^\dagger \in \mathcal{I}_C$.

Proof. Follows from $(*)$ -closure property of the ideal. $\square$

Lemma 0.6 (Constraint consequences remain constrained). If $X \in \mathcal{I}_C$, then for all $A, B \in \mathcal{A}$, the composite expression AXB also belongs to $\mathcal{I}_C$.

Proof. Since $\mathcal{I}_C$ is a left ideal, $AX \in \mathcal{I}_C$. Since $\mathcal{I}_C$ is also a right ideal, $(AX)B \in \mathcal{I}_C$. Therefore $AXB \in \mathcal{I}_C$. $\square$

0.2.8 Non-Degeneracy of the Constraint Ideal

The constraint ideal must not equal the entire algebra. We impose:

Assumption 0.3 (Non-degeneracy). $\mathcal{I}_C \neq \mathcal{A}$.

If $\mathcal{I}_C = \mathcal{A}$, then all elements are "constraint violations," and no non-trivial physical states can exist.

In particular, if a physical state ω were required to annihilate the constraint ideal, then $\omega(\mathbf{1}) = 0$ would follow. For a normalised physical state, however, one expects $\omega(\mathbf{1}) = 1$. Thus, if $\mathcal{I}_C = \mathcal{A}$, normalised physical states cannot exist.

0.2.9 Proper Constraint Systems

A constraint system is called **proper** if its generated constraint ideal is proper:

$$\mathcal{I}_C \subsetneq \mathcal{A}.$$

Equivalently, a proper constraint system does not generate the identity element:

$$\mathbf{1} \notin \mathcal{I}_C.$$

This does not yet prove that physical states exist. It only rules out the immediate obstruction in which the constraints collapse the entire algebra into inconsistency.

> **$\mathbf{1} \notin \mathcal{I}_C$ is necessary for a non-trivial physical sector, but not by itself sufficient.**

Existence of physical states requires an additional positivity-compatible state construction, introduced in the next subsection.

0.2.10 The Constraint Ideal as the Algebraic Inconsistency Sector

The ideal $\mathcal{I}_C$ should be interpreted as the algebraic sector of constraint-generated inconsistency.

This interpretation must be handled carefully. It does not mean that every element of $\mathcal{I}_C$ is a physical object with negative existence or destructive cancellation. It

means that these elements are algebraically generated from violations of primitive consistency requirements.

In this sense, $X \in \mathcal{I}_C$ means: X belongs to the constraint-generated inconsistency class.

The physical state condition then requires this inconsistency structure to vanish in the appropriate expectation-value sense.

0.2.11 Relation to the Constraint Ideal and Physical Equivalence

Although the quotient is not taken as the primitive definition of physicality, the constraint ideal naturally suggests an equivalence relation. Two algebra elements $A, B \in \mathcal{A}$ may be considered constraint-equivalent if:

$$A \sim_C B \quad \Longleftrightarrow \quad A - B \in \mathcal{I}_C.$$

In that case, A and B differ only by an element of the inconsistency sector. This relation is useful later when defining reduced observables, but at the foundational level it remains secondary to the physical state condition.

0.2.12 Guardrails for This Subsection

1. **No geometry assumed.** Constraints are purely algebraic—not spatial or geometric.
2. **No probabilistic meaning yet.** Constraints encode consistency, not probabilities.
3. **No dynamics.** They do not generate evolution; they define admissibility.
4. **No quotient taken yet.** We do not yet identify the physical algebra as $\mathcal{A}/\mathcal{I}_C$; this will be handled later.
5. **No claim of automatic equivalence.** The bridge between configuration-space and algebraic formulations is conditional on a faithful representation-linked model.

0.3 Physical States and Positivity

0.3.1 Purpose of the State Layer

The relational algebra $\mathcal{A}$ provides the primitive language of possible relational expressions. The constraint ideal $\mathcal{I}_C$ identifies the algebraic sector generated by inconsistency. But neither of these structures yet specifies what it means for a relational structure to be physically realised.

To make that distinction, the framework introduces states.

A state is not first understood as a vector in a Hilbert space. Nor is it initially a point in a classical phase space, a field configuration, or a probability distribution over a background spacetime. At the foundational level, a state is a functional on the relational algebra: it assigns values to algebraic expressions in a way compatible with linearity and positivity.

> **A physical state is a positive evaluation of relational structure in which constraint violation vanishes.**

This subsection defines the state concept, explains why positivity is required, and formulates the physical state condition.

0.3.2 Linear Functionals on the Relational Algebra

Definition 0.4 (Linear Functional).

A linear functional on $\mathcal{A}$ is a map $\omega: \mathcal{A} \rightarrow \mathbb{C}$ such that for all $A, B \in \mathcal{A}$ and all $\lambda, \mu \in \mathbb{C}$:

$$\omega(\lambda A + \mu B) = \lambda \omega(A) + \mu \omega(B).$$

The value $\omega(A)$ is interpreted as the state-evaluation of the relational expression A . At this stage, $\omega(A)$ should not automatically be interpreted as a measured expectation value in laboratory spacetime. That interpretation may become appropriate after additional representational and operational structure is introduced. Foundationally, ω is simply the object that evaluates relational expressions.

0.3.3 Why States Are Functionals Rather Than Vectors

The framework does not begin with a Hilbert space. Therefore, it cannot define physical states first as vectors $|\Psi\rangle \in \mathcal{H}$. Instead, the Hilbert-space picture must be derived later from the algebraic state.

This ordering is important. If one starts with a Hilbert space, one has already selected a representation of the relational algebra. That may be useful later, but it is too specific for the primitive layer. The algebraic state ω is more general: it can generate a Hilbert-space representation, but it is not itself defined by one.

The foundational direction is therefore:

$$\omega \quad \longrightarrow \quad (\mathcal{H}_\omega, \pi_\omega, \Omega_\omega),$$

not

$$\mathcal{H} \quad \longrightarrow \quad \omega.$$

The state comes before the represented Hilbert space.

0.3.4 Positivity

Definition 0.5 (Positive Functional).

A linear functional $\omega: \mathcal{A} \rightarrow \mathbb{C}$ is called positive if:

$$\omega(A^\dagger A) \geq 0 \quad \text{for all } A \in \mathcal{A}.$$

Here the expression $A^\dagger A$ is well-defined because $\mathcal{A}$ is a $(*)$ -algebra.

The positivity condition means that every algebraic square has a non-negative state-evaluation. This is the algebraic analogue of the familiar fact that squared magnitudes should not have negative expectation. However, this interpretation should be used only as

intuition. The formal role of positivity is deeper: it makes the state compatible with a Hilbert-space representation and allows the framework to use a Cauchy–Schwarz inequality.

0.3.5 Normalisation

A positive linear functional may also be normalised by requiring:

$$\omega(\mathbf{1}) = 1.$$

When this condition holds, ω is a normalised algebraic state.

For some purely structural arguments, normalisation is not essential. A non-zero positive functional satisfying the physical constraint condition may be enough. However, when ω is to be interpreted probabilistically or as an expectation-value state, normalisation is natural.

Thus, one may distinguish:

- **Positive linear functional:** $\omega(A^\dagger A) \geq 0$ for all A .
- **Normalised positive state:** $\omega(A^\dagger A) \geq 0$ and $\omega(\mathbf{1}) = 1$.

0.3.6 Definition — Algebraic State

Definition 0.6 (Algebraic State).

An algebraic state on $\mathcal{A}$ is a positive linear functional $\omega: \mathcal{A} \rightarrow \mathbb{C}$ satisfying:

$$\omega(A^\dagger A) \geq 0 \quad \text{for all } A \in \mathcal{A}.$$

If additionally $\omega(\mathbf{1}) = 1$, then ω is called a normalised algebraic state.

The set of algebraic states on $\mathcal{A}$ may be denoted $\mathcal{S}(\mathcal{A})$. The physical states will form a distinguished subset of this state space.

0.3.7 The Physical State Condition

A state is physical only if it is compatible with the constraints. Since the primitive constraints are $\hat{C}_\alpha \in \mathcal{A}$, the most direct condition might appear to be:

$$\omega(\hat{C}_\alpha) = 0.$$

However, this linear condition alone is too weak. It can allow cancellations at the level of linear expectation values without ensuring that the constraint itself vanishes in the state.

The stronger and more stable condition is quadratic:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \text{for all } \alpha \in I.$$

This is the foundational physical state condition. It says that each primitive constraint has zero positive square in the state. Since $\hat{C}_\alpha^\dagger \hat{C}_\alpha$ is an algebraic square, positivity ensures:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) \geq 0.$$

Requiring it to be zero therefore means that the constraint has no residual positive magnitude in the state.

0.3.8 Definition — Physical State

Definition 0.7 (Physical State).

A physical state of the relational constraint framework is a positive linear functional $\omega: \mathcal{A} \rightarrow \mathbb{C}$ such that:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \text{for all } \alpha \in I.$$

If normalisation is required, one additionally imposes $\omega(\mathbf{1}) = 1$.

The set of physical states may be denoted $\mathcal{S}_{\mathrm{phys}}(\mathcal{A}, \mathcal{C})$ or simply $\mathcal{S}_{\mathrm{phys}}$.

Thus:

$$\mathcal{S}_{\mathrm{phys}} = \left\{ \omega \in \mathcal{S}(\mathcal{A}) : \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \text{ for all } \alpha \right\}.$$

This definition is the first precise statement of physical admissibility in the framework.

0.3.9 Why Quadratic Vanishing is Stronger Than Linear Vanishing

The condition $\omega(\hat{C}_\alpha) = 0$ does not by itself guarantee that the constraint is absent in the state. It only says that the average value of the constraint vanishes. In many algebraic settings, a quantity can have zero expectation while its square has non-zero expectation.

By contrast:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$$

states that the positive magnitude of the constraint vanishes. Because positivity ensures $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) \geq 0$, zero is the minimal possible value. The constraint has no residual quadratic weight in the state.

This is why the physical state condition uses $\hat{C}_\alpha^\dagger \hat{C}_\alpha$ rather than only $\hat{C}_\alpha$.

> **A physical state does not merely average constraint violation to zero; it eliminates constraint violation in positive norm.**

0.3.10 Positivity and Cauchy–Schwarz

A crucial consequence of positivity is that it induces a Cauchy–Schwarz inequality.

Given a positive linear functional $\omega: \mathcal{A} \rightarrow \mathbb{C}$, define the sesquilinear form:

$$\langle A, B \rangle_\omega := \omega(A^\dagger B).$$

This form is positive semidefinite because $\langle A, A \rangle_\omega = \omega(A^\dagger A) \geq 0$.

Therefore, it satisfies the Cauchy–Schwarz inequality:

$$|\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A) \omega(B^\dagger B).$$

This inequality is the technical reason why quadratic constraint vanishing is powerful. It allows one to show that if a constraint has zero quadratic weight, then insertions of that constraint vanish in state evaluations.

0.3.11 Lemma — Null Constraint Annihilation

Lemma 0.7 (Null Constraint Annihilation).

Let $\omega: \mathcal{A} \rightarrow \mathbb{C}$ be a positive linear functional. If $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$ for some constraint $\hat{C}_\alpha$, then:

$$\omega(\hat{C}_\alpha^\dagger A) = 0 \quad \text{for all } A \in \mathcal{A},$$

and

$$\omega(A^\dagger \hat{C}_\alpha) = 0 \quad \text{for all } A \in \mathcal{A}.$$

Proof.

Apply the Cauchy–Schwarz inequality to $X = \hat{C}_\alpha$ and $Y = A$. Then:

$$|\omega(\hat{C}_\alpha^\dagger A)|^2 \leq \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) \omega(A^\dagger A).$$

By assumption, $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$. Therefore:

$$|\omega(\hat{C}_\alpha^\dagger A)|^2 \leq 0.$$

Since the left-hand side is non-negative, it must be zero:

$$\omega(\hat{C}_\alpha^\dagger A) = 0.$$

Similarly, applying Cauchy–Schwarz to $X = A$ and $Y = \hat{C}_\alpha$:

$$|\omega(A^\dagger \hat{C}_\alpha)|^2 \leq \omega(A^\dagger A) \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0.$$

Therefore:

$$\omega(A^\dagger \hat{C}_\alpha) = 0. \quad \square$$

0.3.12 Extension to the Ideal

The physical state condition is stated on the primitive constraints:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \text{for all } \alpha.$$

However, because the constraint ideal is generated by arbitrary left and right multiplication of those constraints, the effect extends to many ideal-generated expressions.

Lemma 0.8 (Ideal Annihilation).

Let ω be a physical state. Then for any $X \in \mathcal{I}_C$ of the form $X = \sum_i A_i \hat{C}_{\alpha_i} B_i$, we have $\omega(X) = 0$.

Proof.

Using linearity, $\omega(X) = \sum_i \omega(A_i \hat{C}_{\alpha_i} B_i)$.

Using adjoint structure, $\omega(A_i \hat{C}_{\alpha_i} B_i) = \omega(\big((B_i^\dagger \hat{C}_{\alpha_i}^\dagger A_i^\dagger)^\dagger\big)) = \overline{\omega(B_i^\dagger \hat{C}_{\alpha_i}^\dagger A_i^\dagger)}$.

By Lemma 0.7, $\omega(\hat{C}_{\alpha_i}^\dagger A_i^\dagger) = 0$. Repeated application of the Cauchy-Schwarz inequality for positive linear functionals ensures that $\omega(B_i^\dagger \hat{C}_{\alpha_i}^\dagger A_i^\dagger) = 0$ as well. Therefore, each term vanishes, and $\omega(X) = 0$. $\blacksquare$

0.3.13 Interpretation of Zero in the Physical State Condition

The condition $\omega(\hat{C}_{\alpha}^\dagger \hat{C}_{\alpha}) = 0$ should not be interpreted as saying that the universe is empty, that all physical quantities vanish, or that every structure is cancelled by an opposite structure.

It says only that the positive square of each constraint has zero state-evaluation. In other words:

> **Zero means zero constraint violation, not zero existence.**

A physical state may support non-trivial relational structure, non-zero observable values, correlations, causal dependencies, emergent geometry, and effective dynamics. What vanishes is the inconsistency encoded by the primitive constraints.

0.3.14 Existence of Physical States

The properness condition $\mathcal{I}_C \subsetneq \mathcal{A}$ prevents immediate triviality, but it does not by itself guarantee that physical states exist.

The existence of a positive functional satisfying $\omega(\hat{C}_{\alpha}^\dagger \hat{C}_{\alpha}) = 0$ for all α is an additional mathematical question. In concrete models, it may require construction, representation theory, compactness arguments, quotient positivity, or other methods.

Proposition 0.1 (Sufficient Condition for Physical-State Existence).

If the reduced algebra $\mathcal{A}/\mathcal{I}_C$ admits a positive linear functional $\tilde{\omega}$, then the pullback $\omega = \tilde{\omega} \circ q$ (where $q: \mathcal{A} \rightarrow \mathcal{A}/\mathcal{I}_C$ is the quotient map) is a physical state of $\mathcal{A}$.

Proof.

Positivity is inherited by pullback. Since every constraint element maps to zero in the quotient, $\omega(\hat{C}_{\alpha}^\dagger \hat{C}_{\alpha}) = \tilde{\omega}(q(\hat{C}_{\alpha}^\dagger \hat{C}_{\alpha})) = \tilde{\omega}(0) = 0$. $\blacksquare$

Therefore, the foundational definition should be accompanied by the following guardrail:

> **$\mathcal{I}_C \subsetneq \mathcal{A}$ is necessary for non-triviality, but physical states must still be shown or assumed to exist.**

0.3.15 The Physical Sector as a State-Defined Object

The physical sector is first defined not as a subspace of a Hilbert space, but as a subset of the algebraic state space:

$$\mathfrak{S}_{\mathrm{phys}} \subseteq \mathfrak{S}(\mathcal{A}).$$

Explicitly:

$$\mathfrak{S}_{\mathrm{phys}} = \left\{ \omega: \mathcal{A} \rightarrow \mathbb{C}; \omega \text{ positive and } \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \text{ for all } \alpha \right\}.$$

Only after a state $\omega \in \mathfrak{S}_{\mathrm{phys}}$ is chosen does one construct a represented Hilbert-space physical sector, such as $\mathscr{H}_{\mathrm{phys}}^\omega$.

Thus, the order remains:

$$\text{physical state} \rightarrow \text{represented physical Hilbert space}.$$

0.3.16 Guardrails for This Subsection

1. **No Hilbert space yet.** States are defined abstractly—no vectors or operators are assumed.
2. **Positivity is not optional.** It is required for quadratic constraint conditions, Cauchy–Schwarz, and the later GNS representation.
3. **The physical state condition is quadratic:** $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$, not merely linear: $\omega(\hat{C}_\alpha) = 0$.
4. **Zero means vanishing of constraint violation, not literal nothingness.**
5. **Properness of the constraint ideal does not automatically guarantee the existence of physical states.**
6. **Full annihilation of the entire constraint ideal is stronger than primitive quadratic constraint vanishing and should be stated separately when needed.**
7. **The physical sector is first a state-defined object, not a pre-existing Hilbert subspace.**

0.4 GNS Representation and the Derived Master Constraint

0.4.1 Purpose of the Representation Layer

The preceding subsections established the primitive algebraic foundation of the framework: a relational algebra $\mathcal{A}$, a constraint ideal $\mathcal{I}_C$, and physical states ω satisfying the quadratic constraint-vanishing condition.

At this stage, no Hilbert space has been assumed. This is intentional. The framework does not begin with vectors, wavefunctions, operators on a pre-existing Hilbert space, or a background quantum state space. Instead, it begins with algebraic relations and physical state functionals.

The purpose of this subsection is to show how a Hilbert-space representation arises from a physical state rather than being imposed at the beginning. This is done through the Gelfand–Naimark–Segal (GNS) construction.

> **Hilbert space is generated by a physical state; it is not primitive.**

The GNS construction plays a crucial role in the framework. It allows the algebraic foundation to connect with the familiar language of vectors, operators, cyclic states, kernels, and constraint annihilation, while preserving the deeper claim that the algebra-state structure comes first.

This subsection also introduces the **master constraint operator** as a representation-level packaging of the primitive constraints. The master constraint is not primitive. It is a derived object that appears only after a state has been chosen and a representation has been constructed.

0.4.2 Why a Representation Is Needed

The algebraic formulation is the most primitive level of the framework, but many later constructions require represented objects. For example, to speak about a physical Hilbert space, a master constraint operator, kernels of operators, or vectors annihilated by constraints, one needs a representation of $\mathcal{A}$ on some Hilbert space.

However, the representation must not be chosen arbitrarily at the foundation. If one simply postulates a Hilbert space $\mathcal{H}$ and a representation $\pi: \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H})$, then the primitive theory would already depend on a selected representation. That would weaken the algebra-first structure.

The GNS construction avoids this problem. It says that once a positive state ω is given, the Hilbert space associated with that state can be constructed from the algebra itself.

Thus the direction is:

$$(\mathcal{A}, \omega) \quad \longrightarrow \quad (\mathcal{H}_\omega, \pi_\omega, \Omega_\omega).$$

The state chooses the representation. The representation does not define the state at the primitive level.

0.4.3 The State-Induced Inner Product

Let $\omega: \mathcal{A} \rightarrow \mathbb{C}$ be a positive linear functional.

Define a sesquilinear form on $\mathcal{A}$ by:

$$\langle A, B \rangle_\omega := \omega(A^\dagger B), \quad A, B \in \mathcal{A}.$$

This form is linear in its second argument and conjugate-linear in its first argument, following the usual mathematical convention for Hilbert-space inner products.

Because ω is positive:

$$\langle A, A \rangle_\omega = \omega(A^\dagger A) \geq 0.$$

Therefore, $\langle \cdot, \cdot \rangle_\omega$ is a positive semidefinite sesquilinear form.

It may fail to be positive definite. That is, there may exist non-zero algebra elements $A \in \mathcal{A}$ such that:

$$\omega(A^\dagger A) = 0.$$

Such elements have zero length with respect to the state-induced form. They must be quotiented out before a Hilbert space can be formed.

0.4.4 The Null Space of the State

Definition 0.8 (GNS Null Space).

Let ω be a positive linear functional on $\mathcal{A}$. The GNS null space (or simply the null space) associated with ω is:

$$\mathcal{N}_\omega := \{ A \in \mathcal{A} : \omega(A^\dagger A) = 0 \}.$$

Equivalently:

$$\mathcal{N}_\omega = \{ A \in \mathcal{A} : \langle A, A \rangle_\omega = 0 \}.$$

Elements of $\mathcal{N}_\omega$ are algebraically non-zero in $\mathcal{A}$, but have zero state-induced length. In the representation determined by ω , they become null.

Lemma 0.9 (Null Space Is a Left Ideal).

Let ω be a positive linear functional on $\mathcal{A}$. Then $\mathcal{N}_\omega$ is a left ideal of $\mathcal{A}$.

Proof.

Let $A \in \mathcal{N}_\omega$ and $B \in \mathcal{A}$. We must show that $BA \in \mathcal{N}_\omega$, i.e., that $\omega((BA)^\dagger(BA)) = 0$.

Compute:

$$(BA)^\dagger(BA) = A^\dagger B^\dagger B A.$$

Thus:

$$\omega((BA)^\dagger(BA)) = \omega(A^\dagger B^\dagger B A).$$

By the Cauchy–Schwarz inequality for positive linear functionals, if $\omega(A^\dagger A) = 0$, then $\omega(A^\dagger X) = 0$ for all $X \in \mathcal{A}$ (Lemma 0.7). Set $X = B^\dagger B A$. Then:

$$\omega(A^\dagger B^\dagger B A) = 0.$$

Therefore:

$$\omega((BA)^\dagger(BA)) = 0,$$

so $BA \in \mathcal{N}_\omega$. Hence $\mathcal{N}_\omega$ is a left ideal.

□

Remark 0.1. $\mathcal{N}_\omega$ is generally not a two-sided ideal—only a left ideal. This asymmetry is why GNS constructions require quotienting carefully.

0.4.5 The Constraint Ideal Sits Inside the Null Space

****Theorem 0.4 (Constraint Inclusion).****

Let ω be a physical state. Then $\mathcal{I}_C \subseteq \mathcal{N}_\omega$.

Proof.

Take any $X \in \mathcal{I}_C$. By definition, $X = \sum_i A_i \hat{C}_{\alpha_i} B_i$ for some $A_i, B_i \in \mathcal{A}$ and $\alpha_i \in I$.

By Lemma 0.8, $\omega(X) = 0$.

We now show $\omega(X^\dagger X) = 0$. Each term in $X^\dagger X$ contains factors of the form $B^\dagger \hat{C}^\dagger A^\dagger A \hat{C} B$. By repeated application of Lemma 0.7, every such contribution vanishes under ω .

Hence $\omega(X^\dagger X) = 0$, so $X \in \mathcal{N}_\omega$. Therefore $\mathcal{I}_C \subseteq \mathcal{N}_\omega$. $\square$

****Interpretation.**** Constraint violations are null in any physical state. This is stronger than $\omega(X) = 0$; it ensures $\omega(X^\dagger X) = 0$, so violations are completely eliminated at the quadratic level.

0.4.6 Construction of the Pre-Hilbert Space

Because $\mathcal{N}_\omega$ is a left ideal, we can form the quotient vector space:
 $\mathcal{D}_\omega := \mathcal{A} / \mathcal{N}_\omega$.

Elements of $\mathcal{D}_\omega$ are equivalence classes:
 $[A]_\omega := A + \mathcal{N}_\omega$.

****Definition 0.9 (Induced Inner Product).****

Define an inner product on $\mathcal{D}_\omega$ by:

$$\langle [A]_\omega, [B]_\omega \rangle_\omega := \omega(A^\dagger B).$$

****Lemma 0.10 (Well-Definedness).****

The above definition does not depend on the choice of representatives.

Proof.

Let $A' = A + N_1$ and $B' = B + N_2$ with $N_1, N_2 \in \mathcal{N}_\omega$. Then:

$$\omega((A')^\dagger B') = \omega(A^\dagger B) + \text{terms involving } N_1, N_2.$$

All additional terms vanish because $\omega(N^\dagger X) = 0$ and $\omega(X^\dagger N) = 0$ for all $X \in \mathcal{A}$ and $N \in \mathcal{N}_\omega$ (by Lemma 0.7 and its conjugate). Thus:

$$\langle [A']_\omega, [B']_\omega \rangle_\omega = \langle [A]_\omega, [B]_\omega \rangle_\omega. \quad \square$$

****Lemma 0.11 (Positive Definiteness).****

After quotienting by the null space, the inner product becomes positive definite:

$$\langle [A]_\omega, [A]_\omega \rangle_\omega = 0 \quad \Longleftrightarrow \quad [A]_\omega = 0.$$

Proof.

If $\omega(A^\dagger A) = 0$, then $A \in \mathcal{N}_\omega$ by definition, so $[A]_\omega = 0$. $\square$

Therefore, $\mathcal{D}_\omega$ is a pre-Hilbert space.

0.4.7 The Hilbert Space Completion

Definition 0.10 (GNS Hilbert Space).

The GNS Hilbert space $\mathcal{H}_\omega$ is the completion of $\mathcal{D}_\omega = \mathcal{A}/\mathcal{N}_\omega$ under the induced norm:

$$\| [A]_\omega \| := \sqrt{\omega(A^\dagger A)}.$$

This is now a genuine Hilbert space, constructed purely from $(\mathcal{A}, \omega)$.

0.4.8 The Representation Map

Definition 0.11 (GNS Representation).

Define the representation map $\pi_\omega: \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H}_\omega)$ by:

$$\pi_\omega(A)[B]_\omega := [AB]_\omega.$$

This defines the action of the algebra on the pre-Hilbert space by left multiplication. In suitable bounded or completed settings, this extends to:

$$\pi_\omega: \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H}_\omega).$$

Lemma 0.12 (Representation Is Well-Defined).

If $[B]_\omega = [B']_\omega$, then $[AB]_\omega = [AB']_\omega$.

Proof.

If $B' = B + N$ with $N \in \mathcal{N}_\omega$, then $AB' = AB + AN$. By Lemma 0.9, $AN \in \mathcal{N}_\omega$ (since $\mathcal{N}_\omega$ is a left ideal). Thus $[AB']_\omega = [AB]_\omega$. $\square$

Lemma 0.13 (Representation Homomorphism).

For all $A, B \in \mathcal{A}$:

1. $\pi_\omega(A + B) = \pi_\omega(A) + \pi_\omega(B)$,
2. $\pi_\omega(AB) = \pi_\omega(A)\pi_\omega(B)$.

Proof.

Direct from definition:

$$\begin{aligned} \pi_\omega(AB)[C]_\omega &= [(AB)C]_\omega = [A(BC)]_\omega = \\ \pi_\omega(A)[BC]_\omega &= \pi_\omega(A)\pi_\omega(B)[C]_\omega. \quad \square \end{aligned}$$

****Lemma 0.14 (($\cdot$)-Representation Property).**

$$\langle \pi_\omega(A^\dagger) \rangle = \langle \pi_\omega(A) \rangle^\dagger.$$

Proof.

Compute:

$$\langle \pi_\omega(A)[B]_\omega, [C]_\omega \rangle = \langle [AB]_\omega, [C]_\omega \rangle = \langle \omega((AB)^\dagger C) \rangle = \langle \omega(B^\dagger A^\dagger C) \rangle = \langle [B]_\omega, [A^\dagger C]_\omega \rangle.$$

$$\text{Thus } \langle \pi_\omega(A) \rangle^\dagger = \langle \pi_\omega(A^\dagger) \rangle. \quad \square$$

0.4.9 The Cyclic Vector

****Definition 0.12 (Cyclic Vector).**

Define the cyclic vector:

$$\Omega_\omega := [\mathbf{1}]_\omega.$$

****Lemma 0.15 (Cyclicity).**

$$\overline{\langle \pi_\omega(\mathcal{A}) \Omega_\omega \rangle} = \mathcal{H}_\omega.$$

Proof.

$\langle \pi_\omega(A) \Omega_\omega \rangle = [A]_\omega$. All equivalence classes are generated this way. $\square$

0.4.10 State Reconstruction

****Theorem 0.5 (GNS Representation Theorem).**

For all $A \in \mathcal{A}$:

$$\langle \omega(A) \rangle = \langle \Omega_\omega, \pi_\omega(A) \Omega_\omega \rangle.$$

Proof.

$$\langle \Omega_\omega, \pi_\omega(A) \Omega_\omega \rangle = \langle [\mathbf{1}]_\omega, [A]_\omega \rangle = \langle \omega(\mathbf{1}^\dagger A) \rangle = \langle \omega(A) \rangle. \quad \square$$

This equation explains how the familiar expectation-value form arises from the algebraic state.

0.4.11 Constraint Action in the Representation

****Theorem 0.6 (Vacuum Annihilation).**

Let ω be a physical state. Then:

$$\langle \pi_\omega(\hat{C}_\alpha) \Omega_\omega \rangle = 0 \quad \forall \alpha \in I.$$

Proof.

$$\begin{aligned} \langle \pi_\omega(\hat{C}_\alpha) \Omega_\omega \rangle^2 &= \langle \pi_\omega(\hat{C}_\alpha) [\mathbf{1}]_\omega \rangle^2 = \langle [\hat{C}_\alpha]_\omega \rangle^2 = \langle \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) \rangle = 0. \end{aligned}$$

A vector of zero Hilbert norm is the zero vector. $\square$

****Interpretation.**** This is one of the most important structural results:

> ****Constraints appear as annihilation operators on the cyclic vector.****

The familiar statement $\hat{C}_\alpha |\Psi\rangle = 0$ is not primitive in this framework. It is the represented version of the algebraic state condition.

The correct hierarchy is:

$$\begin{aligned} \pi_\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) &= 0 \quad \Longleftrightarrow \quad \pi_\omega(\hat{C}_\alpha) \Omega_\omega = 0. \end{aligned}$$

The vector equation is derived from the state condition, not assumed before it.

0.4.12 The Represented Physical Sector

****Definition 0.13 (Represented Physical Hilbert Space).****

The represented physical sector associated with the GNS representation is:

$$\pi_\omega(\mathcal{H}_{\mathrm{phys}})^\omega := \bigcap_{\alpha \in I} \ker \pi_\omega(\hat{C}_\alpha).$$

Equivalently, it is the simultaneous kernel of all represented primitive constraints.

Because Ω_ω satisfies $\pi_\omega(\hat{C}_\alpha) \Omega_\omega = 0$ for all α , we have:

$$\Omega_\omega \in \mathcal{H}_{\mathrm{phys}}^\omega.$$

****Theorem 0.7 (Ideal-Kernel Correspondence).****

Let $\mathcal{I}_C \subset \mathcal{A}$ be the constraint ideal. Then:

$$\psi \in \mathcal{H}_{\mathrm{phys}}^\omega \quad \Longleftrightarrow \quad \pi_\omega(X)\psi = 0 \quad \text{for all } X \in \mathcal{I}_C.$$

Proof.

($\Rightarrow$) If $\psi \in \mathcal{H}_{\mathrm{phys}}^\omega$, then $\pi_\omega(\hat{C}_\alpha)\psi = 0$ for all α . Since $\mathcal{I}_C$ is generated by combinations $X = \sum_i A_i \hat{C}_{\alpha_i} B_i$, we get $\pi_\omega(X)\psi = \sum_i \pi_\omega(A_i) \pi_\omega(\hat{C}_{\alpha_i}) \pi_\omega(B_i)\psi = 0$.

($\Leftarrow$) If $\pi_\omega(X)\psi = 0$ for all $X \in \mathcal{I}_C$, then in particular $\pi_\omega(\hat{C}_\alpha)\psi = 0$. Thus $\psi \in \ker \hat{\mathfrak{M}}_\omega$. $\blacksquare$

0.4.13 Representation Dependence

The GNS construction depends on the state ω . Different physical states may generate different Hilbert spaces:

$$\mathcal{H}_{\omega_1}, \quad \mathcal{H}_{\omega_2},$$

different representations:

$$\pi_{\omega_1}, \sqsubset \pi_{\omega_2},$$

and different cyclic vectors:

$$\Omega_{\omega_1}, \sqsubset \Omega_{\omega_2}.$$

This is not a defect. It reflects the fact that Hilbert-space structure is not primitive. The primitive layer is the algebra and the state condition.

Therefore, one should not speak of "the" fundamental Hilbert space unless an additional theorem or assumption identifies a preferred representation. The careful phrase is: "the GNS Hilbert space associated with the state ω ."

0.4.14 The Derived Master Constraint Operator

Now that the representation is available, the framework can package the represented constraints into a single positive operator.

Definition 0.14 (Represented Master Constraint Operator).

Let $(\mathcal{H}_\omega, \pi_\omega, \Omega_\omega)$ be the GNS representation. Let $w_\alpha > 0$ be strictly positive weights. Define:

$$\hat{\mathcal{M}}_\omega := \sum_{\alpha \in I} w_\alpha \pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha).$$

Technical Note. If the index set I is infinite, the sum is taken in an appropriate operator topology. For finite sets, it reduces to an ordinary finite sum.

Lemma 0.16 (Positivity of $\hat{\mathcal{M}}_\omega$).

For all $\psi \in \mathcal{H}_\omega$:

$$\langle \psi, \hat{\mathcal{M}}_\omega \psi \rangle \geq 0.$$

Proof.

$$\langle \psi, \hat{\mathcal{M}}_\omega \psi \rangle = \sum_\alpha w_\alpha \|\pi_\omega(\hat{C}_\alpha) \psi\|^2.$$

Each term is non-negative and $w_\alpha > 0$. $\square$

0.4.15 Kernel Equivalence

Theorem 0.8 (Master-Zero Equivalence).

$$\ker \hat{\mathcal{M}}_\omega = \bigcap_{\alpha \in I} \ker \pi_\omega(\hat{C}_\alpha).$$

Proof.

$(\Rightarrow)$ If $\psi \in \ker \hat{\mathcal{M}}_\omega$, then $\langle \psi, \hat{\mathcal{M}}_\omega \psi \rangle = 0$. Since each term is non-negative, $\|\pi_\omega(\hat{C}_\alpha) \psi\|^2 = 0$ for all α . Thus $\pi_\omega(\hat{C}_\alpha) \psi = 0$ for all α .

$(\Leftarrow)$ If $\pi_\omega(\hat{C}_\alpha) \psi = 0$ for all α , then every term in the sum vanishes: $\langle \psi, \hat{\mathcal{M}}_\omega \psi \rangle = 0$. $\square$

****Corollary 0.1.****

$$\mathscr{H}_{\mathrm{phys}}^{\omega} = \ker \hat{\mathfrak{M}}_{\omega}. \quad \mathbb{R}$$

$$\Omega_{\omega} \in \ker \hat{\mathfrak{M}}_{\omega}. \quad \mathbb{R}$$

0.4.16 Why the Master Constraint Is Not Primitive

The master constraint operator $\hat{\mathfrak{M}}_{\omega}$ is a derived object, not a primitive one. Its definition depends on additional choices not present in the primitive algebraic layer:

1. A physical state ω ,
2. The GNS representation π_{ω} ,
3. A Hilbert space $\mathscr{H}_{\omega}$,
4. Weights $w_{\alpha} > 0$,
5. Domain and convergence conditions if the constraint family is infinite.

The primitive data are:

$$\mathcal{A}, \quad \mathcal{I}_C, \quad \omega. \quad \mathbb{R}$$

The master constraint appears only after representation.

The correct order is:

$$\mathcal{A}, \quad \mathcal{I}_C, \quad \omega \quad \longrightarrow \quad (\mathscr{H}_{\omega}, \pi_{\omega}, \Omega_{\omega}) \quad \longrightarrow \quad \hat{\mathfrak{M}}_{\omega}. \quad \mathbb{R}$$

0.4.17 The Effective Classical Master Functional

In an effective classical description, one may introduce configuration-space constraint maps

$C_{\alpha}: \mathcal{K} \rightarrow V_{\alpha}$ and define:

$$\mathfrak{M}(x) = \sum_{\alpha} w_{\alpha} |C_{\alpha}(x)|^2. \quad \mathbb{R}$$

A configuration $x \in \mathcal{K}$ is effective-physical if $\mathfrak{M}(x) = 0$.

****Theorem 0.9 (Classical Zero-Equivalence).****

$$\mathfrak{M}(x) = 0 \quad \Longleftrightarrow \quad C_{\alpha}(x) = 0 \quad \forall \alpha. \quad \mathbb{R}$$

Proof. Each term $w_{\alpha} |C_{\alpha}(x)|^2 \geq 0$. A sum of non-negative terms is zero iff every term is zero. $\square$

****Proposition 0.2.****

The effective master functional $\mathfrak{M}$ is a derived object, not a primitive one. Its definition depends on:

1. A configuration space $\mathcal{K}$,
2. Effective constraint maps C_{α} ,
3. Norms on codomains V_{α} ,
4. Positive weights w_{α} ,
5. A representation/state-to-configuration assignment.

By contrast, the primitive framework requires only $(\mathcal{A}, \mathcal{I}_C, \omega)$.

Corollary 0.2 (Conditional Correspondence).

Assume a representation-linked configuration model $(\mathcal{K}, \mathcal{C}_\alpha, \mathcal{X})$ for $(\mathcal{A}, \omega)$ with the property: for every relevant state or represented vector s , $\mathcal{C}_\alpha(\mathcal{X}(s)) = 0$ if and only if the represented constraint associated with $\hat{\mathcal{C}}_\alpha$ vanishes on s . Then:
 $\mathfrak{M}(\mathcal{X}(s)) = 0 \iff s \text{ is physical.}$

0.4.18 The Complete Foundational Hierarchy

The full foundational hierarchy can now be summarised as:

$$\boxed{\mathcal{A}} \iff \hat{\mathcal{C}}_\alpha \iff \mathcal{I}_C \iff \omega \iff (\mathcal{H}_\omega, \pi_\omega, \Omega_\omega) \iff \hat{\mathfrak{M}}_\omega \iff \mathfrak{M}(x).$$

Each arrow represents a step of additional structure. The ordering matters:

1. $\mathcal{A}$ is the primitive relational algebra.
2. $\hat{\mathcal{C}}_\alpha$ are primitive constraint elements.
3. $\mathcal{I}_C$ is the algebraic inconsistency sector generated by them.
4. ω is a positive physical state functional.
5. $(\mathcal{H}_\omega, \pi_\omega, \Omega_\omega)$ is the GNS representation.
6. $\hat{\mathfrak{M}}_\omega$ is the represented master constraint operator.
7. $\mathfrak{M}(x)$ is an effective classical master functional.

0.4.19 Guardrails for This Subsection

1. **Representation dependence.** $\mathcal{H}_{\text{phys}}^\omega$ depends on ω . Different states yield different Hilbert spaces.
2. **The master operator is derived.** $\hat{\mathfrak{M}}_\omega$ depends on π_ω . It is not foundational.
3. **No dynamics yet.** The kernel defines admissibility, not evolution.
4. **No geometry yet.** This is still purely algebraic/operator structure.
5. **Hilbert space is not primitive.** The GNS construction is derived structure.
6. **No probability interpretation yet.** The inner product exists, but probabilities are not yet introduced.
7. **The effective classical functional is not primitive.** It is a derived, coarse-grained shadow of the algebraic constraint structure.
8. **No ontological reversal.** The effective master functional must not be treated as more fundamental than the algebraic state-constraint structure.
9. **No claim of automatic equivalence.** The bridge between configuration-space and algebraic formulations is conditional on a faithful representation-linked model.
10. **No dynamics from $\mathfrak{M}$.** The effective master functional is a consistency selector, not an evolution law.

11. ****No claim that $\mathcal{K}$ is spacetime.**** At this stage $\mathcal{K}$ is only an effective configuration space; spacetime-like structure will emerge later from correlation geometry and causal/burden structure.

0.5 Foundational Guardrails and Summary

0.5.1 Purpose of the Guardrail Subsection

The preceding subsections established the algebraic foundation of the framework:

$\mathcal{A}, \quad \{\hat{C}_\alpha\}_{\alpha \in I}, \quad \mathcal{I}_C, \quad \omega, \quad (\mathcal{H}_\omega, \pi_\omega, \Omega_\omega), \quad \hat{\mathcal{M}}_\omega, \quad \mathcal{M}(x).$

These objects form a hierarchy. The relational algebra comes first. The constraint ideal is generated inside it. Physical states are positive linear functionals compatible with the constraints. Hilbert space arises only after a state is chosen. The master constraint operator is a representation-level packaging. The classical master functional is an effective or coarse-grained shadow of the represented constraint structure.

Because the framework uses familiar mathematical words—state, operator, constraint, zero, Hilbert space, functional, physical sector—it is easy to misread the foundation as making stronger claims than it actually makes. This subsection records the main guardrails needed to keep the theory theorem-safe.

> ****The foundation must say exactly what it proves, and no more.****

The guardrails below prevent premature interpretation. They also protect later emergence claims by ensuring that space, time, matter, geometry, and dynamics are not smuggled into the starting assumptions.

0.5.2 Guardrail I — Zero Is Constraint Compatibility, Not Nothingness

The most important guardrail concerns the meaning of zero.

The foundational physical state condition is:

$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \text{for all } \alpha.$

In a represented Hilbert-space setting, this becomes:

$\pi_\omega(\hat{C}_\alpha) \Omega_\omega = 0.$

In the master-constraint packaging, it becomes:

$\hat{\mathcal{M}}_\omega \Omega_\omega = 0.$

$$\frac{M}{x} = 0.$$

In all these cases, zero means that **constraint violation vanishes**. It does not mean:

- that reality is empty,
- that no relational structure exists,
- that all observable quantities vanish,
- that the physical state is trivial.

A physical state may contain non-trivial correlations, admissible relational structure, emergent geometry, causal ordering, records, field-like behaviour, and effective dynamics. What vanishes is the forbidden constraint content.

> ****Zero means zero inconsistency, not zero being.****

Throughout the manuscript, the phrase "master zero" should be read as shorthand for "vanishing of total constraint violation." It should not be read as a metaphysical claim that the universe is literally nothing.

— — —

0.5.3 Guardrail II — Zero Is Not Pairwise Cancellation

The master constraint has the positive form:

$$\|\hat{M}\|_{\omega} = \sum_{\alpha} w_{\alpha} \|\hat{C}_{\alpha}\|_{\omega}^{\dagger} \|\hat{C}_{\alpha}\|_{\omega}, \quad w_{\alpha} > 0.$$

For any vector ψ in the appropriate domain:

$$\langle \psi, \hat{\mathcal{M}}_\omega \psi \rangle = \sum_\alpha w_\alpha |\langle \psi | \mathcal{C}_\alpha \rangle|^2.$$

Similarly, the classical effective master functional has the form:

$$\mathbb{E} \frac{M}{n}(x) = \sum_{\alpha \in \mathcal{A}} w_{\alpha} |C_{\alpha}(x)|^2.$$

Every term is non-negative. Therefore, the vanishing of the sum cannot arise from cancellation between positive and negative terms. It can only occur when each individual term vanishes.

Thus:

$$\frac{M}{x} = 0 \quad \Longleftrightarrow \quad C_\alpha(x) = 0 \quad \text{for all } \alpha,$$

and

$$\langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0 \quad \Longleftrightarrow \quad \langle \hat{C}_\alpha \rangle_\psi = 0 \quad \text{for all } \alpha.$$

> ****Master zero is not cancellation; it is simultaneous constraint satisfaction.****

This matters because later parts of the framework may discuss duality, balance, charge-like compensation, source terms, or boundary exchange. Those later ideas must not be retroactively projected onto the foundational zero condition.

At the foundation, zero does not mean:

$$\text{\text{positive content}} + \text{\text{negative content}} = 0.$$

It means:

$$\text{\text{each constraint violation vanishes individually.}}$$

0.5.4 Guardrail III — The Constraint Ideal Is Not Automatically the Physical Observable Algebra

The constraint ideal $\mathcal{I}_C$ is the algebraic sector generated by the primitive constraints. It records inconsistency and its algebraic consequences. It is not itself the algebra of physical observables.

Likewise, the full relational algebra $\mathcal{A}$ is not automatically the physical observable algebra. Elements of $\mathcal{A}$ are primitive relational expressions. Some may become observables, but only after satisfying additional compatibility conditions.

A physical observable must preserve the physical sector, commute with constraints in an appropriate sense, or descend consistently to a quotient by the constraint ideal. Those conditions are not automatic from membership in $\mathcal{A}$.

Thus, one must distinguish:

$$A \in \mathcal{A} \not\sqsubset \text{\text{from}} \sqsubset A \in \mathcal{A}_{\text{\text{obs}}}.$$

The first means that A is a possible relational expression. The second, when defined later, means that A is physically admissible as an observable.

> **$\mathcal{A}$ is the kinematic relational algebra, not yet the physical observable algebra.**

The observable algebra must be constructed later as a quotient, commutant, or kernel-preserving subalgebra.

0.5.5 Guardrail IV — Do Not Quotient Too Early

Once the constraint ideal is defined, it is mathematically natural to consider the quotient:

$$\mathcal{A} / \mathcal{I}_C.$$

This quotient may indeed become useful. It identifies algebra elements that differ by constraint-generated expressions. It may appear later when discussing reduced observables or physical equivalence classes.

However, the quotient is not the primitive definition of physicality in this framework.

The foundation defines physicality through positive states satisfying:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0.$$

This state-based condition is stronger and more structured than simply declaring the constraints to be zero by quotienting. It preserves positivity, supports the GNS construction, and gives a controlled path to Hilbert-space representation.

Therefore, the foundational order is:

$$\mathcal{A} \quad \longrightarrow \quad \mathcal{I}_C \quad \longrightarrow \quad \omega \quad \longrightarrow \quad \mathcal{H}_\omega,$$

not immediately:

$$\mathcal{A} \quad \longrightarrow \quad \mathcal{A}/\mathcal{I}_C.$$

> **The quotient may be useful later, but physicality is first state-defined.**

0.5.6 Guardrail V — Hilbert Space Is Derived, Not Primitive

Hilbert space appears through the GNS construction associated with a positive state:

$$(\mathcal{A}, \omega) \quad \longrightarrow \quad (\mathcal{H}_\omega, \pi_\omega, \Omega_\omega).$$

Therefore, $\mathcal{H}_\omega$ is not part of the primitive foundation. It is generated by the algebra and the state.

This is one of the central structural commitments of the framework. It prevents the theory from starting with a pre-selected quantum state space. Instead, quantum-like representation emerges from the algebraic state structure.

> ** $(\mathcal{A}, \omega)$ first; $\mathcal{H}_\omega$ second.**

Accordingly, statements such as $\hat{C}_\alpha |\Psi\rangle = 0$ should be understood as represented versions of the deeper algebraic condition $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$. The vector condition is useful, but not foundational.

0.5.7 Guardrail VI — The Master Constraint Operator Is Representation-Dependent

The master constraint operator is written as $\hat{\mathfrak{M}}_\omega$, with a subscript ω , because it is built inside the GNS representation associated with the state ω .

It depends on:

1. The chosen state ω .
2. The representation π_ω .
3. The Hilbert space $\mathscr{H}_\omega$.
4. The represented constraint operators $\pi_\omega(\hat{C}_\alpha)$.
5. The positive weights w_α .
6. Domain and convergence assumptions where necessary.

It is therefore not a primitive universal operator existing before representation.

> **$\hat{\mathfrak{M}}_\omega$ packages represented constraints; it does not define the primitive theory.**

The primitive structure remains the algebraic constraint system:

$(\mathcal{A}, \mathcal{I}_C, \omega)$.

0.5.8 Guardrail VII — The Classical Master Functional Is Effective, Not Fundamental

The effective classical master functional:

$\mathfrak{M}(x) = \sum_\alpha w_\alpha |C_\alpha(x)|^2$

belongs to a classical or coarse-grained representation. It requires an effective configuration space $\mathcal{X}$, constraint functions $C_\alpha(x)$, norms, weights, and an interpretation of x .

None of these is part of the bare algebraic foundation.

Therefore, the functional $\mathfrak{M}(x)$ must not be presented as the fundamental starting point. It is a useful effective object that mirrors the positivity of the operator master constraint, but it is downstream of the algebraic structure.

> **$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$ is primitive; $\mathfrak{M}(x) = 0$ is effective.**

This distinction prevents the theory from becoming prematurely classical.

0.5.9 Guardrail VIII — The Configuration Label x Is Not Necessarily a Spacetime Point

In the effective classical functional $\mathcal{M}(x)$, the variable x denotes a configuration label. It does not automatically denote a point in spacetime.

The framework does not begin with a spacetime manifold M , nor with points $p \in M$. If spacetime-like structure appears later, it must be reconstructed from correlation, causal dependency, burden, or other relational structures.

Therefore, it is unsafe to read x as a spacetime coordinate unless a later section has explicitly introduced an emergent manifold or continuum approximation.

> **$x \in \mathcal{X}$ labels an effective configuration, not necessarily a spacetime location.**

This prevents background geometry from being smuggled into the foundation.

0.5.10 Guardrail IX — The Master Constraint Is Not a Hamiltonian

The master constraint operator $\hat{\mathcal{M}}_\omega$ is positive and operator-like. Because of this, it may be tempting to treat it as a Hamiltonian. That would be a mistake.

The equation:

$$\hat{\mathcal{M}}_\omega \psi = 0$$

is a constraint equation. It selects the physical sector. It is not an external-time evolution equation.

It is not of the form:

$$\frac{d}{dt}\psi(t) = H\psi(t).$$

No external time parameter t has been introduced. No Hamiltonian flow has been defined. No primitive time evolution is assumed.

> **The master constraint selects admissible states; it does not generate primitive time evolution.**

Time must emerge later, if it emerges, from the internal structure of admissible relational configurations. This includes causal dependency, burden-weighted depth, clock-rate suppression, and open extension.

0.5.11 Guardrail X — Constraint Satisfaction Is Not Dynamics

More generally, the entire constraint layer should not be confused with dynamics.

The primitive constraints specify admissibility:

> Which relational structures are allowed?

They do not yet specify evolution:

> How does one admissible structure become another?

Those are different questions.

The foundational condition $\omega(\hat{C}_{\alpha}^{\dagger} \hat{C}_{\alpha}) = 0$ answers the first question. It says that the state is compatible with the constraints.

It does not answer the second question. It does not define update rules, causal growth, temporal succession, histories, or physical duration.

Those notions must be introduced later through structures such as:

- causal dependency (prec),
- admissible extension (Ext),
- zero-preserving transformations,
- burden-weighted causal depth,
- open extension sequences.

> **Admissibility is not yet dynamics.**

0.5.12 Guardrail XI — Time Is Not Assumed at the Foundation

The foundation is deliberately timeless in the following precise sense: it does not assume an external background parameter t with respect to which states evolve.

This does not mean that the theory denies change, history, process, or temporal experience. It means only that these notions are not primitive. They must be reconstructed later from relational and constraint-preserving structure.

Thus, at the foundation, one should not write equations such as:

$$\frac{dA}{dt}, \quad \frac{d\omega}{dt}, \quad |\Psi(t)\rangle, \quad$$

unless an effective time parameter has already been introduced.

> **No external time parameter is primitive.**

This preserves the possibility that time emerges rather than being assumed.

0.5.13 Guardrail XII — Space Is Not Assumed at the Foundation

The foundation also does not assume a spatial manifold, metric, topology, or locality relation. There is no primitive background space M , no primitive distance function $d(p,q)$, and no primitive neighbourhood structure.

The relational algebra contains algebraic expressions, not points in space. Later, space-like structure may be reconstructed from correlation profiles, equivalence classes, or metric-like functions derived from physical states.

> **No background spatial geometry is primitive.**

This is necessary if the theory is to explain emergent geometry rather than presuppose it.

0.5.14 Guardrail XIII — Causality Is Not Yet Spacetime Causality

Later sections will introduce causal dependency. But at the foundational level, no spacetime causal cone, Lorentzian metric, or light-speed structure has been assumed.

Therefore, when the manuscript later uses a relation such as $e_i \prec e_j$, it must be introduced as a relational dependency order, not as a pre-existing spacetime causal order.

> **Causality, when introduced, begins as dependency, not as background light-cone structure.**

Only after emergent geometry has been constructed may one compare the dependency relation with spacetime-like causal structure.

0.5.15 Guardrail XIV — Energy Is Not Yet Defined

The framework has introduced positive quantities, such as:

$\omega(\hat{C}_{\alpha}^{\dagger} \hat{C}_{\alpha}), \quad \angle \psi,$
 $\hat{\mathfrak{M}}_{\omega} \psi \rangle, \quad \mathfrak{M}(x).$

These quantities measure constraint violation or its represented/effective analogues. They should not automatically be called energy.

Energy normally requires additional structure, such as:

- time-translation symmetry,
- a Hamiltonian generator,
- stress-energy structure,
- conservation laws,
- effective field dynamics.

None of that has been introduced at the foundational level.

> **Positive constraint cost is not automatically energy.**

Later sections may define burden, source structure, or energy-like quantities, but those must be introduced with their own assumptions and theorem status.

0.5.16 Guardrail XV — The Framework Is Not Yet Quantum Mechanics in Disguise

The use of a $(*)$ -algebra, positivity, GNS representation, and Hilbert space resembles algebraic quantum theory. This resemblance is intentional at the mathematical level.

However, the framework should not be reduced too quickly to standard quantum mechanics.

At the foundation:

- The algebra is relational and constraint-based.
- Hilbert space is derived from states.
- The master constraint selects admissibility.
- No external time, Hamiltonian dynamics, measurement postulates, Born rule, or spacetime-local field algebra has yet been assumed.

> **The foundation is compatible with quantum representation, but not identical to full quantum theory.**

If Born probabilities, measurement records, decoherence, field modes, or particle sectors are introduced later, they require additional arguments.

0.5.17 Guardrail XVI — Existence of Physical States Must Be Established or Assumed

The properness condition $\mathcal{I}_C \not\subset \mathcal{A}$ prevents immediate collapse of the algebra into inconsistency. But it does not by itself prove that physical states exist.

A physical state must be a positive linear functional satisfying:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \forall \alpha.$$

Whether such a functional exists is a non-trivial mathematical question. In a concrete model, it must be proven, constructed, or explicitly assumed.

> **$\frac{S}{\mathrm{phys}} \neq \varnothing$ is an existence condition, not an automatic consequence.**

The manuscript should clearly mark where physical-state existence is assumed and where it is proved.

0.5.18 Guardrail XVII — Infinite Sums Require Analytic Control

Both the master constraint operator and the classical master functional may involve sums over constraints:

$$\sum_{\alpha \in I}.$$

If I is finite, the positivity and zero-equivalence arguments are straightforward.

If I is infinite, one must specify convergence, domains, topology, and operator-theoretic conditions. For the operator master constraint, one must address whether the sum defines a positive self-adjoint or closable operator on a suitable domain. For the classical functional, one must specify whether the non-negative series converges and what it means for all terms to vanish.

> **Finite master sums are immediate; infinite master sums require explicit analytic assumptions.**

This prevents formal notation from silently carrying more rigour than has been supplied.

0.5.19 Guardrail XVIII — Effective Approximations Do Not Replace Exact Definitions

In later applications, it may be useful to define approximate physicality:

$$\frac{M}{\epsilon}(x) \leq \epsilon.$$

This is a valid effective idea, but it is not the foundational definition.

Exact physicality remains:

$$\omega(\hat{C}_{\alpha}^{\dagger} \hat{C}_{\alpha}) = 0 \quad \text{for all } \alpha,$$

or, in an exact effective classical model:

$$\frac{M}{\epsilon}(x) = 0.$$

Approximate physicality belongs to modelling, perturbation, coarse-graining, or numerical analysis.

> ** ϵ -physicality is effective; exact physicality is constraint-zero.**

0.5.20 Guardrail XIX — Later Emergence Claims Must Be Layered

The foundation does not yet prove:

- space,
- time,
- fields,

- particles,
- gravity,
- black holes,
- Born probabilities,
- cosmology.

It prepares the algebraic ground on which such structures may be developed.

Each later emergence claim must state its assumptions and theorem status. For example:

- Correlation geometry requires a suitable correlation kernel.
- Metricity requires triangle-like or multiplicative consistency assumptions.
- Time requires causal ordering or burden-weighted depth.
- Gravity-like behaviour requires a burden-to-metric dictionary.
- Black-hole-like domains require threshold, isolation, and flux-suppression conditions.

> ****No later structure should be smuggled into the foundation.****

This keeps the manuscript honest and makes the emergence programme credible.

0.5.21 Consolidated Foundational Hierarchy

The full foundational hierarchy is:

$$\boxed{\mathcal{A}} \quad \longrightarrow \quad \{\hat{C}_\alpha\} \quad \longrightarrow \quad \mathcal{I}_C \quad \longrightarrow \quad \omega \quad \longrightarrow \quad (\mathcal{H}_\omega, \pi_\omega, \Omega_\omega) \quad \longrightarrow \quad \hat{\mathcal{M}}_\omega \quad \longrightarrow \quad \mathcal{M}(x).$$

Each arrow represents a step of additional structure. The ordering matters:

1. $\mathcal{A}$ is the primitive relational algebra.
2. $\{\hat{C}_\alpha\}$ are primitive constraint elements.
3. $\mathcal{I}_C$ is the algebraic inconsistency sector generated by them.
4. ω is a positive physical state functional.
5. $(\mathcal{H}_\omega, \pi_\omega, \Omega_\omega)$ is the GNS representation.
6. $\hat{\mathcal{M}}_\omega$ is the represented master constraint operator.
7. $\mathcal{M}(x)$ is an effective classical master functional.

This hierarchy is one of the most important organising principles of the manuscript.

0.5.22 Foundational Guardrails — Consolidated Table

#	Guardrail	Statement
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I	Zero is constraint compatibility	Zero means zero inconsistency, not zero being
II	Zero is not pairwise cancellation	Master zero is simultaneous constraint satisfaction
III	$\mathcal{A}$ is kinematic	$\mathcal{A}$ is not automatically the observable algebra
IV	Do not quotient too early	Physicality is first state-defined
V	Hilbert space is derived	$\mathcal{H}_\omega$ is not primitive
VI	Master constraint is representation-dependent	$\widehat{\mathfrak{M}}_\omega$ does not define the primitive theory
VII	Classical functional is effective	$\mathfrak{M}(x)$ is not fundamental
VIII	x is not spacetime	x labels a configuration, not a spacetime point
IX	Master constraint is not Hamiltonian	It selects admissibility, not time evolution
X	Admissibility is not dynamics	Constraint satisfaction does not define evolution
XI	No external time	No background time parameter is primitive
XII	No background space	No spatial geometry is primitive
XIII	Causality is dependency	Causality begins as relational order, not light-cone structure
XIV	Constraint cost is not energy	Positive burden is not automatically energy
XV	Not yet quantum mechanics	The foundation is algebraic, not full QM
XVI	State existence is conditional	$\mathfrak{S}_{\mathrm{phys}} \neq \varnothing$ must be shown or assumed
XVII	Infinite sums need control	Infinite master sums require analytic assumptions
XVIII	Approximate $\neq$ exact	ε -physicality is effective, not foundational
XIX	Emergence is layered	No later structure is smuggled into the foundation

0.5.23 Summary of Section 0

Section 0 has established the canonical algebraic foundation of the Relational Constraint Framework.

****What has been accomplished:****

1. ****The relational algebra**** $\mathcal{A}$ was defined as a unital complex $(*)$ -algebra, providing the primitive language for relational expressions without assuming spacetime, time, geometry, fields, particles, observers, or Hilbert space.
2. ****The constraint ideal**** $\mathcal{I}_C$ was defined as the smallest two-sided $(*)$ -ideal containing all primitive constraints $\{\widehat{C}_\alpha\}$. It gathers all algebraic consequences of the primitive constraints and functions as the algebraic inconsistency sector.
3. ****Physical states**** were defined as positive linear functionals $\omega: \mathcal{A} \rightarrow \mathbb{C}$ satisfying the quadratic constraint-vanishing condition $\omega(\widehat{C}_\alpha^\dagger \widehat{C}_\alpha) = 0$ for all α . This condition says that primitive constraint violations vanish in positive state-evaluation.
4. ****The GNS representation**** was constructed from a physical state, yielding a Hilbert space $\mathcal{H}_\omega$, a representation π_ω , and a cyclic vector

Ω_ω . The physical state condition was shown to imply constraint annihilation of the cyclic vector: $\pi_\omega(\hat{C}_\alpha)\Omega_\omega = 0$.

5. **The master constraint operator** $\hat{\mathfrak{M}}_\omega$ was introduced as a representation-level packaging of the primitive constraints. Its kernel was shown to coincide with the simultaneous kernel of all individual constraints.

6. **The effective classical master functional** $\mathfrak{M}(x)$ was introduced as a derived, coarse-grained description. Its zero-equivalence with the simultaneous satisfaction of all effective constraints was proven. Its status as a derived, non-primitive object was established.

7. **Nineteen foundational guardrails** were recorded to prevent overinterpretation of the zero condition, the Hilbert-space representation, the master constraint, the classical functional, and the scope of the foundation.

What has not been accomplished:

The foundation does not yet contain space, time, particles, fields, gravity, observers, measurement, probability, dynamics, or cosmology. It supplies the constraint-compatible relational ground from which those structures may later be derived.

The foundational commitment is:

> **A physical structure is an admissible relational structure whose constraint violations vanish in the physical state.**

This is the starting point of the relational constraint framework. It is intentionally spare. It does not yet contain space, time, particles, fields, gravity, or observers. It supplies the constraint-compatible relational ground from which those structures may later be derived.

You are right. Section 1 deserves the same depth and rigor as Section 0. Let me expand it properly, starting from the beginning.

Section 1 — The Zero-Preserving Causal Foundation

1.0 Purpose of the Zero-Preserving Causal Foundation

1.0.1 Transition from Admissibility to Structure

Section 0 established the canonical foundation of the framework. It defined a relational algebra $\mathcal{A}$, a family of primitive constraint elements $\{\hat{C}_\alpha\}_{\alpha \in I}$, the generated constraint ideal $\mathcal{I}_C$, and physical states Ω_ω : $\mathcal{A} \rightarrow \mathbb{C}$ satisfying the quadratic constraint-vanishing condition $\Omega_\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$. It then showed how a Hilbert-space representation may be obtained from a physical state by the GNS construction, and how the

represented constraints may be packaged into a positive master constraint operator $\hat{\mathfrak{M}}_\omega$.

The result of Section 0 is an **admissibility criterion**. It tells us what it means for a relational state to be physical:

> **Physicality means constraint-compatible relational evaluation.**

However, admissibility alone is not yet structure. It does not yet tell us what counts as an operation, what counts as an observable, how admissible structures may be transformed, how one admissible event may depend on another, or how causal order can arise without an external time parameter.

The purpose of the present section is to begin that next layer.

1.0.2 What This Section Introduces

Section 1 introduces the first post-foundational structures:

1. **Zero-preserving operations** — transformations that preserve the physical sector.
2. **Physical observables** — operations compatible with constraint admissibility.
3. **Causal dependency** — a strict partial order among admissible events, derived from constraint closure.
4. **The open extension principle** — the universe is an extendible, not a closed, zero-preserving structure.
5. **The two-link principle** — causal links and relational correlation links are structurally distinct.

These structures are still pre-geometric. They do not assume a background spacetime, metric, manifold, field equation, or external clock. They describe how physical admissibility may be preserved, related, and extended within the relational constraint framework.

1.0.3 Why "Zero-Preserving" Comes First

The foundational zero condition identifies the physical sector. In a represented Hilbert-space form, this sector may be written as:

$$\mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega = \bigcap_{\alpha} \ker \pi_\omega(\hat{C}_\alpha).$$

In an effective classical description, the physical configuration space may be written as:

$$\mathcal{X}_{\mathrm{phys}} = \mathfrak{M}^{-1}(0).$$

Once such a physical sector has been identified, the next natural question is:

> **Which operations preserve it?**

An operation that maps a physical state or configuration into an unphysical one cannot be regarded as a physical operation without further qualification. If applying an operation produces constraint violation, then that operation does not preserve admissibility.

Therefore, the first structural requirement beyond the foundation is:

> **Physical operations must preserve the zero sector.**

This is the origin of the term **zero-preserving**.

In the represented Hilbert-space setting, an operator $\hat{O}$ is zero-preserving if:

$$\hat{O} \mid \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega.$$

Equivalently:

$$\hat{O} \mid \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega.$$

In the effective classical setting, a transformation T is zero-preserving if:

$$T(\mathfrak{M}^{-1}(0)) \subseteq \mathfrak{M}^{-1}(0).$$

This condition is not yet dynamics. It is a compatibility condition. It says that an admissible operation must not carry the system out of the physical sector.

1.0.4 Observables as Admissibility-Preserving Operations

The relational algebra $\mathcal{A}$ contains possible relational expressions. But Section 0 already established that not every algebra element should automatically be treated as a physical observable. Membership in $\mathcal{A}$ means only that the expression belongs to the kinematic relational language.

An observable must satisfy more. At minimum, it must be compatible with the constraints. In the present framework, this means that observables must preserve the physical sector, either exactly or in an appropriately reduced or quotient sense.

> **An observable is not merely an algebra element; it is an admissibility-preserving relational operation.**

There may be stronger and weaker versions of this idea. A **strong observable** may commute with the master constraint:

$$[\hat{O}, \hat{\mathfrak{M}}_\omega] = 0.$$

This is sufficient for zero preservation, because if $\psi \in \ker \hat{\mathfrak{M}}_\omega$, then:

$$\hat{\mathfrak{M}}_\omega \hat{O} \psi = \hat{O} \hat{\mathfrak{M}}_\omega \psi = 0.$$

Therefore $\hat{O} \psi \in \ker \hat{\mathfrak{M}}_\omega$.

But exact commutation on the full kinematic representation may be stronger than necessary. A **weak observable** need only satisfy:

$$\hat{O} \setminus \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega.$$

This distinction allows the theory to remain flexible. The physical requirement is preservation of admissibility; global commutation may be a sufficient condition, not always a necessary one.

1.0.5 Why Causal Dependency Must Be Introduced Without External Time

The foundation does not assume external time. Therefore, causal order cannot be introduced as an ordering of events by a background parameter t . There is no primitive equation of the form:

$$\frac{d}{dt}\psi(t) = \cdots,$$

and no pre-existing spacetime in which events are already arranged along temporal curves.

Nevertheless, the theory must eventually account for temporal structure. It must explain how one event may depend on another, how admissible structures may extend, and how a direction of history may arise.

The first step is to define causal order not as temporal succession in a background time, but as ****dependency among zero-preserving structures****.

> ****Causal order begins as admissibility dependency, not as background time order.****

If an admissible event e_j requires the prior closure, satisfaction, or availability of another event e_i , then one may write:

$$e_i \prec e_j.$$

This relation does not yet mean that e_i occurs earlier than e_j in a spacetime coordinate. It means that e_j 's admissibility depends on e_i . Temporal interpretation comes later.

Thus, causal dependency is introduced as an internal relational order:

$$e_i \prec e_j \Leftrightarrow \text{\textit{e_i is a necessary admissibility predecessor of e_j.}}$$

Under suitable assumptions—such as irreflexivity, asymmetry, and transitive closure of dependency—this relation becomes a strict partial order.

This gives the framework a way to speak about direction, precedence, and dependency without assuming a clock.

1.0.6 Events Are Not Spacetime Points

A key guardrail for this section is that an event is not initially a spacetime point.

Let $\mathcal{E}$ denote a set of primitive admissible events or event-like structures. At this level, an event $e \in \mathcal{E}$ should be understood as a localised instance of constraint-compatible relational structure, update, resolution, or admissible occurrence.

It is not yet a point $p \in M$ in a spacetime manifold. It is also not yet a field value at a coordinate, a particle occurrence in a background geometry, or a measurement result recorded by an observer. Those interpretations may become available later, once additional structures have been derived.

For now, an event is a unit of admissible relational structure.

> **Events are primitive admissible relational occurrences, not background spacetime points.**

This distinction prevents the causal foundation from smuggling in spacetime geometry.

1.0.7 The Two-Link Principle

The next major purpose of this section is to distinguish two kinds of connection between events: **causal dependency** and **relational correlation**.

A **causal link** is directed. It has the form:

$$e_i \prec e_j.$$

It says that e_j depends on e_i in the admissibility structure.

A **relational or correlation link**, by contrast, need not be directed. It may express the degree to which two event contexts, observables, or relational regions are correlated in a physical state. A schematic correlation may later be written as:

$$K_\omega(e_i, e_j)$$

or

$$\mathrm{Corr}_\omega(O_i, O_j).$$

This kind of relation supports the later construction of emergent spatial geometry. Strong correlation may correspond to small emergent distance, while weak correlation may correspond to large emergent distance.

But this is not the same as causal dependence. Two structures may be strongly correlated without one being the causal predecessor of the other. Conversely, two events may be causally ordered without being spatially near in an emergent correlation geometry.

The section therefore introduces the **Two-Link Principle**:

> **Causal dependency and relational correlation are distinct structural links.**

This distinction is essential. If causal dependency and relational correlation are collapsed into one relation, then the theory risks identifying space with causality too early. Instead, the framework keeps them separate:

- Causal links support emergent time-like order.
- Correlation links support emergent space-like geometry.

Their later interaction may produce spacetime-like structure, but neither is assumed to be spacetime from the start.

1.0.8 Open Extension Without External Time

The final purpose of this section is to introduce admissible extension.

The foundation defines what it means for a state or structure to be physical. Zero-preserving operations define what it means to remain physical under transformation. Causal dependency defines how admissible events may depend on one another. Open extension then describes the possibility that an admissible relational structure can be enlarged while preserving constraint compatibility.

If $\mathcal{U}_s$ is an admissible relational structure, an extension:

$$\mathcal{U}_s \rightarrow \mathcal{U}_{s+1}$$

is admissible if the extended structure remains physical.

In an effective master-functional notation, this may be written as:

$$\mathcal{M}(\mathcal{U}_{s+1}) = 0.$$

In the algebraic formulation, it means that the extended state or structure remains compatible with the constraint ideal.

The index s should not be interpreted as external time. It is an ordering label for admissible extension. Physical time, if it emerges, must be reconstructed later from causal depth, burden, clock-rate structure, or other internal measures.

> **Extension order is not yet physical time.**

The universe, or a relational structure, may be called **open** if there remains at least one admissible extension:

$$\mathrm{Ext}(\mathcal{U}) \neq \varnothing.$$

This does not yet define dynamics. It defines the possibility of non-final admissible continuation.

1.0.9 What This Section Proves and What It Does Not Prove

This section begins the transition from static admissibility to relational structure. It establishes the conceptual and mathematical ingredients needed for later emergence, but it does not yet prove spacetime, dynamics, gravity, or quantum measurement.

It does establish, or prepare to establish, the following:

- Zero-preserving operations preserve the physical sector.
- Physical observables must be admissibility-compatible.
- Causal dependency may be formulated as a relation among admissible events.
- Under suitable closure assumptions, causal dependency forms a strict partial order.
- Causal links and relational correlation links are distinct.

- Admissible extension can be defined without an external time parameter.

It does not yet establish:

- A spacetime manifold.
- A metric.
- Lorentzian causal cones.
- Clock time.
- Hamiltonian time evolution.
- Field propagation.
- Particles.
- Gravity.

Those belong to later sections.

> **Section 1 constructs the zero-preserving causal skeleton, not full spacetime physics.**

1.0.10 Structural Sequence of Section 1

The section proceeds in the following order.

First, it restates master-zero equivalence in the form needed for the post-foundational layers. This clarifies that the physical sector is the zero sector of the represented or effective master constraint.

Second, it defines zero-preserving transformations and observables. These are operations that preserve physical admissibility.

Third, it introduces causal dependency among admissible events. Under appropriate assumptions, this dependency relation is shown to have partial-order structure.

Fourth, it states the Two-Link Principle, distinguishing causal dependency from relational correlation.

Fifth, it introduces admissible extension and openness, while guarding against interpreting extension order as external time.

The conceptual sequence is:

$$\boxed{\text{zero sector}} \quad \longrightarrow \quad \text{zero-preserving operations} \quad \longrightarrow \quad \text{causal dependency} \quad \longrightarrow \quad \text{open extension}.$$

This is the first bridge from pure admissibility toward emergent temporal and causal structure.

1.0.11 Guardrails for This Section

The following guardrails apply throughout Section 1.

1. **Zero preservation is not dynamics.** It is admissibility preservation.
2. **Observables are not merely arbitrary algebra elements.** They must be compatible with the physical sector.
3. **Causal dependency is not background spacetime causality.** It is an internal dependency relation among admissible structures.
4. **Events are not initially spacetime points.**
5. **Extension order is not external time.**
6. **Relational correlation is not the same as causal dependence.**
7. **This section does not yet define metric distance, duration, mass, energy, fields, or gravity.**

These guardrails protect the emergence programme by ensuring that later structures are not assumed before they are derived.

1.1 Master-Zero Constraint Equivalence

1.1.1 Purpose of This Subsection

Section 0 introduced several equivalent ways of expressing physical admissibility, depending on the level of description being used.

At the primitive algebraic level, physicality is expressed by the quadratic constraint-vanishing condition:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \forall \alpha.$$

At the represented Hilbert-space level, the same condition becomes:

$$\pi_\omega(\hat{C}_\alpha) \psi = 0 \quad \forall \alpha,$$

for vectors ψ in the represented physical sector.

At the master-operator level, the many represented constraints may be packaged into one positive operator:

$$\hat{\mathcal{M}}_\omega = \sum_\alpha w_\alpha \pi_\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha), \quad w_\alpha > 0.$$

The physical sector can then be described by the single zero condition:

$$\hat{\mathcal{M}}_\omega \psi = 0.$$

This subsection records the exact equivalence between these formulations. It is placed at the beginning of the zero-preserving causal foundation because later notions—observables, causal events, admissible extensions, and dependency order—all rely on knowing what it means to preserve the physical zero sector.

> **Master zero is equivalent to simultaneous satisfaction of all included constraints.**

This is a structural theorem about positive sums. It is not a claim about nothingness, cancellation, temporal evolution, or spacetime geometry.

1.1.2 The Physical Zero Sector

Let $(\mathscr{H}_\omega, \pi_\omega, \Omega_\omega)$ be the GNS representation associated with a physical state ω , and let $\hat{\mathfrak{M}}_\omega$ be the represented master constraint operator.

The represented physical sector is defined by:

$$\mathscr{H}_{\mathrm{phys}}^\omega := \ker \hat{\mathfrak{M}}_\omega.$$

This is the simultaneous kernel of all represented primitive constraints:

$$\mathscr{H}_{\mathrm{phys}}^\omega = \bigcap_{\alpha \in I} \ker \pi_\omega(\hat{C}_\alpha).$$

The master-zero formulation asserts that the same sector may also be written as:

$$\mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega,$$

provided the master constraint is a positive weighted sum of represented constraint squares.

1.1.3 Definition — Represented Master Constraint

Definition 1.1 (Represented Master Constraint).

Let $\{\hat{C}_\alpha\}_{\alpha \in I} \subset \mathcal{A}$ be the primitive constraint family. In the representation associated with ω , define:

$$\hat{\mathfrak{M}}_\omega := \sum_{\alpha \in I} w_\alpha \pi_\omega(\hat{C}_\alpha)^{\dagger} \pi_\omega(\hat{C}_\alpha),$$

whenever the sum is well-defined on a common invariant domain.

Technical Note.

- If the index set I is finite, this definition is immediate.
- If I is infinite, one must impose the analytic assumptions required for the sum to define a positive operator on a suitable common domain.
- The clean theorem-safe setting is: assume the master constraint is a well-defined positive operator.

1.1.4 Positivity Identity

For any vector $\psi \in \mathscr{H}_\omega$ in the domain of all relevant operators:

$$\langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = \sum_{\alpha} w_\alpha \langle \psi, \pi_\omega(\hat{C}_\alpha)^{\dagger} \pi_\omega(\hat{C}_\alpha) \psi \rangle = \sum_{\alpha} w_\alpha |\pi_\omega(\hat{C}_\alpha) \psi|^2.$$

Since each term satisfies $|\pi_\omega(\hat{C}_\alpha) \psi|^2 \geq 0$ and each weight satisfies $w_\alpha > 0$, we obtain:

$$\langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle \geq 0.$$

Thus the master constraint is positive.

1.1.5 Theorem — Master-Zero Equivalence

****Theorem 1.1 (Master-Zero Equivalence).****

Let $\hat{\mathfrak{M}}_\omega = \sum_\alpha w_\alpha \pi_\omega(\hat{C}_\alpha)$ with $w_\alpha > 0$ be a well-defined positive master constraint. Then for every vector ψ in the relevant domain:

$$\langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0 \iff \Longleftrightarrow \pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha.$$

Equivalently:

$$\ker \hat{\mathfrak{M}}_\omega = \bigcap_{\alpha \in I} \ker \pi_\omega(\hat{C}_\alpha).$$

Proof.

($\Rightarrow$) Assume $\langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0$. Using the positivity identity:

$$0 = \sum_\alpha w_\alpha |\pi_\omega(\hat{C}_\alpha)\psi|^2.$$

Every summand is non-negative. Since each weight is strictly positive, the only way the sum can vanish is if every summand vanishes:

$$|\pi_\omega(\hat{C}_\alpha)\psi|^2 = 0 \quad \forall \alpha.$$

Therefore $\pi_\omega(\hat{C}_\alpha)\psi = 0$ for all α .

($\Leftarrow$) Assume $\pi_\omega(\hat{C}_\alpha)\psi = 0$ for all α . Then every term in the sum vanishes, so $\hat{\mathfrak{M}}_\omega \psi = 0$. $\square$

1.1.6 Physical-Sector Form

Using the theorem, the physical sector may be written in any of the following equivalent represented forms:

$$\mathcal{H}_{\text{phys}}^\omega = \{ \psi : \pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha \},$$

$$\mathcal{H}_{\text{phys}}^\omega = \bigcap_\alpha \ker \pi_\omega(\hat{C}_\alpha),$$

$$\mathcal{H}_{\text{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega,$$

or, at the level of expectation values:

$$\mathcal{H}_{\text{phys}}^\omega = \{ \psi : \langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0 \}.$$

These are equivalent only under the positivity assumptions stated above.

The most useful form for the next subsection is:

$$\boxed{\mathcal{H}_{\text{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega.}$$

This is the zero sector that physical observables must preserve.

1.1.7 Effective Classical Version

In an effective classical model, one may define a configuration space $\mathcal{X}$ and constraint functions $C_\alpha: \mathcal{X} \rightarrow V_\alpha$. With weights $w_\alpha > 0$, the effective classical master functional is:

$$\mathcal{M}(x) = \sum_\alpha w_\alpha |C_\alpha(x)|^2.$$

The same positive-sum argument gives:

$$\mathcal{M}(x) = 0 \iff C_\alpha(x) = 0 \text{ for all } \alpha.$$

Thus the effective classical physical configuration space is:

$$\mathcal{X}_{\mathrm{phys}} = \mathcal{M}^{-1}(0) = \bigcap_\alpha C_\alpha^{-1}(0).$$

1.1.8 Algebraic Version

At the primitive algebraic level, physicality is stated through the positive state condition:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \text{ for all } \alpha.$$

Once ω is represented through the GNS construction, this condition implies:

$$\pi_\omega(\hat{C}_\alpha)\Omega_\omega = 0 \text{ for all } \alpha.$$

By master-zero equivalence:

$$\Omega_\omega \in \bigcap_\alpha \ker \pi_\omega(\hat{C}_\alpha).$$

Therefore:

$$\Omega_\omega \in \ker \hat{\mathcal{M}}_\omega.$$

Thus the algebraic physical state condition generates the represented master-zero condition for the cyclic vector:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \iff \hat{\mathcal{M}}_\omega \Omega_\omega = 0.$$

The implication runs from algebraic state physicality to represented master zero. The master constraint does not replace the algebraic condition.

1.1.9 Why the Equivalence Matters for Observables

The next subsection will define zero-preserving observables. For that definition to be meaningful, the zero sector must be identified clearly.

Using master-zero equivalence, the represented zero sector can be described as:

$$\ker \hat{\mathcal{M}}_\omega.$$

An operator $\hat{O}$ is then zero-preserving if:

$$\hat{O} \in \ker \hat{\mathcal{M}}_\omega \subseteq \ker \hat{\mathcal{M}}_\omega.$$

Equivalently, if ψ satisfies all represented constraints, then $\hat{O}\psi$ must also satisfy all represented constraints.

That is:

$$\pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha \quad \Longleftrightarrow \quad \pi_\omega(\hat{C}_\alpha)\hat{O}\psi = 0 \quad \forall \alpha.$$

Thus, master-zero equivalence is the bridge from constraint satisfaction to observable admissibility. It tells us exactly what must be preserved.

1.1.10 Strong and Weak Readings of Zero Preservation

The master-zero condition supports two useful levels of observable compatibility.

A **strong compatibility condition** is:

$$[\hat{O}, \hat{\mathfrak{M}}_\omega] = 0.$$

This implies that $\hat{O}$ preserves the master kernel. If $\hat{\mathfrak{M}}_\omega \psi = 0$, then:

$$\hat{\mathfrak{M}}_\omega \hat{O} \psi = \hat{O} \hat{\mathfrak{M}}_\omega \psi = 0.$$

So $\hat{O} \psi \in \ker \hat{\mathfrak{M}}_\omega$.

A **weak compatibility condition** is simply:

$$\hat{O} \in \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega.$$

This does not require $\hat{O}$ to commute with $\hat{\mathfrak{M}}_\omega$ on the full kinematic Hilbert space. It requires only that $\hat{O}$ preserve the physical sector.

The weak condition is the essential one. The strong condition is sufficient, but may not be necessary.

This distinction will matter when defining the observable algebra.

1.1.11 Why Positive Weights Cannot Be Relaxed

The equivalence theorem depends on $w_\alpha > 0$.

If some weight were zero, say $w_\beta = 0$, then the corresponding constraint $\pi_\omega(\hat{C}_\beta)$ would not contribute to the master constraint. A vector could satisfy $\hat{\mathfrak{M}}_\omega \psi = 0$ while still violating $\pi_\omega(\hat{C}_\beta)\psi \neq 0$. Thus, zero weights destroy the equivalence.

If negative weights were allowed, then different constraint terms could cancel in expectation. One could have $\sum_\alpha w_\alpha |\pi_\omega(\hat{C}_\alpha)\psi|^2 = 0$ without each norm being zero. That destroys the master-zero equivalence.

Therefore, the condition $w_\alpha > 0$ is essential.

> **Master-zero equivalence holds for strictly positive weights.**

1.1.12 Infinite Constraint Families

If the index set I is infinite, the master constraint:

$$\hat{\mathfrak{M}}_\omega = \sum_{\alpha \in I} w_\alpha \pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha)$$
 requires additional assumptions.

One must specify:

- the common domain of the constraints,
- the convergence of the positive quadratic form,
- whether the resulting form defines a self-adjoint or closable positive operator,
- whether kernel equivalence holds on the chosen domain.

The theorem remains conceptually the same: a non-negative sum can vanish only if each non-negative term vanishes. But in an infinite-dimensional or unbounded-operator setting, one must state the analytic assumptions precisely.

> **For finite constraint families, the equivalence is immediate; for infinite families, it holds under the assumptions required to make the positive sum well-defined.**

1.1.13 What Master-Zero Equivalence Does Not Say

Master-zero equivalence is often a place where overinterpretation enters. The theorem says only that a positive weighted sum of constraint squares vanishes exactly when every included constraint vanishes.

It does not say:

- that spacetime has emerged,
- that time has begun,
- that causal order is defined,
- that dynamics are present,
- that every physical quantity vanishes,
- that positive and negative quantities cancel,
- that the master constraint is an energy or a Hamiltonian,
- that the master constraint is a time-evolution equation.

> **Zero total constraint violation is equivalent to zero individual constraint violation. Everything beyond that belongs to later layers.**

1.1.14 Role of Master-Zero Equivalence in the Framework

Within the full manuscript, master-zero equivalence plays three roles.

First, it justifies using a single master object to describe many constraints:

$$\hat{C}_\alpha \quad \leadsto \quad \hat{\mathfrak{M}}_\omega$$

Second, it defines the physical zero sector that observables must preserve:

$$\ker \hat{\mathfrak{M}}_\omega$$

Third, it provides the admissibility criterion for effective classical models:

$$\frac{M}{\hbar}(x) = 0.$$

Thus, master-zero equivalence is a bridge between:

- constraint algebra,
- represented Hilbert space,
- and effective classical configuration space.

But it remains a bridge, not the foundation itself.

1.1.15 Guardrails for This Subsection

1. **No pairwise cancellation.** The theorem does not imply $(+X) + (-X) = 0$ or any requirement that physical structure appears as compensating pairs.
2. **No local balancing requirement.** It does not require local compensation or observable pairing.
3. **No dynamics.** The theorem does not describe evolution, time, or causation. It is purely a selection principle.
4. **No claim that $\mathcal{K}$ is spacetime.** The effective configuration space is not yet spacetime.

1.1.16 Summary of Subsection 1.1

Master-zero equivalence states that for a positive master constraint:

$$\begin{aligned} \hat{\frac{M}{\hbar}}_{|\Omega\rangle} &= \sum_{\alpha} w_{\alpha} \, \pi_{|\Omega\rangle}(\hat{C}_{\alpha})^{\dagger} \pi_{|\Omega\rangle}(\hat{C}_{\alpha}), \quad w_{\alpha} > 0, \\ \text{one has:} \\ \langle \psi | \hat{\frac{M}{\hbar}}_{|\Omega\rangle} | \psi \rangle &= 0 \quad \Longleftrightarrow \quad \pi_{|\Omega\rangle}(\hat{C}_{\alpha}) | \psi \rangle = 0 \quad \text{for all } \alpha. \end{aligned}$$

Equivalently:

$$\ker \hat{\frac{M}{\hbar}}_{|\Omega\rangle} = \bigcap_{\alpha} \ker \pi_{|\Omega\rangle}(\hat{C}_{\alpha}).$$

In the effective classical setting:

$$\frac{M}{\hbar}(x) = 0 \quad \Longleftrightarrow \quad C_{\alpha}(x) = 0 \quad \text{for all } \alpha.$$

The meaning of master zero is therefore precise:

$$\boxed{\text{master zero means simultaneous constraint satisfaction.}}$$

It is not nothingness, not cancellation, not time evolution, not energy, and not geometry. It is the positive packaging of constraint compatibility.

This equivalence prepares the next subsection: the definition of zero-preserving observables.

1.2 Zero-Preserving Observables

1.2.1 Purpose of This Subsection

Section 1.1 established that the physical sector is the zero sector of the represented master constraint:

$$\mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega = \bigcap_\alpha \ker \pi_\omega(\hat{C}_\alpha).$$

We now ask: which operations preserve this sector?

Not every represented algebra element should be counted as a physical observable. An element may be well-defined on the larger kinematic structure while failing to preserve the constraint-zero sector. Such an operation would carry physical states into non-physical states.

> **A physical observable must map admissible states to admissible states.**

This makes zero-preserving observables the natural analogue of gauge-compatible or constraint-compatible observables.

This subsection defines the class of operations that preserve physical admissibility, establishes their algebraic closure properties, constructs the algebraic observable sector as a quotient, and distinguishes strong from weak observables.

1.2.2 Kinematic Operations Versus Physical Observables

The relational algebra $\mathcal{A}$ contains possible relational expressions. After choosing a state ω , its GNS representation gives represented algebra elements $\pi_\omega(A)$ for $A \in \mathcal{A}$.

However, not every represented algebra element is automatically a physical observable. An element $A \in \mathcal{A}$ may be kinematically meaningful while still failing to preserve the physical sector. In that case, it belongs to the algebraic language of possible relational expressions, but it does not define a physical observable on admissible states.

Thus, one must distinguish:

$$A \in \mathcal{A} \quad \text{from} \quad A \in \mathcal{A}_{\mathrm{obs}}.$$

The first statement says that A is a possible relational expression. The second says that A is compatible with the constraints in the relevant physical sense.

> **Kinematic operations are allowed expressions; observables are admissibility-preserving operations.**

This prevents the framework from treating the entire algebra as directly measurable or physical before constraint compatibility has been imposed.

1.2.3 Definition — Zero-Preserving Operator

Definition 1.2 (Zero-Preserving Operator).

Fix a physical state ω and its GNS representation $(\mathcal{H}_\omega, \pi_\omega, \Omega_\omega)$. An operator $O \in \mathcal{B}(\mathcal{H}_\omega)$ is called **zero-preserving** if:

$$\pi_\omega(O) \mathcal{H}_\omega^{\text{phys}} \subseteq \mathcal{H}_\omega^{\text{phys}}.$$

Equivalently:

$$\pi_\omega(O) \ker \pi_\omega \subseteq \ker \pi_\omega.$$

In terms of the individual represented constraints, this means:

$$\pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha \quad \Longleftrightarrow \quad \pi_\omega(\hat{C}_\alpha)\hat{O}\psi = 0 \quad \forall \alpha.$$

Since $\psi \in \mathcal{H}_\omega^{\text{phys}}$ means

$$\pi_\omega(\hat{C}_\alpha)\psi = 0 \text{ for all } \alpha, \text{ zero preservation may be written as: } \boxed{\pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha \quad \Longleftrightarrow \quad \pi_\omega(\hat{C}_\alpha)\hat{O}\psi = 0 \quad \forall \alpha.}$$

Thus, $\hat{O}$ is physically admissible if it does not turn a constraint-satisfying vector into a constraint-violating vector.

1.2.4 Definition — Physical Observable

Definition 1.3 (Physical Observable).

A **physical observable** is a zero-preserving operation. When $\hat{O}$ arises from an algebra element $O \in \mathcal{A}$, so that $\hat{O} = \pi_\omega(O)$, we say that O is a **zero-preserving observable representative** in the representation ω if:

$$\pi_\omega(O) \mathcal{H}_\omega^{\text{phys}} \subseteq \mathcal{H}_\omega^{\text{phys}}.$$

The corresponding represented observable class may be denoted:

$$\mathcal{A}^{\text{obs}}_\omega := \{ O \in \mathcal{A} : \pi_\omega(O) \mathcal{H}_\omega^{\text{phys}} \subseteq \mathcal{H}_\omega^{\text{phys}} \}.$$

This definition is representation-dependent because it refers to π_ω and $\mathcal{H}_\omega^{\text{phys}}$. A more intrinsic algebraic definition may be developed later using constraint ideals, normalisers, quotient structures, or state-preserving maps. For the present section, the represented version is sufficient.

1.2.5 Strong Observables

A sufficient condition for zero preservation is **commutation with the master constraint**.

Definition 1.4 (Strong Observable).

An operator $\hat{O}$ is a **strong observable** if:

$$[\hat{O}, \hat{\mathfrak{M}}_\omega] = 0$$

on a common invariant domain containing $\mathscr{H}_{\mathrm{phys}}^\omega$.

Remark 1.1. The strong/weak distinction is explicitly present in the framework: strong observables commute with the master operator, while weak observables need only preserve the physical kernel.

1.2.6 Theorem — Strong Observables Preserve the Physical Kernel

Theorem 1.2 (Strong Observables Preserve the Physical Kernel).

If $[\hat{O}, \hat{\mathfrak{M}}_\omega] = 0$ on a common invariant domain containing $\mathscr{H}_{\mathrm{phys}}^\omega$, then:

$$\hat{O} \restriction \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega.$$

Proof.

Let $\psi \in \ker \hat{\mathfrak{M}}_\omega$. Then $\hat{\mathfrak{M}}_\omega \psi = 0$.

Using the commutator condition:

$$\hat{\mathfrak{M}}_\omega \hat{O} \psi = \hat{O} \hat{\mathfrak{M}}_\omega \psi = \hat{O}(0) = 0.$$

Therefore $\hat{O} \psi \in \ker \hat{\mathfrak{M}}_\omega$. $\square$

Interpretation. Strong commutation is a sufficient algebraic criterion for physicality. This is the main sufficient condition explicitly singled out in the framework.

1.2.7 Weak Observables

The essential physical condition is not global commutation. It is preservation of the physical sector.

Definition 1.5 (Weak Observable).

A **weak observable** is an operator $\hat{O}$ such that:

$$\hat{O} \restriction \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega,$$

without necessarily satisfying $[\hat{O}, \hat{\mathfrak{M}}_\omega] = 0$ on the full kinematic Hilbert space.

This distinction matters because there may be operators that fail to commute with the master constraint away from the physical sector, but nevertheless act consistently on physical states.

In that case, demanding global commutation would exclude admissible operations unnecessarily.

The hierarchy is:

$$\boxed{\text{commutation with constraints} \rightarrow \text{commutation with master constraint} \rightarrow \text{zero preservation}}$$

where the reverse implications need not hold.

Thus, weak zero preservation is the minimal condition for physical admissibility.

1.2.8 Closure Under Composition

Theorem 1.3 (Closure Under Composition).

If $\hat{O}_1$ and $\hat{O}_2$ are zero-preserving operators on $\mathcal{H}_{\text{phys}}^\omega$, then the composite operator $\hat{O}_2 \hat{O}_1$ is also zero-preserving, provided the composition is well-defined on the physical sector.

Proof.

Let $\psi \in \mathcal{H}_{\text{phys}}^\omega$. Since $\hat{O}_1$ is zero-preserving:

$$\hat{O}_1 \psi \in \mathcal{H}_{\text{phys}}^\omega.$$

Since $\hat{O}_2$ is zero-preserving:

$$\hat{O}_2(\hat{O}_1 \psi) \in \mathcal{H}_{\text{phys}}^\omega.$$

Therefore:

$$(\hat{O}_2 \hat{O}_1) \psi \in \mathcal{H}_{\text{phys}}^\omega \quad \forall \psi \in \mathcal{H}_{\text{phys}}^\omega.$$

So $\hat{O}_2 \hat{O}_1$ is zero-preserving. $\square$

Corollary 1.1. The set of zero-preserving observables on $\mathcal{H}_{\text{phys}}^\omega$ is closed under composition.

1.2.9 Closure Under Linear Combination

Theorem 1.4 (Closure Under Linear Combination).

If $\hat{O}_1$ and $\hat{O}_2$ are zero-preserving operators, and $\lambda, \mu \in \mathbb{C}$, then $\lambda \hat{O}_1 + \mu \hat{O}_2$ is zero-preserving, provided the linear combination is well-defined on the physical sector.

Proof.

Let $\psi \in \mathcal{H}_{\text{phys}}^\omega$. Since $\hat{O}_1$ and $\hat{O}_2$ are zero-preserving:

$$\hat{O}_1 \psi \in \mathcal{H}_{\text{phys}}^\omega, \quad \hat{O}_2 \psi \in \mathcal{H}_{\text{phys}}^\omega.$$

Since $\mathcal{H}_{\text{phys}}^\omega$ is a linear subspace:

$$\lambda \hat{O}_1 \psi + \mu \hat{O}_2 \psi \in \mathcal{H}_{\text{phys}}^\omega.$$

But:

$$(\lambda \hat{O}_1 + \mu \hat{O}_2) \psi = \lambda \hat{O}_1 \psi + \mu \hat{O}_2 \psi.$$

Therefore:

$$(\lambda \hat{O}_1 + \mu \hat{O}_2) \psi \in \mathcal{H}_{\text{phys}}^\omega.$$

Hence $\lambda \hat{O}_1 + \mu \hat{O}_2$ is zero-preserving. $\square$

1.2.10 Adjoint Closure Is Not Automatic

One must be careful about adjoints. If $\hat{O}$ preserves the physical sector:

$$\hat{O} : \mathcal{H}_{\text{phys}}^\omega \rightarrow \mathcal{H}_{\text{phys}}^\omega$$

it does not automatically follow that:

$\hat{O} \in \mathcal{H}_{\mathrm{phys}}^\omega \subseteq \mathcal{H}_{\mathrm{phys}}^\omega$.

Adjoint closure may require stronger conditions, such as **reduction** of the physical subspace rather than mere invariance.

A closed subspace $\mathcal{H}_{\mathrm{phys}}^\omega$ is said to **reduce** $\hat{O}$ if both:

$\hat{O} \in \mathcal{H}_{\mathrm{phys}}^\omega \subseteq \mathcal{H}_{\mathrm{phys}}^\omega$

and

$\hat{O}^\dagger \in \mathcal{H}_{\mathrm{phys}}^\omega \subseteq \mathcal{H}_{\mathrm{phys}}^\omega$.

If this holds, then the restriction of $\hat{O}$ to the physical sector behaves more cleanly as a physical operator.

> **Zero-preserving operators are closed under composition and linear combination, but adjoint closure requires an additional reduction condition.**

This matters if the physical observable algebra is intended to be a $(*)$ -algebra.

1.2.11 The Algebraic Observable Sector

The framework should also express physical observables directly at the algebraic level, not only inside a chosen Hilbert-space representation.

Definition 1.6 (Algebraically Zero-Preserving Element).

Define:

$\widetilde{\mathcal{A}}_{\mathrm{obs}} := \{ A \in \mathcal{A} : [A, \hat{C}_\alpha] \in \mathcal{I}_C \text{ for all } \alpha \}$.

An element of $\widetilde{\mathcal{A}}_{\mathrm{obs}}$ is called an **algebraically zero-preserving observable**.

Interpretation. Such an element need not commute exactly with each primitive constraint, but it must commute **modulo the constraint ideal**. That is, any failure of exact commutation is itself purely constraint-generated and therefore null in the physical sector.

This is the algebraic analogue of the Dirac observable concept: an observable is an element whose commutator with each constraint vanishes on the physical sector, even if it does not vanish identically on the kinematic algebra.

1.2.12 Closure of the Algebraic Observable Sector

Theorem 1.5 (Algebraic Closure).

Assume $\mathcal{I}_C$ is a two-sided $(*)$ -ideal. Then

$\widetilde{\mathcal{A}}_{\mathrm{obs}}$ is a unital $(*)$ -subalgebra of $\mathcal{A}$.

Proof.

Let $A, B \in \widetilde{\mathcal{A}}_{\{\mathrm{obs}\}}$.

(i) Closure under addition. For every α :

$$[A + B, \hat{C}_\alpha] = [A, \hat{C}_\alpha] + [B, \hat{C}_\alpha].$$

Since each term lies in $\mathcal{I}_C$, their sum also lies in $\mathcal{I}_C$. Hence $A + B \in \widetilde{\mathcal{A}}_{\{\mathrm{obs}\}}$.

(ii) Closure under scalar multiplication. For any $\lambda \in \mathbb{C}$:

$$[\lambda A, \hat{C}_\alpha] = \lambda [A, \hat{C}_\alpha] \in \mathcal{I}_C.$$

(iii) Closure under multiplication. Using the commutator identity $[AB, C] = A[B, C] + [A, C]B$:

$$[AB, \hat{C}_\alpha] = A[B, \hat{C}_\alpha] + [A, \hat{C}_\alpha]B.$$

Because $[B, \hat{C}_\alpha] \in \mathcal{I}_C$ and $[A, \hat{C}_\alpha] \in \mathcal{I}_C$, and $\mathcal{I}_C$ is a two-sided ideal, both terms lie in $\mathcal{I}_C$. Therefore $AB \in \widetilde{\mathcal{A}}_{\{\mathrm{obs}\}}$.

(iv) Unit. $[\mathbf{1}, \hat{C}_\alpha] = 0 \in \mathcal{I}_C$, so $\mathbf{1} \in \widetilde{\mathcal{A}}_{\{\mathrm{obs}\}}$.

(v) Closure under adjoint. Because $\mathcal{I}_C$ is a $(*)$ -ideal:

$$[A^\dagger, \hat{C}_\alpha] = -[A, \hat{C}_\alpha]^\dagger.$$

If the constraint set is $(*)$ -stable or the ideal is generated in a $(*)$ -closed way, the right-hand side lies in $\mathcal{I}_C$. Hence $A^\dagger \in \widetilde{\mathcal{A}}_{\{\mathrm{obs}\}}$.

Thus $\widetilde{\mathcal{A}}_{\{\mathrm{obs}\}}$ is a unital $(*)$ -subalgebra of $\mathcal{A}$. $\blacksquare$

Interpretation. This proposition supplies one of the missing local theorem completions: it makes the observable sector into a genuine algebraic object rather than a loose heuristic. The quotient form proposed in earlier drafts can now be introduced rigorously.

1.2.13 Physical Observable Algebra Modulo Constraints

Definition 1.7 (Physical Observable Algebra).

The **physical observable algebra** is defined by:

$$\mathcal{A}_{\{\mathrm{obs}\}} := \widetilde{\mathcal{A}}_{\{\mathrm{obs}\}} / \mathcal{I}_C.$$

Interpretation. Two algebraically zero-preserving observables are considered physically equivalent if they differ by an element of the constraint ideal, i.e., by something null in the physical sector.

This is the algebraic version of the familiar idea that physical observables are equivalence classes of kinematic expressions that preserve the constraint surface, modulo expressions that vanish on it.

1.2.14 From Algebraic to Represented Observables

Theorem 1.6 (From Algebraic to Represented Observables).

Let $A \in \widetilde{\mathcal{A}}_{\mathrm{obs}}$. Then $\pi_{\omega}(A)$ preserves $\mathcal{H}_{\mathrm{phys}}^{\omega}$.

Proof.

Let $\psi \in \mathcal{H}_{\mathrm{phys}}^{\omega}$. Then $\pi_{\omega}(\hat{C}_{\alpha})\psi = 0$ for all α by Section 0.6. Now:

$$\pi_{\omega}(\hat{C}_{\alpha})\pi_{\omega}(A)\psi = \pi_{\omega}(\hat{C}_{\alpha}A)\psi = \pi_{\omega}(A\hat{C}_{\alpha} + [\hat{C}_{\alpha}, A])\psi.$$

The first term vanishes because $\pi_{\omega}(\hat{C}_{\alpha})\psi = 0$. For the second term, since $A \in \widetilde{\mathcal{A}}_{\mathrm{obs}}$:

$$[A, \hat{C}_{\alpha}] \in \mathcal{I}_C.$$

By the ideal-kernel correspondence (Theorem 0.7), every represented element of $\mathcal{I}_C$ annihilates vectors in $\mathcal{H}_{\mathrm{phys}}^{\omega}$. Hence:

$$\pi_{\omega}([\hat{C}_{\alpha}, A])\psi = 0.$$

Therefore:

$$\pi_{\omega}(\hat{C}_{\alpha})\pi_{\omega}(A)\psi = 0 \quad \forall \alpha,$$

so $\pi_{\omega}(A)\psi \in \mathcal{H}_{\mathrm{phys}}^{\omega}$. $\square$

Interpretation. This proposition ties the algebraic observable algebra directly to the representation-level physical observables: algebraic zero-preservation descends to physical-kernel preservation.

1.2.15 Strong Observables Form a $(*)$ -Subalgebra

Theorem 1.7 (Strong Observable Algebra).

Let:

$$\mathcal{O}_{\mathrm{strong}} := \{ \hat{O} \in \mathcal{B}(\mathcal{H}_{\omega}) : [\hat{O}, \hat{\mathfrak{M}}_{\omega}] = 0 \}.$$

Then $\mathcal{O}_{\mathrm{strong}}$ is a unital $(*)$ -subalgebra of $\mathcal{B}(\mathcal{H}_{\omega})$.

Proof.

Let $\hat{O}_1, \hat{O}_2 \in \mathcal{O}_{\mathrm{strong}}$.

- **Addition:**

$$[\hat{O}_1 + \hat{O}_2, \hat{\mathfrak{M}}_{\omega}] = [\hat{O}_1, \hat{\mathfrak{M}}_{\omega}] + [\hat{O}_2, \hat{\mathfrak{M}}_{\omega}] = 0.$$

- **Scalar multiplication:** immediate.

- **Product:**

$$\begin{aligned} \hat{O}_1 \hat{O}_2, \hat{\mathfrak{M}}_\omega] &= \hat{O}_1 [\hat{O}_2, \\ \hat{\mathfrak{M}}_\omega] + [\hat{O}_1, \hat{\mathfrak{M}}_\omega] \hat{O}_2 &= 0. \end{aligned}$$

- **Unit:**

$$[\mathbf{1}, \hat{\mathfrak{M}}_\omega] = 0.$$

- **Adjoint:** Since $\hat{\mathfrak{M}}_\omega$ is self-adjoint:

$$\begin{aligned} [\hat{O}^\dagger, \hat{\mathfrak{M}}_\omega] &= -[\hat{O}, \\ \hat{\mathfrak{M}}_\omega]^\dagger &= 0. \end{aligned}$$

Thus $\mathcal{O}_{\text{strong}}$ is a unital $(*)$ -subalgebra. $\blacksquare$

Interpretation. The strong observable sector is algebraically stable and can be treated as a natural distinguished subalgebra inside the represented theory.

1.2.16 Effective Classical Zero-Preserving Transformations

In an effective classical description, the physical configuration space is:

$$\mathcal{X}_{\text{phys}} = \mathfrak{M}^{-1}(0).$$

A transformation $T: \mathcal{X} \rightarrow \mathcal{X}$ is **zero-preserving** if:

$$T(\mathcal{X}_{\text{phys}}) \subseteq \mathcal{X}_{\text{phys}}.$$

Equivalently:

$$\mathfrak{M}(x) = 0 \quad \Longleftrightarrow \quad \mathfrak{M}(Tx) = 0.$$

Using the classical master-zero equivalence, this can be written as:

$$C_\alpha(x) = 0 \quad \forall \alpha \quad \Longleftrightarrow \quad C_\alpha(Tx) = 0 \quad \forall \alpha.$$

Thus, a classical transformation is admissible if it maps constraint-satisfying configurations to constraint-satisfying configurations.

Again, this is not dynamics by itself. It is a preservation condition. A dynamical law would require a specified rule selecting a particular family of transformations or trajectories.

1.2.17 Approximate Zero Preservation

In effective or coarse-grained models, one may allow approximate physicality:

$$\mathfrak{M}(x) \leq \epsilon.$$

A transformation T may then be called **approximately zero-preserving** if there exists a tolerance control such that:

$$\mathfrak{M}(x) \leq \epsilon \quad \Longleftrightarrow \quad \mathfrak{M}(Tx) \leq \epsilon'$$

for some acceptable ϵ' .

A stronger approximate condition is:

$$\frac{M}{T}(x) \leq \frac{M}{T}(x) + \delta,$$

where δ measures how much constraint error the transformation may introduce.

These notions are useful in coarse-grained or numerical settings, but they are not part of the exact foundational definition. Exact zero preservation remains:

$$\frac{M}{T}(x) = 0 \quad \Longleftrightarrow \quad \frac{M}{T}(Tx) = 0.$$

> **Approximate zero preservation is effective; exact zero preservation is foundational.**

1.2.18 Observables and Measurement Language

The word "observable" can be misleading if read too operationally too soon. At this stage, a zero-preserving observable is not yet a full measurement procedure. It is not yet associated with a detector, record, Born probability, outcome set, or observer.

It is simply an admissibility-compatible operation or quantity.

Later sections may develop records, classical outcomes, probabilities, or measurement-like structures. Those require additional assumptions and arguments.

For now, the correct meaning is:

$$\boxed{\text{Observable means constraint-compatible relational operation, not yet measurement outcome.}}$$

This keeps the early framework free from premature measurement postulates.

1.2.19 Observables and Causal Structure

Zero-preserving observables prepare the ground for causal dependency.

If admissible events or structures are represented by zero-preserving operations, then one can begin to ask when one such operation depends on another. For example, an event e_j may be admissible only after another event e_i has occurred, closed, or been incorporated into the constraint-compatible structure.

This leads to a causal relation:

$$e_i \prec e_j.$$

But before causal dependency can be defined, the framework must know which operations and event-like structures are admissible. Zero preservation provides that criterion.

Thus, the conceptual order is:

$$\boxed{\text{physical sector} \quad \longrightarrow \quad \text{zero-preserving observables} \quad \longrightarrow \quad \text{causal dependency}.}$$

1.2.20 Guardrail — Zero Preservation Is Not Conservation of Energy

Because zero-preserving operations preserve the master-zero sector, one might be tempted to interpret them as conserving energy. This is not justified.

The master constraint is not energy. It measures constraint violation. Preserving its zero sector means preserving admissibility, not conserving an energy quantity.

An operator may preserve $\ker \hat{\mathfrak{M}}_\omega$ without defining a time symmetry or a conserved Hamiltonian.

> **Zero preservation means preservation of physical admissibility, not energy conservation.**

Energy-like notions require additional structures such as time translation, burden, stress, flux, or effective dynamics.

1.2.21 Guardrail — Zero Preservation Is Not Time Evolution

A zero-preserving operator is not automatically a time-evolution operator.

It may be a candidate physical transformation, but no external time parameter has been introduced. To call a family of zero-preserving operators a dynamics, one would need additional structure, such as:

$U(s_2, s_1)$,

a composition law:

$U(s_3, s_2)U(s_2, s_1) = U(s_3, s_1)$,

and an interpretation of the parameter s .

Even then, s would initially be an ordering or extension parameter, not necessarily physical time.

> **Zero-preserving does not mean time-evolving.**

It only means that physicality is not destroyed.

1.2.22 Guardrail — Strong Commutation Is Sufficient, Not Necessary

The condition $[\hat{O}, \hat{\mathfrak{M}}_\omega] = 0$ is useful and easy to verify, but it is stronger than the minimal requirement.

The minimal requirement is:

$\hat{O} \in \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega$.

An operator can satisfy the second condition without satisfying the first. Therefore, the manuscript should avoid defining all observables solely by global commutation unless that stronger choice is intended.

> **Physical observability requires zero preservation; full commutation is one sufficient route.**

1.2.23 Guardrail — The Observable Algebra May Be Representation-Dependent

Because the physical sector $\mathscr{H}_{\mathrm{phys}}^\omega$ depends on ω , the represented observable class may also depend on ω .

Thus, one should write $\mathcal{A}_{\mathrm{obs}}^\omega$ or $\mathcal{O}_{\mathrm{zp}}^\omega$ when the representation matters.

A representation-independent observable algebra may require an additional construction, such as an ideal normaliser or quotient algebra. That construction should be stated separately.

> **Represented observables are defined relative to a represented physical sector.**

1.2.24 Summary of Subsection 1.2

A zero-preserving observable is an operation that maps physical states to physical states.

In the represented Hilbert-space setting:

$\boxed{\hat{O} \text{ is zero-preserving}} \quad \Longleftrightarrow \quad \hat{O} \in \ker \hat{\mathcal{M}}_\omega \subseteq \ker \hat{\mathcal{M}}_\omega. \}$

Equivalently:

$\pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha \quad \Longleftrightarrow \quad \pi_\omega(\hat{C}_\alpha)\hat{O}\psi = 0 \quad \forall \alpha.$

A strong sufficient condition is:

$[\hat{O}, \hat{\mathcal{M}}_\omega] = 0,$

or more strongly:

$[\hat{O}, \pi_\omega(\hat{C}_\alpha)] = 0 \quad \forall \alpha.$

But the minimal physical condition is kernel preservation.

At the algebraic level, the observable sector is:

$\widetilde{\mathcal{A}}_{\mathrm{obs}} = \{ A \in \mathcal{A} : [A, \hat{C}_\alpha] \in \mathcal{I}_C \quad \forall \alpha \},$

and the physical observable algebra is:

$\mathcal{A}_{\mathrm{obs}} = \widetilde{\mathcal{A}}_{\mathrm{obs}} / \mathcal{I}_C.$

Algebraic zero-preserving observables descend to represented physical observables.

In an effective classical description:

$\boxed{T \text{ is zero-preserving}} \quad \Longleftrightarrow \quad T(\mathcal{M}^{-1}(0)) \subseteq \mathcal{M}^{-1}(0).$

The conceptual role of this subsection is:

$\boxed{\text{The physical sector is not merely defined; it must be preserved by admissible operations.}}$

This prepares the next subsection, where admissible event-like structures can be ordered by causal dependency.

1.3 Causal Dependency as a Strict Partial Order

1.3.1 Purpose of This Subsection

Section 1.1 established that the physical sector is the zero sector of the master constraint. Section 1.2 defined zero-preserving observables as operations that preserve this sector.

The next structural question is:

> **How does causal order arise before time?**

In a background-independent relational theory, time cannot be assumed as an external, continuous parameterising real line along which events evolve. Instead, temporal progression must emerge directly from the internal structural arrangements of the events themselves.

The Relational Constraint Framework establishes this by declaring that the causal dependency relation $\prec$ among primitive events $e \in \mathcal{E}$ constitutes a fundamental partial order. Events are not organised by an external clock; rather, their sequencing is defined by structural necessity—specifically, how the resolution of a constraint defect at one node conditions the boundary conditions for subsequent defects.

This subsection defines causal dependency, introduces the closure-based refinement, and proves that the dependency relation forms a strict partial order.

1.3.2 Why Causality Must Be Introduced Without External Time

The foundation does not assume external time. Therefore, causal order cannot be introduced as an ordering of events by a background parameter t . There is no primitive equation of the form:

$$\frac{d}{dt}\psi(t) = \dots,$$

and no pre-existing spacetime in which events are already arranged along temporal curves.

Nevertheless, the theory must eventually account for temporal structure. It must explain how one event may depend on another, how admissible structures may extend, and how a direction of history may arise.

The first step is to define causal order not as temporal succession in a background time, but as **dependency among zero-preserving structures**.

> **Causal order begins as admissibility dependency, not as background time order.**

If an admissible event e_j requires the prior closure, satisfaction, or availability of another event e_i , then one may write:

$e_i \prec e_j$

This relation does not yet mean that e_i occurs earlier than e_j in a spacetime coordinate. It means that e_j 's admissibility depends on e_i . Temporal interpretation comes later.

1.3.3 Definition — Primitive Event

Definition 1.8 (Primitive Event).

Let $\mathcal{E}$ be a set whose elements are called **primitive zero-preserving events**. At this level, an event $e \in \mathcal{E}$ is not a spacetime point. It is not assumed to be a coordinate location, a field value on a manifold, or an observation made by an external observer. Rather, an event is a localised instance of admissible relational structure, update, resolution, or admissible occurrence.

Interpretation. A primitive event is the minimal unit of structurally admissible change in the zero-preserving relational framework.

1.3.4 Definition — Constraint Closure

Definition 1.9 (Constraint Closure).

For each event $e \in \mathcal{E}$, let $\Gamma(e)$ denote its **constraint closure set**: the set of primitive zero-preserving structures required for e to be admissible as part of a zero-consistent history.

Remark 1.2. The closure notation $\Gamma(e)$ and its role in defining causal order are explicitly present in the framework's mature formulation. At this stage, $\Gamma(e)$ is treated as a primitive closure map sufficient to define causal dependency. The exact mathematical form of $\Gamma(e)$ may vary by model: it may be a set of prerequisite events, a substructure of a causal network, a closure domain, a collection of constraint dependencies, or an algebraic support profile.

1.3.5 Definition — Causal Dependency Relation

Definition 1.10 (Causal Dependency).

For events $e_i, e_j \in \mathcal{E}$, define:

$e_i \prec e_j$

to mean that e_j requires e_i for zero-closure.

In the closure-based refinement:

$$\boxed{e_i \prec e_j} \quad \Longleftrightarrow \quad \Gamma(e_i) \subseteq \Gamma(e_j) \quad \text{and} \quad \Gamma(e_i) \text{ is necessary for } \Gamma(e_j).$$

****Interpretation.**** Causality is thus not taken as a primitive time ordering. Instead:

$$\boxed{\text{causal precedence} = \text{dependency of zero-closure}}$$

This is one of the distinctive structural moves of the framework. Causal order is not imposed from outside; it is derived from the constraint-closure structure.

1.3.6 Why Dependency Implies Distinction

If e_j depends on e_i , that does not mean $e_i = e_j$. In fact, for primitive causality, dependency should imply non-identity.

If $e_i \prec e_j$, then e_i plays the role of a necessary condition for e_j . That means e_j is not identical to e_i , because the dependent structure includes an additional closure requirement not exhausted by e_i .

So we add the axiom:

$$e_i \prec e_j \quad \rightarrow \quad e_i \neq e_j$$

This directly gives irreflexivity. Because if we suppose $e \prec e$, then by the distinction rule $e \neq e$, which is impossible. Therefore $\neg(e \prec e)$.

1.3.7 Theorem — Irreflexivity from Proper Closure Extension

****Theorem 1.8 (Irreflexivity from Proper Closure Extension).****

Let $\mathcal{E}$ be a set of minimal zero-preserving events. Suppose each event $e \in \mathcal{E}$ has a constraint closure $\Gamma(e)$. Define $e_i \prec e_j$ only if $\Gamma(e_i) \subsetneq \Gamma(e_j)$. Then $\prec$ is irreflexive.

Proof.

For any event e , we have $\Gamma(e) = \Gamma(e)$. A set cannot be a proper subset of itself:

$$\Gamma(e) \not\subsetneq \Gamma(e).$$

Therefore the defining condition for $e \prec e$ cannot hold. Hence $\neg(e \prec e)$.

□

1.3.8 Theorem — Transitivity from Closure Inclusion and Necessity Composition

****Theorem 1.9 (Transitivity from Closure Inclusion).****

Let $e_1, e_2, e_3 \in \mathcal{E}$. Suppose $e_1 \prec e_2$ and $e_2 \prec e_3$. Then $e_1 \prec e_3$, provided closure necessity composes.

Proof.

By Definition 1.10:

$$\Gamma(e_1) \subsetneq \Gamma(e_2) \quad \text{and} \quad \Gamma(e_2) \subsetneq \Gamma(e_3).$$

By transitivity of proper subset inclusion:

$$\Gamma(e_1) \subsetneq \Gamma(e_3).$$

By the necessity composition assumption: if e_1 is necessary for e_2 , and e_2 is necessary for e_3 , then e_1 is necessary for e_3 .

Therefore $e_1 \prec e_3$. $\square$

Assumption 1.1 (Necessity Composition).

If $e_1 \prec e_2$ and $e_2 \prec e_3$, then the necessity relation composes: the necessity of $\Gamma(e_1)$ for $\Gamma(e_2)$ together with the necessity of $\Gamma(e_2)$ for $\Gamma(e_3)$ implies the necessity of $\Gamma(e_1)$ for $\Gamma(e_3)$.

This assumption is the minimal extra hypothesis required to complete the transitivity proof.

1.3.9 Theorem — Causal Dependency Is a Strict Partial Order

Theorem 1.10 (Causal Dependency Is a Strict Partial Order).

Under Definition 1.10 and Assumption 1.1, the relation $\prec$ on $\mathcal{E}$ is a strict partial order. That is, for all $e, e_1, e_2, e_3 \in \mathcal{E}$:

- Irreflexivity:** $\neg(e \prec e)$
- Transitivity:** if $e_1 \prec e_2$ and $e_2 \prec e_3$, then $e_1 \prec e_3$
- Asymmetry:** if $e_1 \prec e_2$, then $\neg(e_2 \prec e_1)$

Proof.

(i) Irreflexivity. Proved in Theorem 1.8.

(ii) Transitivity. Proved in Theorem 1.9.

(iii) Asymmetry. Suppose both $e_1 \prec e_2$ and $e_2 \prec e_1$. Then Definition 1.10 requires both $\Gamma(e_1) \subsetneq \Gamma(e_2)$ and $\Gamma(e_2) \subsetneq \Gamma(e_1)$, which is impossible. Hence asymmetry holds.

Therefore $\prec$ is a strict partial order. $\square$

1.3.10 Dependency Order Without Total Ordering

A strict partial order need not compare every pair of events.

There may exist events $e_i, e_j \in \mathcal{E}$ such that neither $e_i \prec e_j$ nor $e_j \prec e_i$ holds. Such events are **dependency-incomparable**.

Definition 1.11 (Causal Incomparability).

Two events $e_i, e_j \in \mathcal{E}$ are **causally incomparable** if:
 $e_i \not\prec e_j \wedge e_j \not\prec e_i$.

Definition 1.12 (Antichain).

An **antichain** $\Sigma \subset \mathcal{E}$ is a set of mutually incomparable events:
 $\forall e_i, e_j \in \Sigma, e_i \not\prec e_j \wedge e_j \not\prec e_i$.

An antichain serves as a candidate emergent spatial cross-section, since its elements are not causally ordered relative to each other.

> **Causal dependency gives partial order, not necessarily total sequence.**

This allows the framework to support parallel admissible developments without imposing a universal external clock.

1.3.11 Causal Order Versus External Time

The relation $e_i \prec e_j$ should not be read as $t(e_i) < t(e_j)$ for some primitive time function t .

At this stage, no such time function has been defined. Instead, $e_i \prec e_j$ means that e_i is structurally prior to e_j in the dependency profile.

A time function, if later introduced, would have to be compatible with the causal order. For example, one might later define a monotone function $T: \mathcal{E} \rightarrow \mathbb{R}$ such that:

$$e_i \prec e_j \quad \Longleftrightarrow \quad T(e_i) < T(e_j).$$

But such a function is not assumed here. It would be a derived or effective ordering parameter.

> **Causal dependency precedes time labelling.**

1.3.12 Causal Depth

Once a partial order exists, one may define a notion of causal depth.

For causally related events $e_i \prec e_j$, define:

$$L_C(e_i, e_j) := \max \{ n \mid e_i \prec e_1 \prec e_2 \prec \cdots \prec e_n \prec e_j \}.$$

This is the length of the longest dependency chain between e_i and e_j .

Causal depth is an ordering measure, but it is not yet physical time. It gives a pre-temporal skeleton from which temporal structure may later be built.

Definition 1.13 (Causal Depth).

The **causal depth** between e_i and e_j is the maximum number of intermediate dependency steps in any chain from e_i to e_j .

Later, proper time may be reconstructed by weighting causal depth by burden, clock factors, or other relational quantities. But the partial order provides the skeleton on which such later temporal structure may be built.

> **Causal depth is order-theoretic; proper time requires further structure.**

1.3.13 Causal Chains and Admissible Histories

A finite causal chain:

$$e_0 \prec e_1 \prec \cdots \prec e_n$$

may be interpreted as an **admissible dependency history**, provided each event in the chain is zero-compatible and each dependency relation is admissibility-preserving.

Such a chain is not yet a trajectory through spacetime. It is a sequence of dependency inclusions.

A history in this framework is therefore initially a chain or network of admissible dependency relations, not a path parameterised by external time.

> **A history is a structured chain of admissibility dependencies.**

Later, if emergent time and geometry are constructed, such histories may acquire temporal or spatial interpretation.

1.3.14 Cycles and Inconsistency

If causal dependency is strict, cycles are forbidden.

A causal cycle would have the form:

$$e_1 \prec e_2 \prec \cdots \prec e_n \prec e_1.$$

By transitivity, this would imply $e_1 \prec e_1$, contradicting irreflexivity.

Therefore, strict causal dependency excludes dependency cycles.

> **Strict causal dependency is acyclic.**

This acyclicity is not imposed by external time. It follows from the strict partial-order structure.

1.3.15 Causal Dependency and Zero Preservation

Causal dependency must be compatible with zero preservation.

If $e_i \prec e_j$, then the transition or dependency relation from e_i to e_j must not introduce constraint violation. Otherwise, the dependency would not relate admissible structures within the physical sector.

In an operator-like representation, one may think of an event transition as being mediated by a zero-preserving operation $\hat{O}_{ij}$. Then one requires:

$$\hat{O}_{ij} \setminus, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega.$$

In an effective classical form, if an event transition is represented by a map T_{ij} , then:

$$T_{ij}(\frac{1}{M} \cdot 0) \subseteq \frac{1}{M} \cdot 0.$$

> **Causal links must be zero-compatible links.**

Thus, causal dependency should be understood as dependency inside the admissible sector, not as arbitrary relation among kinematic possibilities.

1.3.16 Causal Dependency Is Not Correlation

It is also important to distinguish causal dependency from correlation.

The relation $e_i \prec e_j$ is directed. It says that e_j depends on e_i .

A correlation relation, by contrast, may be symmetric or non-directed. For example, two event contexts may have strong correlation in a state ω , expressed schematically as $K_\omega(e_i, e_j) \approx 1$, without either one being a causal predecessor of the other.

> **Causal dependency is directed; correlation need not be.**

This distinction will become central in the next subsection, where the Two-Link Principle separates causal links from relational correlation links.

1.3.17 Local Finiteness

For two causally related events $e_i \prec e_j$, define the **causal interval**:

$$I(e_i, e_j) := \{ e_k \in E \mid e_i \prec e_k \prec e_j \}.$$

A locally finite causal structure satisfies:

$$|I(e_i, e_j)| < \infty.$$

This condition provides a discrete pre-geometric analogue of finite causal separation. It is useful for a first model, but is not assumed forever.

1.3.18 Guardrails for This Subsection

1. **Causality is dependency, not time.** The relation $\prec$ is a structural dependency relation, not an external time order.
2. **Extension is not evolution in time.** The extension index s labels admissible growth steps; it is not physical temporal duration.
3. **The partial-order theorem is completed under an explicit extra assumption.** Assumption 1.1 (necessity composition) is introduced to finish the transitivity proof.
4. **Causal depth is not proper time.** Although causal depth can be defined from a partial order, it is not yet physical duration.
5. **Causal dependency is pre-geometric.** It does not rely on light cones, metric signature, finite propagation speed, or a manifold.

1.3.19 Summary of Subsection 1.3

Causal dependency provides the first internal ordering structure of the framework.

Events are admissible relational occurrences, not spacetime points. A causal relation $e_i \prec e_j$ means that e_j 's admissibility depends on e_i .

Using dependency profiles $\Gamma(e)$, one defines:

$$e_i \prec e_j \iff \Gamma(e_i) \subseteq \Gamma(e_j) \text{ and } \Gamma(e_i) \neq \Gamma(e_j).$$

Under this definition, $\prec$ is a strict partial order: irreflexive, asymmetric, and transitive.

This gives the framework a pre-temporal causal skeleton. It supports dependency chains, causal depth, and admissible histories, but it does not yet define spacetime, metric geometry, external time, or proper duration.

The central result is:

$$\boxed{\text{Causal order can be reconstructed as strict admissibility dependency before time is introduced.}}$$

1.4 The Two-Link Principle

1.4.1 Purpose of This Subsection

The previous subsection introduced causal dependency as a strict partial order among admissible events. In that setting, $e_i \prec e_j$ means that the admissibility of e_j depends on e_i . This relation is directed, order-like, and pre-temporal. It gives the framework a way to speak about causal structure without assuming an external time parameter.

However, causal dependency is not the only kind of relation needed by the theory.

The framework must also explain how space-like structure, distance, locality, and geometric neighbourhoods can emerge. These cannot be obtained merely by saying that one event depends on another. Spatial proximity is not the same thing as causal priority. Two structures may be spatially or relationally close without either one being causally prior to the other. Conversely, one event may be causally upstream of another without being nearby in the eventual emergent geometry.

This motivates the **Two-Link Principle**.

> **The framework requires two distinct relational links: causal dependency and relational correlation.**

Causal links support order and eventual time-like structure. Relational-correlation links support proximity and eventual space-like structure. The two may interact, but they must not be collapsed into one primitive relation.

1.4.2 The Problem with a Single-Link Theory

A theory that uses only one kind of link faces a structural ambiguity.

If every link is interpreted causally, then strong correlation or adjacency would automatically imply causal dependence. This would make it difficult to describe simultaneous, mutually correlated, or space-like related structures. It would also risk turning spatial closeness into causal precedence.

If every link is interpreted correlationally, then dependency and direction are lost. The theory would have difficulty explaining ordered histories, admissibility chains, causal depth, or irreversible extension.

Thus, a single relation cannot safely do both jobs.

The framework therefore separates:

- **directed admissibility dependency** from
- **state-dependent relational correlation**.

This distinction is foundational for later emergence. It prevents space from being reduced too quickly to causal order and prevents causality from being reduced to mere correlation.

> **Correlation is not causation, and causal dependency is not spatial distance.**

1.4.3 Causal Links

A **causal link** is a directed dependency relation between admissible events:

$e_i \prec e_j$.

It means that e_i is a necessary admissibility predecessor of e_j . In the dependency-profile formulation, this may be represented by proper inclusion:

$\Gamma(e_i) \subsetneq \Gamma(e_j) \quad \text{and} \quad \Gamma(e_i) \text{ is necessary for } \Gamma(e_j)$.

A causal link has the following features:

1. It is **directed**.
2. It is **asymmetric** in the strict case.
3. It supports **chains** ($e_0 \prec e_1 \prec \dots \prec e_n$).
4. It contributes to **causal depth**.
5. It is associated with **admissibility dependency** rather than metric nearness.

Thus, if $e_i \prec e_j$, then e_j lies downstream of e_i in the dependency structure.

This does not yet mean that e_i occurs earlier than e_j in physical clock time. It means that e_j 's admissibility requires e_i .

> **Causal links encode dependency before clock time is defined.**

1.4.4 Relational-Correlation Links

A **relational-correlation link** expresses state-dependent association, similarity, mutual constraint sensitivity, or correlation between event contexts or observables.

Given a physical state ω and localisable observables or event-context representatives O_i, O_j , one may define a correlation kernel schematically as:

$$K_\omega(e_i, e_j) \quad \text{or} \quad K_\omega(O_i, O_j).$$

This kernel measures how strongly the two relational contexts are associated in the state ω . The exact definition of K_ω belongs to the later correlation-geometry section. At this stage, only its structural role matters.

A relational-correlation link has the following features:

1. It need not be **directed**.
2. It may be **symmetric**.
3. It may vary with the **physical state ω** .
4. It supports emergent notions of **proximity**.
5. It is **not by itself** a causal dependency relation.

If $K_\omega(e_i, e_j)$ is large, the two event contexts are strongly correlated. Later, one may define an emergent distance by a relation such as:

$$d_\omega(e_i, e_j) = -\lim_{\epsilon \rightarrow 0} \log K_\omega(e_i, e_j),$$

under suitable assumptions.

But this does not mean $e_i \prec e_j$ or $e_j \prec e_i$. It only means that the two contexts are strongly linked in the correlation structure.

> **Relational-correlation links encode proximity-like structure before space is defined.**

1.4.5 Statement of the Two-Link Principle

Principle 1.1 (Two-Link Principle).

The framework contains two structurally distinct kinds of link:

$$\boxed{\text{Causal links:}} \quad e_i \prec e_j$$

and

$$\boxed{\text{Relational-correlation links:}} \quad K_\omega(e_i, e_j).$$

The first is directed and dependency-based. The second is correlation-based and may be symmetric or state-dependent.

The principle may be stated compactly as:

$\boxed{\text{Causal dependency and relational correlation are distinct primitive post-foundational structures.}}$

They are "post-foundational" because both are introduced only after the physical zero sector has been defined. They are not more primitive than constraint compatibility. But once physical admissibility is available, both kinds of link are needed.

1.4.6 The Core Friction: Asymmetry vs. Symmetry

The fundamental friction between these two layers is mathematical and ontological.

The causal network forms an acyclic, directed graph tracking the irreversible flow of constraint enforcement. The relational network forms a weighted, undirected graph tracking instantaneous statistical or algebraic dependencies.

If the relational network operated independently of the causal order, non-local quantum correlations would instantly collapse the causal history into a trivial cliquish structure, rendering the concepts of time, sequence, and evolution meaningless. Conversely, if the causal network existed without relational correlations, events would remain isolated nodes incapable of generating macroscopic spatial intervals or stable field configurations.

The framework resolves this tension through the **master zero constraint**. The master constraint dictates that the algebraic ideal generated by the constraints must annihilate physical states, mapping relational correlations directly onto the causal channels. This guarantees that a relational correlation between two space-like separated domains can only be established if their respective ancestral histories intersect in a manner permitted by the causal propagator network.

1.4.7 Why Causal Links Support Time-Like Structure

Causal links generate ordered chains:

$$e_0 \prec e_1 \prec \dots \prec e_n.$$

Such chains support causal depth:

$$L_C(e_0, e_n) = \max\{n \mid e_0 \prec e_1 \prec \dots \prec e_n\}.$$

This gives the framework an order-theoretic skeleton from which temporal structure may later be derived.

However, causal depth alone is not proper time. To obtain something resembling physical duration, later sections must introduce additional weights, such as burden, reconciliation cost, lapse factors, or local clock-rate suppression.

Still, causal links are the correct starting point for time-like emergence because they encode direction and dependency.

> **Time-like structure begins with directed causal dependency.**

1.4.8 Why Relational-Correlation Links Support Space-Like Structure

Relational-correlation links support proximity-like relations.

If two event contexts are strongly correlated in a physical state, then they may be close in an emergent relational geometry. If they are weakly correlated, they may be far apart.

This motivates the later definition of correlation distance:

$$d_{\omega}(A, B) = -\ell_0 \log K_{\omega}(A, B).$$

Under additional assumptions, such as symmetry, diagonal normalisation, and multiplicative consistency, this distance can become a pseudometric or approximate metric.

But none of that belongs to causal order alone. A causal partial order does not directly define spatial distance. It can say that one event depends on another, but it cannot by itself say how far apart two non-comparable events are.

Therefore, a separate correlation link is needed.

> **Space-like structure begins with state-dependent relational correlation.**

1.4.9 Independence of the Two Links

The two kinds of link are logically independent.

It is possible to have causal dependency without strong correlation:

$$e_i \prec e_j \quad \text{while} \quad K_{\omega}(e_i, e_j) \quad \text{is small.}$$

This would represent a case where e_j depends on e_i , but the two event contexts are not close in the emergent correlation geometry.

It is also possible to have strong correlation without causal dependency:

$$K_{\omega}(e_i, e_j) \approx 1 \quad \text{while neither} \quad e_i \prec e_j \quad \text{nor} \quad e_j \prec e_i.$$

This would represent a case where two contexts are relationally close or highly correlated, but neither is an admissibility predecessor of the other.

Therefore:

$$\boxed{e_i \prec e_j \not\Rightarrow K_{\omega}(e_i, e_j) \text{ is large.}}$$

and

$$\boxed{K_{\omega}(e_i, e_j) \text{ large} \not\Rightarrow e_i \prec e_j.}$$

This independence is necessary for the framework to support both causal ordering and spatial neighbourhoods.

1.4.10 Interaction Between the Two Links

Although causal links and relational-correlation links are distinct, they may interact.

For example, the causal propagation of admissible structure may change correlation profiles. If an event e_i is causally upstream of e_j , then the admissibility of e_j may inherit, transform, or constrain correlations associated with e_i .

Conversely, strong relational correlation may affect which causal extensions are low-burden or admissible. If two regions are highly correlated, extending one may impose constraint pressure on the other.

Thus, the two links are not isolated. They form a coupled relational structure.

A schematic combined structure may be written as:

$(\mathcal{E}, \prec, K_\omega)$,

where:

- $\mathcal{E}$ is the event set,
- $\prec$ is the causal dependency relation,
- K_ω is the state-dependent correlation kernel.

This combined structure is the **pre-spacetime relational skeleton**.

> **Causal and correlation links may interact, but they remain conceptually distinct.**

1.4.11 The Two-Link Graph

One may visualise the combined structure as a graph with two types of edges.

A causal edge is directed:

$e_i \rightarrow e_j$.

A relational-correlation edge may be undirected or weighted:

$e_i \rightleftharpoons e_j$,

with weight $K_\omega(e_i, e_j)$.

Thus the event structure is not a single graph but a **layered graph**:

$\mathcal{G}_\omega = (\mathcal{E}, \mathcal{C}, \mathcal{R}_\omega)$,

where:

- $\mathcal{C} = \{(e_i, e_j) : e_i \prec e_j\}$ is the causal edge set,
- $\mathcal{R}_\omega$ is the correlation-link structure determined by the state ω .

This is only a schematic representation. The true mathematical object may be a partially ordered set equipped with a kernel, a category with correlation weights, a network of local algebras, or another relational structure. But the two-link distinction remains.

1.4.12 Cosmological Implication: Common Closure

The Two-Link Principle suggests a cosmological application.

In standard cosmology, the horizon problem asks why widely separated regions of space appear homogeneous despite insufficient time for ordinary causal interaction. Inflation is the standard leading response, in which regions now widely separated were once close enough to be in causal contact before rapid expansion.

In the present framework, this is reformulated as the **Primordial Common-Closure Hypothesis**.

Regions that are spatially distant in the later emergent geometry may nevertheless share common causal-constraint ancestry in the pre-geometric order:

$$C_0 \prec A, \quad C_0 \prec B, \quad d_\omega(A, B) \gg \ell_0.$$

Thus spatial separation at late times does not imply primordial constraint independence.

This may provide a pre-geometric alternative to inflation: large-scale uniformity may reflect inherited common constraint values rather than late spatial communication.

> **Large spatial distance does not imply primordial causal-constraint independence.**

This is not yet a proof of CMB uniformity. It is a structural possibility required by the two-link separation.

1.4.13 Avoiding Causal Overreach

A common error would be to treat every correlation as a causal link. In this framework, that is not allowed.

If $K_\omega(e_i, e_j)$ is large, one may infer strong relational association, but not directed dependency. To assert causal dependency, one must show $e_i \prec e_j$ or $e_j \prec e_i$ through the dependency structure.

> **High correlation does not imply causal precedence.**

This guardrail is especially important when later discussing non-local correlations, entanglement-like structure, or long-range relational coherence.

1.4.14 Avoiding Geometric Overreach

The reverse error is to treat every causal link as a spatial neighbourhood relation.

If $e_i \prec e_j$, then e_i is causally upstream of e_j . But this does not by itself imply that e_i and e_j are close in emergent space.

Spatial closeness requires a correlation-geometric measure, such as $K_\omega(e_i, e_j)$ or its associated distance.

> **Causal dependency does not by itself define spatial distance.**

This guardrail prevents the causal partial order from being overused as a metric.

1.4.15 Guardrails for This Subsection

1. **Two links means two structural roles, not two substances.** The principle distinguishes dependency and correlation. Both may be carried by the same underlying relational algebraic structure.
2. **Correlation links are state-dependent.** The notation K_ω includes the state ω for a reason. Correlation is evaluated in a state. Different physical states may produce different correlation structures.
3. **The causal order is not automatically Lorentzian.** Although causal dependency may later approximate a Lorentzian causal order, this is not assumed here.
4. **Causal dependency and relational correlation are distinct until related by a model.** The framework does not automatically identify spatial separation with causal independence.

1.4.16 Summary of Subsection 1.4

The Two-Link Principle separates two essential structures:

$\boxed{\text{causal dependency: } e_i \prec e_j}$

and

$\boxed{\text{relational correlation: } K_\omega(e_i, e_j)}$

Causal dependency is directed, order-like, and supports time-like emergence. Relational correlation is state-dependent, proximity-like, and supports space-like emergence.

The two links are independent but may interact. Strong correlation does not automatically imply causal precedence, and causal dependency does not automatically imply spatial closeness.

The combined pre-spacetime relational structure may be represented schematically as:

$\boxed{(\mathcal{E}, \prec, K_\omega)}$

This structure gives the framework enough expressive power to later reconstruct both temporal and spatial features without assuming either at the foundation.

1.5 The Open Extension Principle

1.5.1 Purpose of This Subsection

The preceding subsections established three post-foundational structures:

1. **The physical zero sector** — the set of admissible states satisfying the master constraint.
2. **Zero-preserving observables** — operations that preserve admissibility.
3. **Causal dependency** — a strict partial order among admissible events based on constraint closure.

The Two-Link Principle then distinguished causal dependency from relational correlation.

The next question is:

> **How does the admissible relational structure continue, grow, or unfold?**

The foundation defines what it means for a state or structure to be physical. Zero-preserving operations define what it means to remain physical under transformation. Causal dependency defines how admissible events may depend on one another. But none of these yet says whether the admissible structure is complete, extendible, or dynamically closed.

The **Open Extension Principle** addresses this gap. It states that a physical universe is not merely a closed solution to the constraint equations, but an **extendible** zero-preserving structure: one that admits further admissible continuations.

> **A physical relational structure is open if it admits further zero-preserving extension.**

This principle allows the framework to speak about growth, continuation, history, and non-finality without treating time as primitive. Extension is not yet physical time. It is an admissibility relation between relational structures.

1.5.2 Why Extension Is Needed

A constraint-compatible structure may be physically admissible, but admissibility alone does not say whether the structure is complete, extendible, or dynamically closed.

Suppose a relational structure $\mathcal{U}$ satisfies the physical constraints. One may then ask:

> **Can $\mathcal{U}$ be enlarged while remaining physical?**

If the answer is yes, then the structure is not terminal. It has possible admissible continuations.

This matters because a framework that only defines physicality as a static zero condition risks producing a frozen picture. The open extension layer prevents that by introducing the possibility of constraint-compatible continuation.

The key point is that continuation must preserve physicality. An arbitrary enlargement of a relational structure is not acceptable. Only an extension that remains within the zero sector counts as physically admissible.

> **Extension is not arbitrary growth; it is zero-preserving enlargement.**

1.5.3 Relational Structures

Definition 1.14 (Relational Structure).

A **relational structure** is a tuple:

$$\mathcal{U}_s = (\mathcal{E}_s, \prec_s, \mathcal{R}_{\{\omega, s\}}),$$

where:

- $\mathcal{E}_s$ is a set of primitive zero-preserving events at extension stage s ,
- $\prec_s$ is the causal dependency partial order on $\mathcal{E}_s$,
- $\mathcal{R}_{\{\omega, s\}}$ is the relational/correlation structure determined by the physical state ω at stage s .

The index s is an **extension label**, not a physical time coordinate. It orders stages of admissible growth.

Interpretation. Depending on the level of description, $\mathcal{U}_s$ may represent:

- a finite event network,
- a partial causal structure,
- a constraint-compatible algebraic state,
- a physical configuration in an effective model,
- or a coarse-grained relational universe.

The abstract requirement is that $\mathcal{U}_s$ belongs to the physical sector. In effective notation:

$$\mathfrak{M}(\mathcal{U}_s) = 0.$$

In algebraic notation, it means that the state associated with $\mathcal{U}_s$ satisfies:

$$\omega(\hat{C}_{\alpha}^{\dagger} \hat{C}_{\alpha}) = 0 \quad \forall \alpha.$$

1.5.4 Definition — Extension

Definition 1.15 (Extension).

An **extension** of a relational structure $\mathcal{U}_s$ is a structure $\mathcal{U}_{s+1}$ such that $\mathcal{U}_{s+1}$ contains, continues, refines, or enlarges $\mathcal{U}_s$ in the relevant relational sense.

We write:

$$\mathcal{U}_s \hookrightarrow \mathcal{U}_{s+1} \quad \text{or} \quad \mathcal{U}_s \rightarrow \mathcal{U}_{s+1}.$$

The notation does not yet denote time evolution. It denotes structural inclusion, continuation, or extension.

The relation $\mathcal{U}_s \rightarrow \mathcal{U}_{s+1}$ may mean, depending on context:

1. Adding a new admissible event to $\mathcal{E}_s$.
2. Adding a new causal dependency link to $\prec_s$.
3. Adding a new correlation relation to $\mathcal{R}_{\{\omega, s\}}$.
4. Refining a coarse-grained structure.
5. Extending a partial relational configuration.
6. Enlarging a constraint-compatible event network.
7. Continuing an admissible history by one or more dependency steps.

The common requirement is that $\mathcal{U}_{s+1}$ preserves the physical admissibility of $\mathcal{U}_s$.

1.5.5 Definition — Admissible Extension

Definition 1.16 (Admissible Extension).

An extension $\mathcal{U}_s \rightarrow \mathcal{U}_{s+1}$ is called **admissible** if the extended structure remains physical.

In effective master-functional notation:

$$\boxed{\mathcal{U}_s \rightarrow \mathcal{U}_{s+1} \text{ is admissible}} \quad \Longleftrightarrow \quad \mathcal{M}(\mathcal{U}_{s+1}) = 0.$$

In represented Hilbert-space notation, if the extension is implemented by an operator or map $\hat{E}: \mathcal{H}_\omega \rightarrow \mathcal{H}_\omega$, then admissibility requires:

$$\hat{E} \mid \mathcal{H}_{\mathrm{phys}}^\omega \subseteq \mathcal{H}_{\mathrm{phys}}^\omega.$$

That is:

$$\hat{E} \mid \ker \hat{\mathcal{M}}_\omega \subseteq \ker \hat{\mathcal{M}}_\omega.$$

In algebraic state language, an extension from a physical state ω_s to an extended state ω_{s+1} is admissible if:

$$\omega_{s+1}(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \forall \alpha.$$

Thus, in every formulation, admissible extension means:

$$\boxed{\text{extension while preserving constraint-zero physicality.}}$$

1.5.6 Extension Is Not External Time Evolution

The notation $\mathcal{U}_s \rightarrow \mathcal{U}_{s+1}$ is useful, but it must be interpreted carefully.

The label s is **not** a physical time coordinate. It is an ordering label for extension steps.

At this stage, the framework has not introduced:

- a primitive time parameter t ,
- a derivative $\frac{d}{dt}$,
- a Hamiltonian H ,
- or a Schrödinger-type equation $\frac{d}{dt}\psi(t) = H\psi(t)$.

Therefore, extension should not be described as ordinary time evolution.

Instead, $\mathcal{U}_s \rightarrow \mathcal{U}_{s+1}$ means that an admissible structure is extended to another admissible structure. The extension index s labels stages of admissible growth, not instants of physical time.

> **Extension order is not yet physical time.**

Physical time may later be reconstructed from causal chains, extension depth, burden weighting, clock suppression, or other relational quantities. But it is not assumed here.

1.5.7 Definition — Extension Set

Definition 1.17 (Extension Set).

For an admissible structure $\mathcal{U}_s$, define its **extension set** by:

$$\mathrm{Ext}(\mathcal{U}_s) := \{ \mathcal{U}_{s+1} : \mathcal{U}_s \text{ to } \mathcal{U}_{s+1} \text{ is admissible} \}.$$

Equivalently, in effective notation:

$$\mathrm{Ext}(\mathcal{U}_s) = \{ \mathcal{U}_{s+1} : \mathcal{U}_s \text{ to } \mathcal{U}_{s+1} \text{ and } \frac{M}{\mathcal{U}_{s+1}} = 0 \}.$$

This set contains all possible zero-preserving continuations of $\mathcal{U}_s$.

If $\mathrm{Ext}(\mathcal{U}_s) = \varnothing$, then $\mathcal{U}_s$ is **terminal** with respect to the chosen extension rules.

If $\mathrm{Ext}(\mathcal{U}_s) \neq \varnothing$, then $\mathcal{U}_s$ remains **open**.

1.5.8 Definition — Open Relational Structure

Definition 1.18 (Open Relational Structure).

An admissible relational structure $\mathcal{U}_s$ is called **open** if it has at least one admissible extension:

$$\boxed{\mathrm{Ext}(\mathcal{U}_s) \neq \varnothing}.$$

It is called **extension-terminal** if:

$$\boxed{\mathrm{Ext}(\mathcal{U}_s) = \varnothing}.$$

Openness therefore means that the structure is not complete relative to the admissibility rules. There remains at least one way to continue the relational structure without generating constraint violation.

This definition does not assume that all possible extensions occur. It only says that the admissible continuation space is non-empty.

1.5.9 The Open Extension Principle

Postulate 1B (Open Extension Principle).

A physical universe is not merely a zero-consistent partial order, but an **extendible** zero-consistent partial order. That is, a universe is a structure $\mathcal{U}_s = (\mathcal{E}_s, \prec_s)$ such that:

$$\frac{M}{\mathcal{U}_s} = 0$$

and there exists at least one admissible extension $\mathcal{U}_s \text{ to } \mathcal{U}_{s+1}$ with:

$$\frac{M}{\mathcal{U}_{s+1}} = 0.$$

A compact statement is:

$\boxed{\text{A physical relational structure may continue only through admissible zero-preserving extensions.}}$

A stronger cosmological version may be stated as:

$\boxed{\text{The universe is open if its current admissible relational structure has non-empty extension set.}}$

That is:

$\boxed{\mathrm{Ext}(\mathcal{U}) \neq \varnothing.}$

This should be treated as a principle or assumption unless a specific model proves non-empty extension.

1.5.10 Classification of Structures

The open extension principle classifies relational structures into three types:

1. Closed Solution.

A structure $\mathcal{U}$ is a closed solution if:

$\mathrm{Ext}(\mathcal{U}) = \varnothing.$

This is a completed mathematical object. It satisfies the constraints, but has no admissible continuation. It is like a finished proof or a closed book.

2. Open Universe.

A structure $\mathcal{U}$ is an open universe if:

$\mathrm{Ext}(\mathcal{U}) \neq \varnothing.$

This is an ongoing, dynamically unfinished structure. It satisfies the constraints, and there remains at least one way to extend it while preserving constraint compatibility.

3. Maximal Universe.

A structure $\mathcal{U}$ is maximal if it has exhausted all possible extensions. It is treated as a "completed history," not a "living universe." A maximal universe is the limit of an infinite chain of admissible extensions.

> **The open extension principle states that a physical universe is an open structure, not a closed solution.**

This distinction is philosophically important. The framework does not treat the universe as a static block. It treats it as an ongoing, extendible, zero-preserving process.

1.5.11 Extension and Causal Dependency

Admissible extension naturally interacts with causal dependency.

If an extension adds a new event e_{new} to an existing structure containing events $\{e_0, \dots, e_n\}$, then the new event must be placed consistently with the causal dependency relation.

For example, an extension may add dependency links:

$e_i \prec e_{\mathrm{new}}$

for some existing events e_i . These links indicate that the admissibility of e_{new} depends on prior closure supplied by those earlier events.

The extension is admissible only if the resulting dependency structure remains consistent. In particular, if causal dependency is strict, the extension must not introduce cycles such as:

$e_{\mathrm{new}} \prec e_i \prec e_{\mathrm{new}}$.

Thus, admissible extension must preserve both:

- **constraint-zero compatibility**, and
- **causal-order consistency**.

> $\mathcal{U}_s \rightarrow \mathcal{U}_{s+1}$ is admissible only if it preserves zero-compatibility and causal acyclicity.

The exact causal consistency conditions depend on the chosen model of $\prec$.

1.5.12 Extension and the Two-Link Structure

If the relational structure is represented as $(\mathcal{E}, \prec, K_\omega)$, then an extension may modify one or more of these components:

$(\mathcal{E} \rightarrow \mathcal{E}', \prec \rightarrow \prec', K_\omega \rightarrow K'_\omega)$

For example, an admissible extension may:

1. Add new events to $\mathcal{E}$.
2. Add new causal links to $\prec$.
3. Update correlation weights in K_ω .
4. Refine equivalence classes of relational proximity.
5. Alter correlation profiles in response to new admissible structure.

The extension is acceptable only if the resulting structure remains physical and internally consistent.

Thus, for a two-link relational skeleton, admissible extension requires:

$(\mathcal{E}, \prec, K_\omega) \rightarrow (\mathcal{E}', \prec', K'_\omega)$

with preservation of:

- zero compatibility,
- causal consistency,
- well-defined correlation structure.

1.5.13 Lemma — Zero-Preservation Under Extension

****Lemma 1.2 (Zero-Preservation Under Extension).****

If $\mathcal{U}_s \rightarrow \mathcal{U}_{s+1}$ is an admissible extension in the sense of Definition 1.16, then $\mathcal{U}_{s+1}$ remains in the physically admissible class.

Proof.

By definition of admissible extension, $\frac{M}{\mathcal{U}_{s+1}} = 0$. Therefore $\mathcal{U}_{s+1}$ satisfies the zero-preserving condition and remains physically admissible. $\square$

****Interpretation.**** This lemma is conceptually simple but structurally important: extension does not move the system out of the physical sector.

1.5.14 Extension-History Equivalence

The extension index ss is a gauge-like construction parameter. Different extension sequences that generate the same causal dependency records should be physically equivalent.

Suppose two incomparable events e_a, e_b (neither depends on the other) are added in different orders:

- Sequence 1: e_a then e_b .
- Sequence 2: e_b then e_a .

If neither $e_a \prec e_b$ nor $e_b \prec e_a$ holds, then both histories are physically equivalent:

$$(e_a, e_b) \sim_{\{\mathrm{hist}\}} (e_b, e_a).$$

****Definition 1.19 (Extension-History Equivalence).****

Two extension histories are ****physically equivalent**** if they generate the same causal dependency records. If $e_a \not\prec e_b$ and $e_b \not\prec e_a$, then:

$$\boxed{(e_a, e_b) \sim_{\{\mathrm{hist}\}} (e_b, e_a).}$$

This equivalence relation prevents the extension order from becoming a physically observable preferred frame. It ensures that only the resulting causal order $(\mathcal{E}, \prec)$ and internal relational quantities are observable, not the construction sequence ss .

> **** ss = gauge extension parameter, not time.****

1.5.15 Branching Extensions

An admissible structure may have more than one possible extension:

$$\mathrm{Ext}(\mathcal{U}_s) = \{ \mathcal{U}'_1, \mathcal{U}'_2, \mathcal{U}'_3, \dots \}.$$

This means that the future continuation of the structure is not represented by a single predetermined chain at the foundational level. Instead, there may be multiple admissible continuations.

This branching should not be interpreted too quickly as quantum branching, many-worlds ontology, stochastic collapse, or classical indeterminism. It simply means that, relative to the extension rules, more than one zero-preserving continuation is available.

> **Branching extension means multiple admissible continuations, not yet a probability interpretation.**

Probability, Born measure, records, or selection rules require additional structure.

1.5.16 Extension Chains

An **extension chain** is a sequence:

$$\mathcal{U}_0 \rightarrow \mathcal{U}_1 \rightarrow \mathcal{U}_2 \rightarrow \cdots$$

such that each step is admissible:

$$\mathcal{U}_{s+1} \in \mathrm{Ext}(\mathcal{U}_s).$$

That is, $\mathcal{M}(\mathcal{U}_s) = 0$ for all s in an effective master-functional notation.

An extension chain may be interpreted as an **admissible history** of relational growth. But again, it is not yet a trajectory through external time.

The chain parameter s labels extension steps. It does not yet measure physical duration.

> **An extension chain is an admissible history, not yet a timed trajectory.**

1.5.17 Extension Depth

Given an extension chain $\mathcal{U}_0 \rightarrow \mathcal{U}_1 \rightarrow \cdots \rightarrow \mathcal{U}_n$, one may define an **extension depth**:

$$D_{\mathrm{ext}}(\mathcal{U}_n) = n$$

relative to the chosen initial structure $\mathcal{U}_0$ and the chosen chain.

If there are multiple chains reaching the same structure, one may define maximal or minimal extension depth.

These quantities are order-theoretic. They are not yet physical time.

Physical duration may later require weighting extension depth by burden, cost, clock factors, or causal structure.

1.5.18 Admissible Extension as Zero-Preserving Map

If extension is represented by a map $E_s: \mathcal{U}_s \mapsto \mathcal{U}_{s+1}$, then admissibility requires:

$$\mathcal{U}_s \in \mathcal{U}_{\mathrm{phys}} \quad \Longrightarrow \quad E_s(\mathcal{U}_s) \in \mathcal{U}_{\mathrm{phys}}.$$

In effective notation:

$$\frac{M(\mathcal{U}_s)}{M(E_s(\mathcal{U}_s))} = 0 \quad \Longleftrightarrow \quad \frac{M}{E_s(\mathcal{U}_s)} = 0.$$

This is exactly zero preservation.

Therefore, admissible extension is a special case of zero-preserving transformation:

$$\boxed{\text{admissible extension} \subseteq \text{zero-preserving transformation}.}$$

The difference is interpretive: a general zero-preserving transformation may act within a fixed structure, while an extension enlarges or continues the structure.

1.5.19 Open Extension and Novelty

The open extension principle allows the framework to speak about **novelty** in a controlled way.

A new event, relation, or correlation may appear in an extension $\mathcal{U} \rightarrow \mathcal{U}'$ if it is not already contained in $\mathcal{U}$, but the resulting structure remains admissible.

Thus, novelty is not unconstrained creation. It is constrained extension.

> **New relational content is physical only if it enters through admissible extension.**

This gives a formal role to emergence without abandoning constraint compatibility.

1.5.20 Terminal Structures

A structure $\mathcal{U}$ is extension-terminal if:

$$\text{Ext}(\mathcal{U}) = \varnothing.$$

This means that no allowed extension preserves the required constraints under the chosen rules.

Terminality may arise for several reasons:

1. All possible extensions generate constraint violation.
2. Causal consistency would be broken by any new event.
3. The extension rules impose maximal closure.
4. The structure is complete relative to the chosen model.

However, terminality is model-relative. Changing the allowed extension rules or enlarging the class of admissible structures may change whether $\mathcal{U}$ is terminal.

> **Terminality is relative to the specified extension rules.**

1.5.21 Open Universe Condition

If $\mathcal{U}$ denotes the relational structure of the universe at a given extension stage, then the universe is open if:

$$\boxed{\mathrm{Ext}(\mathcal{U}) \neq \varnothing.}$$

This condition means that at least one admissible continuation exists.

It does not specify:

- which continuation occurs,
- a probability distribution over continuations,
- an external time parameter,
- that all possible continuations are realised.

It simply states non-terminality of the current admissible relational structure.

> **The open universe condition is: non-empty admissible extension set.**

1.5.22 Relation to Dynamics

The open extension principle is close to dynamics, but it is not yet dynamics.

A dynamics would require more than the existence of admissible extensions. It would require a rule selecting, weighting, or generating actual extensions.

For example, one might later introduce:

- a least-burden extension principle,
- a transition amplitude,
- a variational rule,
- a stochastic selection law,
- a causal growth law,
- or an extremal reconciliation principle.

None of these has been defined yet.

> **Open extension gives possibility of continuation, not a full dynamical law.**

This distinction protects the framework from claiming more than has been built.

1.5.23 Relation to Emergent Time

Although extension is not time, it prepares the ground for time.

An extension chain $\mathcal{U}_0 \rightarrow \mathcal{U}_1 \rightarrow \dots$ provides an ordering of admissible continuation. A causal chain $e_0 \prec e_1 \prec \dots$ provides an ordering of dependency.

Later, these orderings may be combined with burden or clock factors to define a proper-time-like quantity.

For example, one might later define a duration along a causal chain by:

$$\tau[\gamma] = \sum_{e \in \gamma} \frac{\epsilon}{1 + \lambda B(e)},$$
where $B(e)$ is a local burden factor.

But that belongs to a later section.

At this stage:

> **Extension order is a precursor to time, not time itself.**

1.5.24 Guardrails for This Subsection

1. **Openness is not proof of actual growth.** The condition $\mathrm{Ext}(\mathcal{U}) \neq \varnothing$ says that admissible continuations exist. It does not prove that any particular continuation is actualised.
2. **Extension is not background time.** The sequence $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2, \dots$ should not be interpreted as states at times $t_0, t_1, t_2, \dots$ unless a time map has been defined.
3. **Branching is not yet probability.** If $|\mathrm{Ext}(\mathcal{U})| > 1$, there are multiple admissible continuations. This does not automatically define probabilities.
4. **Extension rules must be specified.** The set $\mathrm{Ext}(\mathcal{U})$ depends on what counts as an allowed extension. If the extension rules are too permissive, inadmissible or ill-defined continuations may appear. If they are too restrictive, the theory may artificially become terminal.

1.5.25 Summary of Subsection 1.5

The Open Extension Principle introduces the possibility of admissible continuation without assuming external time.

An extension $\mathcal{U} \rightarrow \mathcal{U}'$ is admissible if the extended structure remains physical:

$$\boxed{\mathrm{mathfrak{M}}(\mathcal{U}') = 0}$$

in effective notation, or equivalently if the extension preserves the physical zero sector.

The set of admissible extensions is:

$$\boxed{\mathrm{Ext}(\mathcal{U}) = \{ \mathcal{U}' : \mathcal{U} \rightarrow \mathcal{U}' \text{ is admissible} \}}$$

A structure is open if:

$$\boxed{\mathrm{Ext}(\mathcal{U}) \neq \varnothing}$$

This principle supports extension chains, admissible histories, branching continuation, and non-terminal relational structure. But it does not yet provide a full dynamics, a probability rule, or physical time.

The central result of this subsection is:

$$\boxed{\text{Physical continuation means zero-preserving extension.}}$$

This completes the zero-preserving causal foundation at the level of:

- master-zero equivalence,
- observables,
- causality,
- two-link separation,
- and open extension.

The next subsection introduces the emergent causal propagation speed and Lorentz compatibility, which bridge the gap between the causal skeleton and relativistic structure.

1.6 Emergent Causal Propagation Speed

1.6.1 Purpose of This Subsection

The framework has established a causal dependency order $\prec$ and an open extension principle. However, a causal order by itself does not impose a finite speed limit on propagation. A partial order merely states dependencies; it does not specify how fast influence can propagate across the relational network.

Standard physics requires a finite invariant speed c . In the Relational Constraint Framework, this speed cannot be postulated as a property of a background spacetime metric. Instead, it must emerge from the interplay between causal depth and relational distance.

The purpose of this subsection is to derive an emergent causal speed ceiling $c_{\{\mathrm{RCF}\}}$ from the primitive structures of the theory: relational distance, causal depth, and an elementary reconciliation interval.

1.6.2 The Causal Reconciliation Principle

Before defining the speed, we must understand *why* there is a speed limit at all.

In this framework, physical reality is defined by constraint compatibility. When a relational structure extends (grows via the Open Extension Principle), it must do so while preserving the zero-constraint condition. However, constraint satisfaction cannot happen instantaneously across an infinite network. The relational structure must "reconcile" new extensions with the existing causal dependency structure.

> **The effective speed of light and the rate of time are bounded by the degree to which relational structures can reconcile with causal dependency.**

This is the **Causal Reconciliation Principle**. It is not a new primitive axiom, but a deeper grounding of the existing causal, relational, and burden-based structure. When reconciliation is efficient, causal propagation is fast and time advances normally. When burden or constraint loading increases, reconciliation slows, the causal ceiling tightens, and proper time is suppressed.

1.6.3 Relational Distance

To measure how far a causal influence has propagated, we need a notion of spatial separation. At this stage, the full correlation geometry has not been developed, but we can use the primitive correlation kernel K_ω .

Definition 1.20 (Relational Distance).

For two events $e_i, e_j \in \mathcal{E}$ with associated local observables O_i, O_j , the **relational distance** is:

$$d_\omega(e_i, e_j) := -\ell_0 \log K_\omega(O_i, O_j),$$
 where $\ell_0 > 0$ is a fundamental length scale.

Strong correlation ($K_\omega \approx 1$) means small relational distance. Weak correlation means large relational distance.

1.6.4 Causal Depth and Elementary Reconciliation Time

Definition 1.21 (Causal Depth).

For causally related events $e_i \prec e_j$, the causal depth $L_C(e_i, e_j)$ is the maximum number of intermediate dependency steps in any chain from e_i to e_j .

Definition 1.22 (Elementary Reconciliation Interval).

Let $\epsilon > 0$ be the **elementary reconciliation interval**: the minimal internal time cost required for a single primitive causal step to reconcile with the constraint structure.

Definition 1.23 (Causal-Depth Time).

The unburdened causal time between e_i and e_j is:

$$\tau_C(e_i, e_j) := \epsilon \cdot L_C(e_i, e_j).$$

This is not yet full physical proper time, which will later include burden-weighting. It is the minimal structural time cost of traversing the causal network.

1.6.5 Definition — Effective Causal Speed

Definition 1.24 (Effective Causal Speed).

For a causal propagation from e_i to e_j , the effective causal speed is:

$$v_C(e_i, e_j) := \frac{d_\omega(e_i, e_j)}{\tau_C(e_i, e_j)} = \frac{d_\omega(e_i, e_j)}{\epsilon \cdot L_C(e_i, e_j)}.$$

1.6.6 Bounded Relational Step Length

If the relational distance d_ω could grow arbitrarily large for a single causal step ($L_C = 1$), then the effective speed v_C would be unbounded. To obtain a finite speed limit, we must assume that a single primitive causal dependency step can only bridge a finite amount of relational distance.

Assumption 1.2 (Bounded Relational Step Length).

There exists a maximum relational distance $\ell_C > 0$ that a single primitive causal link can bridge. For any primitive causal link $e_a \prec e_b$:

$$d_\omega(e_a, e_b) \leq \ell_C.$$

Interpretation. This is the core locality assumption of the framework. It states that causal influence cannot jump arbitrarily far across the correlation geometry in a single reconciliation step. The constant ℓ_C is the maximum "range" of a single primitive zero-preserving update.

1.6.7 Theorem — Emergent Causal Speed Limit

Theorem 1.11 (Emergent Causal Speed Limit).

Let $(\mathcal{E}, \prec, \mathcal{R}_\omega)$ be an RCF causal-relational structure satisfying the bounded relational step assumption. Then any causal propagation from e_i to e_j satisfies:

$$v_C(e_i, e_j) \leq c_{\{\mathrm{RCF}\}},$$

where:

$$c_{\{\mathrm{RCF}\}} = \frac{\ell_C}{\epsilon}.$$

Proof.

Let $e_i = e_0 \prec e_1 \prec \dots \prec e_N = e_j$ be a causal chain connecting e_i to e_j . By the bounded relational step assumption (Assumption 1.2):

$$d_\omega(e_k, e_{k+1}) \leq \ell_C$$

for every elementary causal step k .

By the triangle inequality for the correlation distance (which holds under the metricity assumptions to be formalized in Section 2):

$$d_\omega(e_i, e_j) \leq \sum_{k=0}^{N-1} d_\omega(e_k, e_{k+1}) \leq \sum_{k=0}^{N-1} \ell_C = N \ell_C.$$

The causal depth of this chain is $L_C = N$. The minimal causal-time cost of N elementary steps is:

$$\tau_C = N \epsilon.$$

Therefore, the effective speed is bounded by:

$$v_C(e_i, e_j) = \frac{d_\omega(e_i, e_j)}{\tau_C} \leq \frac{N \ell_C}{N \epsilon} = \frac{\ell_C}{\epsilon}.$$

Defining $c_{\{\mathrm{RCF}\}} := \ell_C / \epsilon$, we obtain:

$$v_C(e_i, e_j) \leq c_{\{\mathrm{RCF}\}}. \quad \square$$

1.6.8 Interpretation of the Speed Limit

The theorem proves the key point:

> **Bounded causal link length plus a finite reconciliation tick implies an emergent causal speed limit.**

This is not yet a derivation of the physical speed of light c from first principles, as the constants ℓ_C and ϵ are model parameters. However, it provides the internal causal-speed mechanism.

In the continuum limit, if the framework recovers standard relativistic physics, one would identify:

$$c \rightarrow c_{\mathrm{RCF}} = \frac{\ell_C}{\epsilon}.$$

1.6.9 Burden-Corrected Causal Speed

The speed limit c_{RCF} is the structural ceiling for causal propagation. However, the framework also includes constraint burden, which slows down local reconciliation.

The local proper time contribution was defined as:

$$\Delta \tau(e_k) = \frac{\epsilon}{1 + \lambda B(e_k)}.$$

Thus, the burdened causal time along a chain is:

$$\tau[\gamma] = \sum_{k=0}^{N-1} \frac{\epsilon}{1 + \lambda B(e_k)}.$$

This means burden changes the clock experienced along the chain. We must distinguish two different speeds:

1. **Gauge Causal Speed:** Using the unburdened extension tick:

$$v_{\mathrm{gauge}} \leq \frac{\ell_C}{\epsilon} = c_{\mathrm{RCF}}.$$

This is the structural causal propagation bound.

2. **Local Proper Causal Speed:** Using burden-weighted proper time:

$$v_{\mathrm{proper}} = \frac{d\omega}{\tau[\gamma]}.$$

Since higher burden reduces $\Delta \tau$, proper-time speed can appear distorted relative to local clocks.

This distinction is crucial: the causal-speed limit belongs to the causal structure, while observed clock rates belong to the burden-weighted temporal structure. This fits the framework's separation between causal order and emergent local time.

In compact form, the Causal Reconciliation Principle reads:

$$c_{\mathrm{eff}} \sim \frac{\ell_0}{\tau_{\mathrm{rec}}}, \quad d\tau \sim \frac{d\sigma}{1 + \lambda B}, \quad \tau_{\mathrm{rec}} \uparrow \Rightarrow c_{\mathrm{eff}} \downarrow, \quad d\tau \downarrow.$$

Where reconciliation is efficient, causal propagation is fast and time advances normally.

When burden increases, reconciliation slows, the causal ceiling tightens, and proper time is suppressed.

1.7 Emergent Direction and Lorentz Compatibility

1.7.1 Purpose of This Subsection

The Emergent Causal Speed Limit theorem establishes a finite maximum propagation speed c_{RCF} . However, finite speed alone is not enough to recover special relativity. An anisotropic medium can have a finite maximum speed without possessing Lorentz symmetry.

To recover Lorentzian structure, the framework must reconstruct the notion of **direction** from the emergent correlation geometry, and show that Lorentz compatibility appears when the least-burden speed is direction-independent.

1.7.2 Distance Profiles and Emergent Direction

In standard geometry, direction is represented by tangent vectors. But in this framework, we do not begin with a manifold. We begin with localisable observables and their correlation distances.

Definition 1.25 (Distance Profile).

For each localisable observable $A \in \mathcal{A}_{\mathrm{loc}}$, define its **distance profile** as the function:

$$\Phi_A(X) := d_{\omega}(A, X),$$

where X ranges over all other localisable observables.

Definition 1.26 (Emergent Direction).

Given two distinguishable observables A and B , the **direction** from A towards B is represented by the profile difference:

$$\vec{D}_{A \rightarrow B} := \Phi_B - \Phi_A.$$

Explicitly, for any probe X :

$$\vec{D}_{A \rightarrow B}(X) = d_{\omega}(B, X) - d_{\omega}(A, X).$$

A direction is thus not an absolute vector. It is the observable-dependent change in the correlation-distance profile induced by shifting from one correlation class to another.

Definition 1.27 (Direction Equivalence).

Two moves $A \rightarrow B$ and $A \rightarrow C$ define the same emergent direction at A if their profile differences agree up to positive scaling over a sufficiently small neighbourhood:

$$\vec{D}_{A \rightarrow B} \approx \lambda \vec{D}_{A \rightarrow C}, \quad \lambda > 0.$$

The equivalence class $[A \rightarrow B]_{\mathrm{dir}}$ is an **emergent direction** at A . The collection of such classes defines an emergent tangent cone $T_A^{\mathrm{em}} \mathcal{X}_{\omega}$.

1.7.3 Directional Isotropy

The emergent propagation ceiling may depend on the direction of the relational displacement. Let $\hat{n}$ denote an emergent direction. The speed ceiling might be a function:

$$c_{\star}(\hat{n}).$$

****Definition 1.28 (Directional Isotropy).****

The emergent propagation ceiling is ****direction-independent**** if:

$$c_{\star}(\hat{n}) = c_{\star} \quad \forall \hat{n}.$$

Directional isotropy means that the least-burden reconciliation speed is the same in all emergent relational directions. This removes preferred spatial axes from the leading-order propagation law.

1.7.4 Theorem — Directional Isotropy Implies Lorentz Form

****Theorem 1.12 (Directional Isotropy Implies Lorentz Form).****

If the emergent propagation ceiling is independent of direction and the continuum limit is regular, then the effective interval can be written in Lorentz form:

$$\boxed{ds^2 = -c_{\star}^2 d\tau^2 + d\ell_{\omega}^2.}$$

Proof sketch.

Once the speed bound $c_{\star}$ is direction-independent (isotropic), the only remaining first-order invariant compatible with the causal structure is a quadratic interval with one time-like and spatially isotropic part.

The sign choice is fixed by the causal ordering already built into the relational time rule: time-like separation must correspond to burden-weighted causal evolution ($d\tau$), while space-like separation corresponds to relational displacement ($d\ell_{\omega}$).

Hence, the effective geometry takes the Lorentzian form $ds^2 = -c_{\star}^2 d\tau^2 + d\ell_{\omega}^2$. $\blacksquare$

1.7.5 Emergent Mass-Shell Relation

Once the effective geometry is Lorentz-compatible, the dynamics of excitations must align with relativistic dispersion relations.

****Theorem 1.13 (Emergent Mass Shell, Conditional Form).****

Assume the effective geometry is Lorentz-compatible with invariant interval $ds^2 = -c_{\star}^2 d\tau^2 + d\ell_{\omega}^2$, and assume the excitation carries a conserved burden-weighted four-momentum p_{μ} . Then the leading-order dispersion relation is:

$$\boxed{p_{\mu} p^{\mu} = -m_{\mathrm{eff}}^2 c_{\star}^2.}$$

Proof sketch.

In a Lorentzian effective geometry, the only scalar quadratic invariant built from four-momentum is $p_{\mu} p^{\mu}$. Conservation of the effective propagation law forces this invariant to remain fixed along free evolution. The fixed value is naturally identified with the squared effective rest mass.

The burden sector shifts the effective inertial response, so the mass parameter m_{eff} is not primitive, but emergent from the burdened relational excitation. Therefore, the free effective dynamics lies on a mass shell. $\blacksquare$

1.7.6 Status of Lorentz Symmetry

It is crucial to maintain theorem-safe status language regarding Lorentz symmetry:

1. **The speed bound** ($v_C \leq c_{\mathrm{RCF}}$) is a **conditional theorem** (proven from bounded relational step length and finite reconciliation tick).
2. **Directional isotropy** ($c_{\star}(\hat{n}) = c_{\star}$) is a **theorem target or regime assumption** (it emerges if the coarse-grained correlation geometry is sufficiently homogeneous).
3. **Lorentz symmetry** is an **effective continuum theorem target** (it follows if isotropy and regularity hold).

The framework does not assume Lorentz symmetry at the foundational algebraic level. It derives it as an effective continuum limit of the relational causal structure.

1.8 Guardrails and Summary of Section 1

1.8.1 Guardrails for Section 1

1. **Zero preservation is not dynamics.** It is admissibility preservation.
2. **Observables are not merely arbitrary algebra elements.** They must be compatible with the physical sector.
3. **Causal dependency is not background spacetime causality.** It is an internal dependency relation among admissible structures.
4. **Events are not initially spacetime points.**
5. **Extension order is not external time.**
6. **Relational correlation is not the same as causal dependence.**
7. **This section does not yet define metric distance, duration, mass, energy, fields, or gravity.**
8. **Causal speed is a conditional theorem.** Directional isotropy is a regime assumption. Lorentz symmetry is a theorem target.

1.8.2 Summary of Section 1

Section 1 established the zero-preserving causal foundation:

1. **Master-zero equivalence** was restated: $\ker \hat{\mathfrak{M}}_{\omega} = \bigcap_{\alpha} \ker \pi_{\omega}(\hat{C}_{\alpha})$.
2. **Zero-preserving observables** were defined as operations preserving the physical kernel. Strong observables commute with $\hat{\mathfrak{M}}_{\omega}$; weak observables need only preserve the kernel. The algebraic observable sector $\widetilde{\mathcal{A}}_{\mathrm{obs}}$ was defined and shown to be a unital (*)-subalgebra.
3. **Causal dependency** was introduced as a strict partial order via constraint closure inclusion. Irreflexivity, transitivity, and asymmetry were proven.

4. **The Two-Link Principle** was stated: causal links ($\prec$) and relational-correlation links ($\mathcal{R}_\omega$) are structurally distinct.
5. **The Open Extension Principle** was formalised: a universe is an extendible zero-preserving partial order, not a closed static block. Extension-history equivalence was defined to prevent the extension label from becoming a preferred time frame.
6. **Emergent causal propagation speed** was derived: $v_C \leq c_{\mathrm{RCF}} = \ell_C / \epsilon$, grounded in the Causal Reconciliation Principle.
7. **Directional isotropy** was shown to imply an effective Lorentzian interval $ds^2 = -c_{\star}^2 d\tau^2 + d\ell_\omega^2$, leading to an emergent mass-shell relation.

The conceptual sequence is:

$\boxed{\text{zero sector}} \rightarrow \text{zero-preserving operations} \rightarrow \text{causal dependency} \rightarrow \text{open extension} \rightarrow \text{causal speed} \rightarrow \text{Lorentz compatibility}.$

This completes the first bridge from pure admissibility toward emergent temporal and causal structure.

Section 2 — Correlation Geometry and Emergent Space

2.0 Purpose of Correlation Geometry and Emergent Space

2.0.1 Transition from Causal Structure to Spatial Structure

Section 1 developed the first post-foundational structures: the physical zero sector, zero-preserving observables, causal dependency, the Two-Link Principle, and open extension. These gave the framework a pre-geometric causal skeleton and an emergent speed limit.

However, causal order alone is not space. A relation such as $e_i \prec e_j$ tells us that e_j depends on e_i . It does not tell us how far apart two events are. It does not define a metric, topology, dimension, locality, or spatial region. It also does not define distance between events that are not causally comparable.

Thus, after causal dependency has been separated from relational correlation by the Two-Link Principle, the next task is to develop the correlation side.

The purpose of this section is to show how space-like structure can be reconstructed from state-dependent relational correlation.

> **Space is not assumed as a background arena; it is reconstructed from correlation structure.**

The framework does not begin with points in a manifold. It begins with admissible relational structure. What later appears as spatial distance is to be derived from how strongly physical observables, events, or localisable relational contexts are correlated in a physical state.

2.0.2 Why Correlation Is the Correct Starting Point for Emergent Space

If space is not primitive, then spatial nearness cannot be defined by coordinate distance. There is no initial metric $d(p,q)$, no background manifold M , and no prior statement that two things are close because their coordinate labels differ only slightly.

Instead, nearness must be reconstructed from internal relational data.

Correlation provides the natural candidate. If two admissible relational contexts are strongly correlated in a physical state, then they behave as though they are close in the emergent structure. If they are weakly correlated, they behave as though they are far apart. This suggests that spatial distance should be a derived measure of correlation decay.

The basic intuition is:

- > **Strong correlation corresponds to small emergent separation.**
- > **Weak correlation corresponds to large emergent separation.**

This does not yet prove that correlation distance is a metric. It only identifies the structural bridge from relational information to space-like geometry. Metricity requires additional assumptions, which will be stated explicitly in later subsections.

2.0.3 Relation to the Two-Link Principle

This section develops the second link introduced by the Two-Link Principle.

The causal link was:

$$e_i \prec e_j.$$

It meant that e_j depends on e_i for zero-closure. This belongs to the causal/dependency layer.

The relational link is:

$$K_\omega(e_i, e_j) \equiv \mathrm{Corr}_\omega(O_{e_i}, O_{e_j}).$$

It measures the strength of relational association between two event contexts in a physical state ω .

The present section develops this relational layer into a geometric structure. It defines a correlation distance from K_ω , establishes conditions under which this distance becomes a pseudometric, constructs emergent points as equivalence classes of perfectly correlated observables, and introduces approximate metricity under coarse-graining.

2.0.4 What This Section Establishes

This section establishes the following:

1. **The correlation kernel** K_ω is defined on localisable zero-preserving observables.

2. **Correlation distance** $d_\omega = -\ell_0 \log K_\omega$ is defined.
3. Under **multiplicative correlation consistency**, d_ω is shown to be a pseudometric.
4. **Emergent points** are defined as equivalence classes of perfectly correlated observables.
5. The quotient of the observable sector by this equivalence relation yields a **metric space** of emergent points.
6. **Approximate metricity** is established under coarse-graining, providing the bridge from microscopic relational structure to macroscopic smooth geometry.
7. **Observable-dependent direction** is reconstructed from distance profiles.
8. **Dimensional closure** is proven: $D=3$ is the unique spatial dimension where relational inference closes without degeneracy or redundancy.

The theorem-safe structure is:

$$\boxed{\text{kernel assumptions} \quad \longrightarrow \quad \text{correlation distance} \quad \longrightarrow \quad \text{pseudometric or approximate metric} \quad \longrightarrow \quad \text{emergent point set}.}$$

2.1 The Correlation Kernel

2.1.1 Localisable Observables

The construction begins with a class of observables that can play the role of local probes.

Definition 2.1 (Localisable Observable Sector).

Let $\mathcal{A}_{\mathrm{loc}} \subseteq \mathcal{A}_{\mathrm{obs}}$ denote a chosen class of **localisable zero-preserving observables**. These are observables whose mutual correlation structure will be used to define emergent geometry.

Interpretation. At this stage, "localisable" should be understood operationally: not as already embedded in a pre-existing space, but as belonging to the class of observables suitable for later spatial interpretation. An element $A \in \mathcal{A}_{\mathrm{loc}}$ is not an operator at a spacetime point. It is a relational probe whose pattern of correlation with other probes can be used to reconstruct a location-like equivalence class.

> **Localisable** does not mean already located in background space. It means capable of supporting emergent localisation through correlation structure.

2.1.2 Definition — Correlation Kernel

Definition 2.2 (Correlation Kernel).

Let ω be a physical state. A **correlation kernel** is a map:

$K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \rightarrow (0, 1]$ assigning to each pair $A, B \in \mathcal{A}_{\mathrm{loc}}$ a number $K_\omega(A, B)$ that measures the strength of relational association between A and B in the state ω .

The range condition $0 < K_\omega(A, B) \leq 1$ means that the kernel is strictly positive and normalised above by one.

****Interpretation.****

- $K_\omega(A, B) = 1$ represents maximal correlation or indistinguishability for the purposes of the chosen kernel.
- Values closer to zero represent weaker correlation.

This range condition is chosen so that the logarithmic distance $d_\omega(A, B) = -\elll_0 \log K_\omega(A, B)$ will be non-negative and finite.

2.1.3 Why the Kernel Is Normalised

The upper bound $K_\omega(A, B) \leq 1$ is required because the later distance formula uses $-\log K_\omega(A, B)$.

If $0 < K_\omega(A, B) \leq 1$, then:

$$-\log K_\omega(A, B) \geq 0,$$

and therefore:

$$-\log K_\omega(A, B) \geq 0.$$

Thus, normalisation guarantees non-negative correlation distance.

The strict lower bound $K_\omega(A, B) > 0$ avoids infinite distance at this stage. If one wishes to allow exactly vanishing correlations, then the extended value $d_\omega(A, B) = +\infty$ may be introduced later. For the basic metricity results, it is cleaner to begin with strictly positive kernels.

2.1.4 Possible Concrete Forms of the Kernel

The abstract theory does not require a single universal formula for $K_\omega(A, B)$. Different models may define the kernel differently, provided the required structural properties are satisfied.

Possible forms include normalised expectation overlaps, covariance-based correlations, state-overlap measures, mutual constraint-response functions, or other relational association measures.

For example, in a represented setting one might use a normalised connected correlation of the schematic form:

$$K_\omega(A, B) \sim \frac{\omega(A^\dagger B)}{\sqrt{\omega(A^\dagger A) \omega(B^\dagger B)}},$$

when the denominator is non-zero and the expression is appropriately bounded.

The theorem-safe point is not that one of these formulas is uniquely fundamental. The point is:

> **Any kernel used for geometry must satisfy the assumptions required by the later distance theorems.**

Thus, the correlation kernel is an input structure at this stage. Its admissibility is judged by its properties.

2.1.5 Minimal Kernel Assumptions

For the first layer of correlation geometry, the following assumptions are natural.

Assumption 2.1 (Positivity).

$$K_{\omega}(A, B) > 0 \quad \text{for all } A, B \in \mathcal{A}_{\mathrm{loc}}.$$

This ensures that $\log K_{\omega}(A, B)$ is defined and finite.

Assumption 2.2 (Upper Normalisation).

$$K_{\omega}(A, B) \leq 1 \quad \text{for all } A, B \in \mathcal{A}_{\mathrm{loc}}.$$

This ensures non-negative logarithmic distance.

Assumption 2.3 (Diagonal Normalisation).

$$K_{\omega}(A, A) = 1 \quad \text{for all } A \in \mathcal{A}_{\mathrm{loc}}.$$

This ensures that every observable has zero distance from itself.

Assumption 2.4 (Symmetry).

$$K_{\omega}(A, B) = K_{\omega}(B, A) \quad \text{for all } A, B \in \mathcal{A}_{\mathrm{loc}}.$$

This ensures that the induced distance is symmetric.

Not all of these assumptions are needed immediately. Positivity and upper normalisation are enough to define a non-negative distance. Diagonal normalisation, symmetry, and multiplicative consistency are needed for pseudometric structure.

2.1.6 Correlation Profiles

The kernel assigns to each observable $A \in \mathcal{A}_{\mathrm{loc}}$ a **correlation profile**:

Definition 2.3 (Correlation Profile).

The correlation profile of A is the function:

$$\Pi_{\omega}(A): \mathcal{A}_{\mathrm{loc}} \rightarrow (0, 1], \quad \Pi_{\omega}(A)(X) = K_{\omega}(A, X).$$

The profile $\Pi_{\omega}(A)$ records how A correlates with every other localisable observable. This profile is more informative than any single pairwise correlation. It gives a relational signature for the emergent location of A .

Two observables may be considered to represent the same emergent location if their correlation profiles are indistinguishable. This motivates the later equivalence relation:

$$A \equiv_{\omega} B \quad \Longleftrightarrow \quad \Pi_{\omega}(A) = \Pi_{\omega}(B),$$

or explicitly:

$$A \equiv_{\omega} B \quad \Longleftrightarrow \quad K_{\omega}(A, X) = K_{\omega}(B, X) \\ \quad \text{for all } X \in \mathcal{A}_{\mathrm{loc}}.$$

> **Emergent location is encoded by correlation profile.**

2.1.7 State Dependence and Changing Geometry

Because the kernel depends on ω , the emergent geometry may change when the physical state changes.

If $K_{\omega}(A, B) \neq K_{\omega'}(A, B)$, then the correlation distance induced by ω may differ from that induced by ω' .

Thus, geometry is not a fixed background. It is tied to the relational state.

> **The same algebra may support different emergent geometries in different physical states.**

This does not mean geometry is arbitrary. The state must be physical, and the kernel must satisfy the required structural assumptions. But it does mean geometry is derived from relational-state structure rather than imposed independently.

2.1.8 Kernel Compatibility with Zero Preservation

The kernel should be built from physical or zero-preserving data.

If A and B are not compatible with the physical sector, then their correlation may not correspond to a physical relation. Therefore, the domain of the kernel is restricted to $\mathcal{A}_{\mathrm{loc}}$, the class of localisable zero-preserving observables.

This ensures that correlation geometry is constructed from admissible relational probes, not arbitrary kinematic expressions.

> **$K_{\omega}(A, B)$ is physically meaningful only for admissible localisable observables A, B .**

2.1.9 Guardrails for This Subsection

1. **Correlation is pre-geometric.** The kernel K_{ω} is not yet a distance, metric, or coordinate chart.
2. **Localisable does not mean spatially local in advance.** The term "localisable" is operational and framework-internal, not assumed from prior geometry.
3. **The kernel is not a background metric.** It is evaluated in a physical state and defined on relational observables, not on pre-existing spatial points.

2.2 Correlation Distance

2.2.1 Purpose of This Subsection

The previous subsection introduced the correlation kernel $K_\omega(A, B)$, defined on localisable observables $A, B \in \mathcal{A}_{\mathrm{loc}}$. The kernel measures the strength of relational association between two admissible relational probes in a physical state ω . It is not itself a distance. It behaves in the opposite direction from distance: strong correlation should correspond to nearness, while weak correlation should correspond to separation.

The purpose of this subsection is to convert correlation strength into an emergent distance-like quantity.

> **Emergent distance measures the cost of relational decorrelation.**

2.2.2 From Correlation Strength to Separation

Let $K_\omega(A, B)$ satisfy $0 < K_\omega(A, B) \leq 1$.

The value $K_\omega(A, B) = 1$ represents maximal correlation between A and B . Values closer to zero represent weaker correlation.

A distance-like quantity should satisfy:

- $K_\omega(A, B) = 1 \Rightarrow d_\omega(A, B) = 0$,
- $K_\omega(A, B) \searrow 0 \Rightarrow d_\omega(A, B) \nearrow \infty$.

A natural way to achieve this is to define distance by the negative logarithm of the correlation:

$d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$,
where $\ell_0 > 0$ is a fixed length scale.

The negative logarithm converts high correlation into small distance and low correlation into large distance.

2.2.3 Definition — Correlation Distance

Definition 2.4 (Correlation Distance).

Let $K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \rightarrow (0, 1]$ be a normalised positive correlation kernel. The **correlation distance** associated with K_ω is the function:

$\boxed{d_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \rightarrow [0, \infty)}$

defined by:

$\boxed{d_\omega(A, B) = -\ell_0 \log K_\omega(A, B), \quad \ell_0 > 0.}$

Here ℓ_0 sets the emergent distance scale.

The quantity $d_\omega(A, B)$ is interpreted as the emergent separation between the localisable relational probes A and B as induced by the physical state ω .

This definition should be read carefully. The function d_ω is not assumed to be a metric at this stage. It is a **candidate distance**. Its metric properties depend on additional assumptions about K_ω .

2.2.4 Role of the Scale ℓ_0

The constant $\ell_0 > 0$ converts the dimensionless logarithmic correlation cost into a quantity with units of length, or more generally with units of emergent separation.

If the kernel K_ω is dimensionless, then $-\log K_\omega(A, B)$ is also dimensionless. Multiplying by ℓ_0 assigns the resulting distance a scale.

The value of ℓ_0 may be interpreted differently depending on the model. It may represent:

- a fundamental relational length scale,
- a coarse-graining scale,
- a correlation-decay scale,
- or a conversion factor between logarithmic correlation cost and emergent distance.

At this stage, no particular physical value is assigned to ℓ_0 . It is simply required to be positive.

2.2.5 Non-Negativity of Correlation Distance

Theorem 2.1 (Non-Negativity).

If $0 < K_\omega(A, B) \leq 1$ and $\ell_0 > 0$, then:

$$\boxed{d_\omega(A, B) \geq 0.}$$

Proof.

Since $0 < K_\omega(A, B) \leq 1$, we have $\log K_\omega(A, B) \leq 0$. Therefore $-\log K_\omega(A, B) \geq 0$. Since $\ell_0 > 0$, it follows that $-\ell_0 \log K_\omega(A, B) \geq 0$. Thus $d_\omega(A, B) \geq 0$. $\square$

2.2.6 Zero Distance and Perfect Correlation

Theorem 2.2 (Zero Distance and Perfect Correlation).

$$d_\omega(A, B) = 0 \iff K_\omega(A, B) = 1.$$

Proof.

If $K_\omega(A, B) = 1$, then $d_\omega(A, B) = -\ell_0 \log 1 = 0$.

Conversely, if $d_\omega(A, B) = 0$, then $-\ell_0 \log K_\omega(A, B) = 0$. Since $\ell_0 > 0$, this implies $\log K_\omega(A, B) = 0$, so $K_\omega(A, B) = 1$. $\square$

Interpretation.

Perfect correlation implies zero emergent separation. However, zero emergent separation does not necessarily imply algebraic identity $A = B$. Two distinct observables may be perfectly correlated or indistinguishable by the chosen kernel, even if they are algebraically distinct.

Thus, at this level, d_ω is expected at best to be a **pseudometric** until one quotients by zero-distance equivalence classes.

> **Zero correlation distance means emergent indistinguishability, not necessarily algebraic equality.**

2.2.7 Weak Correlation and Large Separation

If the correlation kernel becomes small, then the distance becomes large.

For example, if $K_\omega(A, B) = e^{-L/\ell_0}$, then:
 $d_\omega(A, B) = -\ell_0 \log(e^{-L/\ell_0}) = L$.

This shows that exponential decay of correlation corresponds to linear growth of distance.

More generally, as $K_\omega(A, B) \rightarrow 0^+$, we have $\log K_\omega(A, B) \rightarrow -\infty$, and therefore $d_\omega(A, B) \rightarrow +\infty$.

> **Vanishing correlation corresponds to infinite emergent separation.**

In the present definition, the kernel is restricted to strictly positive values, so the distance remains finite. If exact zero correlations are allowed later, then d_ω should be treated as an extended distance that may take the value $+\infty$.

2.2.8 Why Logarithmic Distance Is Natural

The logarithm is used because correlations often combine multiplicatively, whereas distances combine additively.

Suppose that correlations satisfy an approximate factorisation relation:
 $K_\omega(A, C) \approx K_\omega(A, B) \cdot K_\omega(B, C)$.

Taking negative logarithms gives:
 $-\log K_\omega(A, C) \approx -\log K_\omega(A, B) - \log K_\omega(B, C)$.

Multiplying by ℓ_0 :
 $d_\omega(A, C) \approx d_\omega(A, B) + d_\omega(B, C)$.

Thus, multiplicative composition of correlations becomes additive composition of distances.

> **Logarithmic distance turns multiplicative correlation decay into additive geometry.**

The logarithm is the unique (up to scale) mapping that converts multiplicative consistency into additive triangle inequality.

2.2.9 Symmetry of Correlation Distance

Theorem 2.3 (Symmetry).

If $K_\omega(A, B) = K_\omega(B, A)$, then:

$$\boxed{d_\omega(A, B) = d_\omega(B, A).}$$

Proof.

Since $K_\omega(A, B) = K_\omega(B, A)$, we have:

$$d_\omega(A, B) = -\ell_0 \log K_\omega(A, B) = -\ell_0 \log K_\omega(B, A) = d_\omega(B, A). \quad \blacksquare$$

2.2.10 Diagonal Vanishing

Theorem 2.4 (Diagonal Vanishing).

If $K_\omega(A, A) = 1$, then:

$$\boxed{d_\omega(A, A) = 0.}$$

Proof.

$$d_\omega(A, A) = -\ell_0 \log K_\omega(A, A) = -\ell_0 \log 1 = 0. \quad \blacksquare$$

2.2.11 Triangle Inequality Is Not Automatic

The correlation distance does not automatically satisfy the triangle inequality. The triangle inequality would require:

$$d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C).$$

Substituting the logarithmic definition gives:

$$-\ell_0 \log K_\omega(A, C) \leq -\ell_0 \log K_\omega(A, B) - \ell_0 \log K_\omega(B, C).$$

Dividing by $-\ell_0$ (which reverses the inequality):

$$\log K_\omega(A, C) \geq \log K_\omega(A, B) + \log K_\omega(B, C).$$

Exponentiating:

$$K_\omega(A, C) \geq K_\omega(A, B) \cdot K_\omega(B, C).$$

Thus, the triangle inequality for d_ω is equivalent to a **multiplicative consistency condition** for K_ω .

> **The triangle inequality requires multiplicative correlation consistency.**

This condition will be treated in the next subsection.

2.2.12 Correlation Distance as a Premetric

Before imposing all metricity assumptions, it is safest to call d_ω a **correlation premetric** or **candidate distance**.

It already has a clear interpretation:

- Large correlation $\rightarrow$ small separation.
- Small correlation $\rightarrow$ large separation.

But without symmetry, diagonal normalisation, and multiplicative consistency, it may not satisfy all metric axioms.

> **$d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ is a non-negative correlation-derived separation under $0 < K_\omega \leq 1$.**

Additional assumptions are required before calling it a pseudometric or metric.

2.2.13 Guardrails for This Subsection

1. **Distance is derived, not primitive.** The function $d_\omega(A, B)$ is not assumed at the beginning of the theory. It is derived from $K_\omega(A, B)$, which is itself evaluated in a physical state ω .
2. **Not yet a true metric.** Separation fails until quotienting.
3. **No coordinates.** We still have no points, no manifold, no dimension.
4. **State dependence.** Distance depends on ω .

2.3 Metricity Under Multiplicative Correlation Consistency

2.3.1 Purpose of This Subsection

The previous subsection defined the correlation distance $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ and showed that it is non-negative when $0 < K_\omega \leq 1$. It also showed that zero distance corresponds to perfect correlation, not necessarily literal equality of observables.

However, non-negativity alone is not enough to make d_ω a metric or even a pseudometric. To obtain metric-like structure, additional properties must be imposed on the kernel. In particular, one needs diagonal normalisation, symmetry, and a condition ensuring the triangle inequality.

The purpose of this subsection is to state and prove the main conditional theorem:

> **Multiplicative consistency of correlation implies additive consistency of distance.**

This is the central mathematical bridge by which a correlation structure becomes a geometry-like structure.

2.3.2 From Correlation to Metric Structure

A distance function d on a set X is a **metric** if it satisfies:

1. $d(x, y) \geq 0$
2. $d(x, y) = 0 \iff x = y$
3. $d(x, y) = d(y, x)$
4. $d(x, z) \leq d(x, y) + d(y, z)$

A **pseudometric** satisfies the same conditions except that distinct elements may have zero distance: $d(x, y) = 0$ does not necessarily imply $x = y$.

In correlation geometry, pseudometricity is the natural first target. This is because two algebraically distinct observables may be perfectly correlated, or indistinguishable by the correlation kernel, and therefore have zero correlation distance.

Thus, the theorem-safe claim should not be that d_ω is automatically a metric on $\mathcal{A}_{\mathrm{loc}}$. The correct claim is:

> **d_ω is a pseudometric under suitable kernel assumptions.**

A true metric may then be obtained later by quotienting by zero-distance equivalence classes.

2.3.3 Kernel Assumptions for Pseudometricity

Let $K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \rightarrow (0, 1]$ be a correlation kernel.

For $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ to be a pseudometric, we assume the following:

Assumption 2.5 (Multiplicative Correlation Consistency).

For all $A, B, C \in \mathcal{A}_{\mathrm{loc}}$:

$$\boxed{K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C).}$$

This is the condition that ensures the triangle inequality for d_ω .

Interpretation.

The condition $K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C)$ says that the direct correlation between A and C is at least as large as the correlation obtained by passing through an intermediate probe B .

Equivalently, correlation should not decay more severely along the direct relation than along a two-step relational path. If correlation from A to B and from B to C is strong, then A and C cannot be arbitrarily decorrelated.

This is precisely the multiplicative condition needed for additive distances.

2.3.4 Theorem — Correlation Distance Is a Pseudometric

Theorem 2.5 (Correlation Distance Is a Pseudometric).

Let $K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \rightarrow (0, 1]$ be a correlation kernel satisfying:

1. $K_\omega(A, A) = 1$ (diagonal normalisation),
2. $K_\omega(A, B) = K_\omega(B, A)$ (symmetry),
3. $K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C)$ (multiplicative consistency).

Let $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ with $\ell_0 > 0$.

Then d_ω is a **pseudometric** on $\mathcal{A}_{\mathrm{loc}}$.

That is, for all $A, B, C \in \mathcal{A}_{\mathrm{loc}}$:

1. $d_\omega(A, B) \geq 0$,
2. $d_\omega(A, A) = 0$,
3. $d_\omega(A, B) = d_\omega(B, A)$,
4. $d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C)$.

Proof.

Non-negativity.

Since $0 < K_\omega(A, B) \leq 1$, we have $\log K_\omega(A, B) \leq 0$. Therefore $-\log K_\omega(A, B) \geq 0$. Since $\ell_0 > 0$, we obtain $d_\omega(A, B) \geq 0$.

Diagonal vanishing.

By diagonal normalisation, $K_\omega(A, A) = 1$. Therefore $d_\omega(A, A) = -\ell_0 \log 1 = 0$.

Symmetry.

By kernel symmetry, $K_\omega(A, B) = K_\omega(B, A)$. Therefore $d_\omega(A, B) = d_\omega(B, A)$.

Triangle inequality.

By multiplicative correlation consistency:

$$K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C).$$

Taking logarithms (which preserves the inequality since $\log$ is monotonically increasing):

$$\log K_\omega(A, C) \geq \log K_\omega(A, B) + \log K_\omega(B, C).$$

Multiplying by $-\ell_0$ (which is negative, so it reverses the inequality):

$$-\ell_0 \log K_\omega(A, C) \leq -\ell_0 \log K_\omega(A, B) - \ell_0 \log K_\omega(B, C).$$

Using the definition of d_ω :

$$d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C).$$

Since non-negativity, diagonal vanishing, symmetry, and the triangle inequality all hold, d_ω is a pseudometric. $\square$

2.3.5 Why the Result Is a Pseudometric Rather Than a Metric

The theorem does not prove that d_ω is a metric on $\mathcal{A}_{\mathrm{loc}}$.

The missing condition is **identity of indiscernibles**:

$$d_\omega(A, B) = 0 \quad \Longleftrightarrow \quad A = B.$$

Correlation distance gives instead:

$$d_\omega(A, B) = 0 \quad \Longleftarrow \quad K_\omega(A, B) = 1.$$

Perfect correlation does not necessarily imply algebraic equality. Two distinct observables may be perfectly correlated or indistinguishable by the kernel.

Therefore, d_ω may assign zero distance to distinct elements. This is exactly why the correct theorem is pseudometricity.

The stronger statement $d_\omega(A, B) = 0 \rightarrow A = B$ requires an additional separation assumption: $K_\omega(A, B) = 1 \rightarrow A = B$. If such an assumption is imposed, then d_ω becomes a metric. But without it, the theorem-safe result is only pseudometric.

> **Correlation distance is naturally pseudometric before quotienting.**

2.3.6 Metric Quotient

Given a pseudometric d_ω , define an equivalence relation:

$$A \sim_\omega B \quad \Longleftrightarrow \quad d_\omega(A, B) = 0.$$

Equivalently:

$$A \sim_\omega B \quad \Longleftrightarrow \quad K_\omega(A, B) = 1.$$

The quotient set:

$$\mathcal{X}_\omega := \mathcal{A}_{\mathrm{loc}} / \sim_\omega$$

then carries an induced metric, provided the distance between equivalence classes is well-defined.

Definition 2.5 (Quotient Distance).

Define:

$$\widetilde{d}_\omega([A]_\omega, [B]_\omega) := d_\omega(A, B).$$

For this to be well-defined, one needs the usual compatibility condition: if $A \sim_\omega A'$ and $B \sim_\omega B'$, then $d_\omega(A, B) = d_\omega(A', B')$.

For pseudometric spaces, this compatibility holds. Therefore, the quotient by zero-distance equivalence produces a genuine metric space.

2.3.7 Theorem — Metric on Correlation Equivalence Classes

****Theorem 2.6 (Metric on Correlation Equivalence Classes).****

Let d_ω be the pseudometric from Theorem 2.5. Define $A \sim_\omega B$ iff $d_\omega(A, B) = 0$. Let $\mathcal{X}_\omega = \mathcal{A} / \sim_\omega$. Define $\tilde{d}_\omega([A]_\omega, [B]_\omega) = d_\omega(A, B)$. Then $\tilde{d}_\omega$ is a well-defined metric on $\mathcal{X}_\omega$.

Proof.

****Well-definedness.****

Suppose $A \sim_\omega A'$ and $B \sim_\omega B'$. Then $d_\omega(A, A') = 0$ and $d_\omega(B, B') = 0$. By the triangle inequality:

$$d_\omega(A, B) \leq d_\omega(A, A') + d_\omega(A', B') + d_\omega(B', B) = d_\omega(A', B').$$

Similarly:

$$d_\omega(A', B') \leq d_\omega(A', A) + d_\omega(A, B) + d_\omega(B, B') = d_\omega(A, B).$$

Therefore $d_\omega(A, B) = d_\omega(A', B')$. Hence $\tilde{d}_\omega$ is well-defined.

****Identity of indiscernibles.****

Suppose $\tilde{d}_\omega([A]_\omega, [B]_\omega) = 0$. Then $d_\omega(A, B) = 0$, so $A \sim_\omega B$. Hence $[A]_\omega = [B]_\omega$.

Non-negativity, symmetry, and the triangle inequality descend directly from the pseudometric properties of d_ω . Thus $\tilde{d}_\omega$ is a metric. $\blacksquare$

2.3.8 Interpretation of Emergent Points

An emergent point is not a basic atom of space. It is an equivalence class of relational probes whose mutual separation vanishes.

If $[A]_\omega \in \mathcal{X}_\omega$, then $[A]_\omega$ contains all localisable observables that represent the same correlation location in the state ω .

Thus, a point is defined by indistinguishability within the correlation geometry. This differs from ordinary manifold-based thinking. In a manifold model, one starts with points and then defines fields or observables at those points. Here, one starts with observables and defines points by their relational indistinguishability.

> ****Points are not primitive; points are correlation-equivalence classes.****

2.3.9 Guardrails for This Subsection

- **The quotient is not optional.**** Without quotienting, d_ω remains only a pseudometric and does not define true points.
- **Perfect correlation is not algebraic equality.**** The relation $A \sim_\omega B$ means zero emergent separation, not literal equality in $\mathcal{A}$.

3. **Metric space does not yet imply manifold.** At this stage $\mathcal{X}_\omega$ is only a metric space, not yet a smooth manifold or continuum spacetime.
4. **State dependence remains.** Both the quotient relation and the emergent metric depend on the physical state ω .

2.4 Coarse-Graining and Approximate Metricity

2.4.1 Purpose of This Subsection

The previous subsections established that, under exact multiplicative correlation consistency, the correlation distance d_ω is a pseudometric on the localisable observable sector $\mathcal{A}_{\mathrm{loc}}$, and that quotienting by perfect correlation yields a genuine metric space $(\mathcal{X}_\omega, \widetilde{d}_\omega)$.

However, exact metricity may be too strong at the microscopic relational level. The primitive correlation structure may be noisy, irregular, fluctuating, scale-dependent, or only approximately multiplicative. The physically relevant claim is often not that the microscopic structure is exactly metric, but rather that the coarse-grained structure is approximately metric.

The purpose of this subsection is to establish the theorem-safe bridge from exact microscopic metricity to approximate macroscopic metricity.

> **Smooth metric geometry need not be exact microscopically; it may emerge approximately after coarse-graining.**

This allows the framework to avoid overclaiming. It does not assert that every microscopic correlation kernel is exactly metric. Instead, it shows that if correlations satisfy an approximate multiplicative consistency condition, then the logarithmic distance satisfies an approximate triangle inequality.

2.4.2 Why Approximate Metricity Is Needed

Exact metric spaces are highly structured. They require strict identities and inequalities to hold for every triple of points or observables. But a pre-geometric relational substrate may not satisfy these conditions exactly.

There are several reasons exact metricity may fail.

First, the microscopic correlation kernel may fluctuate. The value $K_\omega(A, B)$ may depend sensitively on fine relational details that do not survive coarse-graining.

Second, localisable observables may not be perfectly sharp. The distinction between two neighbouring relational probes may be blurred.

Third, correlations may compose only approximately. Instead of exact multiplicative consistency:

$K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C)$,
one may have a controlled error.

Fourth, the emergent point set itself may only become stable after grouping many microscopic relational probes into effective classes.

For these reasons, the appropriate physical statement is often not "the microscopic structure is exactly metric," but rather "the coarse-grained structure is approximately metric."

> **The framework should treat exact metricity as an ideal limiting case and approximate metricity as the physically robust bridge to emergent space.**

2.4.3 Coarse-Grained Correlation Kernels

Let $K_\omega(A, B)$ be a microscopic or fine-grained correlation kernel. A **coarse-grained kernel** may be denoted $K_\omega^{\mathrm{cg}}(A, B)$, where "cg" indicates coarse-graining.

The coarse-grained kernel may be obtained by averaging, smoothing, restricting to stable probe classes, integrating out microscopic fluctuations, or replacing exact representatives with equivalence classes. The precise construction is model-dependent and must be specified in any concrete application.

For the abstract theorem, the exact coarse-graining mechanism is not required. What matters is that the resulting kernel satisfies controlled approximate consistency conditions.

Thus, we assume a kernel:

$K_\omega^{\mathrm{cg}}: \mathcal{A}_{\mathrm{loc}}^{\mathrm{cg}} \times \mathcal{A}_{\mathrm{loc}}^{\mathrm{cg}} \rightarrow (0, 1]$,
where $\mathcal{A}_{\mathrm{loc}}^{\mathrm{cg}}$ is a class of coarse-grained localisable probes.

To avoid excessive notation, this subsection will often write K_ω instead of K_ω^{cg} , with the understanding that the approximate theorem applies to the effective or coarse-grained kernel.

2.4.4 Approximate Multiplicative Correlation Consistency

Exact multiplicative consistency requires:

$K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C)$.

Approximate consistency allows a controlled deficit:

Assumption 2.6 (Approximate Multiplicative Consistency).

There exists $\epsilon \geq 0$ such that for all $A, B, C \in \mathcal{A}_{\mathrm{loc}}$:

$$\boxed{K_\omega(A, C) \geq e^{-\epsilon / \ell_0} K_\omega(A, B) K_\omega(B, C).}$$

When $\epsilon = 0$, this reduces to exact multiplicative consistency:

$$K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C).$$

When $\epsilon > 0$, the direct correlation between A and C is allowed to be smaller than the product of the two-step correlations by a controlled factor $e^{-\epsilon / \ell_0}$.

Thus, the approximate condition says that correlation may fail exact multiplicative consistency, but only by a bounded logarithmic error.

2.4.5 Interpretation of the Error Term

The factor $e^{-\epsilon / \ell_0}$ is chosen because the distance is logarithmic.

If the kernel violates exact consistency by this multiplicative factor, the distance violates the triangle inequality by an additive amount ϵ .

This is the same product-to-sum principle used in the exact case.

The interpretation is:

$$\boxed{\epsilon \text{ measures the allowed triangle-defect in emergent distance.}}$$

If ϵ is small compared to the scale of the distances being studied, then the geometry behaves approximately like a metric geometry.

If ϵ is large, then the metric interpretation becomes weak or unreliable.

Thus, approximate metricity is meaningful only when the error term is controlled relative to the scale of interest.

2.4.6 Theorem — Approximate Triangle Inequality

Theorem 2.7 (Approximate Triangle Inequality).

Let $K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \rightarrow (0, 1]$ be a normalised positive correlation kernel, and let $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ with $\ell_0 > 0$. Assume that for some $\epsilon \geq 0$, the kernel satisfies approximate multiplicative consistency:

$$K_\omega(A, C) \geq e^{-\epsilon / \ell_0} K_\omega(A, B) K_\omega(B, C).$$

Then:

$$\boxed{d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C) + \epsilon.}$$

Proof.

Starting from approximate multiplicative consistency:

$$K_\ell(A, C) \geq e^{-\ell \epsilon / \ell_0} K_\ell(A, B) K_\ell(B, C).$$

Taking logarithms (which preserves the inequality since $\log$ is monotonically increasing):

$$\log K_\ell(A, C) \geq -\frac{\ell \epsilon}{\ell_0} + \log K_\ell(A, B) + \log K_\ell(B, C).$$

Multiplying by $-\ell_0$ (which is negative, so it reverses the inequality):

$$-\ell_0 \log K_\ell(A, C) \leq \ell \epsilon - \ell_0 \log K_\ell(A, B) - \ell_0 \log K_\ell(B, C).$$

Using the definition $d_\ell(X, Y) = -\ell_0 \log K_\ell(X, Y)$:

$$d_\ell(A, C) \leq d_\ell(A, B) + d_\ell(B, C) + \ell \epsilon. \quad \blacksquare$$

2.4.7 Exact Metricity as the Zero-Error Limit

If $\epsilon = 0$, the approximate consistency condition becomes:

$$K_\ell(A, C) \geq K_\ell(A, B) K_\ell(B, C).$$

The approximate triangle inequality becomes:

$$d_\ell(A, C) \leq d_\ell(A, B) + d_\ell(B, C).$$

Thus, the exact pseudometric theorem is recovered as the zero-error limit of the approximate theorem.

This gives a clean hierarchy:

$$\boxed{\epsilon = 0} \quad \Longleftrightarrow \quad \text{exact pseudometricity;} \\ \boxed{\epsilon > 0} \quad \Longleftrightarrow \quad \text{approximate metricity with triangle defect } \epsilon.$$

The exact case is therefore not a separate conceptual mechanism. It is the limiting case of controlled approximate correlation consistency.

2.4.8 Coarse-Graining Maps Exact to Approximate

The key structural role of coarse-graining is to convert exact microscopic metricity into approximate macroscopic metricity.

At the fine level, the correlation kernel may satisfy exact multiplicative consistency, yielding an exact pseudometric d_ℓ . When the kernel is coarse-grained, the exact consistency may be lost, but the approximate consistency may hold with a bounded error ϵ .

This is because coarse-graining averages over microscopic fluctuations. The averaged kernel may not satisfy the exact multiplicative law, but if the fluctuations are bounded, the violation is controlled.

Thus, the coarse-grained distance satisfies:

$\bar{d}_\omega(A, C) \leq \bar{d}_\omega(A, B) + \bar{d}_\omega(B, C) + \epsilon$, where ϵ is the coarse-graining error.

> **Coarse-graining maps exact microscopic metricity to approximate macroscopic metricity.**

The approximation error is not arbitrary. It is controlled by the microscopic fluctuation scale.

2.4.9 Stability Under Coarse-Graining

A coarse-grained geometry is useful only if it is stable. Small changes in microscopic correlation data should not radically change macroscopic distances or neighbourhoods.

Since $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$, small relative changes in K_ω produce additive changes in distance. If:

$K'_\omega(A, B) = e^{\{-\Delta(A, B) / \ell_0\}}, K_\omega(A, B)$, then:

$d'_\omega(A, B) = d_\omega(A, B) + \Delta(A, B)$.

Thus, logarithmic distance makes multiplicative kernel perturbations additive in geometry.

This provides a useful stability interpretation:

$\boxed{\text{Multiplicative correlation errors become additive distance errors.}}$

If the perturbation $\Delta(A, B)$ is bounded by some $\Delta_{\max}$, then the coarse-grained distance changes by at most $\Delta_{\max}$.

2.4.10 Emergent Smoothness as a Further Condition

Approximate metricity is not yet smoothness. A space may satisfy an approximate triangle inequality and still fail to resemble a smooth manifold.

Smoothness requires additional large-scale regularity, such as:

- stable dimension,
- local Euclidean behaviour,
- differentiable coordinate charts,
- continuum approximability.

Thus, the route from approximate correlation geometry to smooth space requires further assumptions:

$\text{approximate metricity} \quad \rightarrow \quad \text{stable neighbourhoods} \quad \rightarrow \quad \text{coarse regions} \quad \rightarrow \quad \text{continuum approximation} \quad \rightarrow \quad \text{smooth manifold-like structure}.$

Only the first step is treated in this subsection.

> **Approximate metricity is necessary for smooth geometry, but not sufficient by itself.**

2.4.11 Guardrails for This Subsection

1. **Approximate geometry is not exact geometry.** The coarse-grained regime is controlled, not exact.
2. **The effective space is state-dependent.** It inherits its structure from K_ω .
3. **Coarse-graining is not quotienting.** Quotienting removes zero-distance degeneracy; coarse-graining aggregates fine structure.
4. **No manifold is assumed yet.** This subsection only prepares the conditions under which manifold-like behaviour can later emerge.

2.5 Observable-Dependent Direction

2.5.1 Purpose of This Subsection

A metric space provides distance, but not direction. To recover full geometric structure—tangent spaces, gradients, directional derivatives—the framework must reconstruct the notion of **direction** from the correlation geometry.

In ordinary differential geometry, direction is represented by tangent vectors in $T_x \Sigma$. But in this framework, we do not begin with a manifold Σ , nor with points $p \in \Sigma$, nor with a pre-existing tangent bundle.

We begin with localisable observables $A \in \mathcal{A}_{\mathrm{loc}}$ and their correlation distances $d_\omega(A, B)$. Direction must therefore be reconstructed from the correlation structure.

> **Direction is not a primitive geometric notion. It is an observable-dependent equivalence class of local correlation-profile changes.**

2.5.2 Distance Profiles

Definition 2.6 (Distance Profile).

For each localisable observable $A \in \mathcal{A}_{\mathrm{loc}}$, define its **distance profile**:

$$\Phi_A: \mathcal{A}_{\mathrm{loc}} \rightarrow [0, \infty), \quad \Phi_A(X) = d_\omega(A, X).$$

The profile Φ_A records the full pattern of relational separations from A to all other probes. This profile is the relational signature of A 's emergent location.

2.5.3 Relative Displacement Profile

To reach direction rather than mere localisation, we compare how two points sit relative to all others.

Definition 2.7 (Relative Displacement Profile).

For $A, B \in \mathcal{A}_{\mathrm{loc}}$, define the **relative displacement profile**:

$$\Delta_{A \rightarrow B}: \mathcal{A}_{\mathrm{loc}} \rightarrow \mathbb{R}, \quad \Delta_{A \rightarrow B}(X) = d_{\omega}(B, X) - d_{\omega}(A, X).$$

The function $\Delta_{A \rightarrow B}$ records how the distance profile changes when one moves from A to B . It is the minimal purely metric-correlation object from which a notion of directionality can be reconstructed without assuming vectors or coordinates in advance.

2.5.4 Basic Properties of Displacement Profiles

Proposition 2.1 (Antisymmetry).

For all $A, B, X \in \mathcal{A}_{\mathrm{loc}}$:

$$\Delta_{A \rightarrow B}(X) = -\Delta_{B \rightarrow A}(X).$$

Proof.

$$\Delta_{A \rightarrow B}(X) = d_{\omega}(B, X) - d_{\omega}(A, X) = -(d_{\omega}(A, X) - d_{\omega}(B, X)) = -\Delta_{B \rightarrow A}(X). \quad \square$$

Proposition 2.2 (Vanishing on the Diagonal).

For every $A \in \mathcal{A}_{\mathrm{loc}}$:

$$\Delta_{A \rightarrow A}(X) = 0 \quad \text{for all } X.$$

Proof.

$$\Delta_{A \rightarrow A}(X) = d_{\omega}(A, X) - d_{\omega}(A, X) = 0. \quad \square$$

2.5.5 Direction Equivalence

Two displacements from the same base point should be considered the same direction if they deform the surrounding distance profile in the same way, up to positive rescaling.

Definition 2.8 (Direction Equivalence).

Let $A, B, C \in \mathcal{A}_{\mathrm{loc}}$ with $B \neq A$ and $C \neq A$. The ordered pairs (A, B) and (A, C) determine the same **emergent direction** at A if there exists $\lambda > 0$ such that:

$$\Delta_{A \rightarrow B}(X) \approx \lambda \Delta_{A \rightarrow C}(X) \quad \text{for all } X \in \mathcal{A}_{\mathrm{loc}}.$$

Two displacements from the same base point determine the same direction if they deform the surrounding distance profile in the same pattern up to positive rescaling.

2.5.6 Emergent Direction Space

Definition 2.9 (Emergent Direction Space).

For a fixed $A \in \mathcal{A}_{\mathrm{loc}}$, define $\mathfrak{D}_A$ to be the set of equivalence classes of ordered pairs (A, B) with $B \neq A$, modulo the direction equivalence of Definition 2.8.

The elements of $\mathfrak{D}_A$ are called **emergent directions** based at A .

This gives the first pre-vectorial notion of directional structure inside the emergent metric space.

2.5.7 Emergent Tangent Cone

Definition 2.10 (Emergent Tangent Cone).

The **emergent tangent cone** at A is:

$$T_A^{\mathrm{em}} \mathcal{X}_\omega := \frac{D}{A}.$$

The collection of all such classes defines a pre-differential directional structure extracted from relational metric profiles.

2.5.8 Theorem — Direction Requires Correlation Distinguishability

Theorem 2.8 (Direction Requires Distinguishability).

If $A \sim_\omega B$ (i.e., $d_\omega(A, B) = 0$), then $\Delta_{A \rightarrow B}(X) = 0$ for all X . Therefore $A \rightarrow B$ defines no non-trivial spatial direction.

Proof.

If $A \sim_\omega B$, then $d_\omega(A, B) = 0$. For any X , the triangle inequality gives:

$$d_\omega(A, X) \leq d_\omega(A, B) + d_\omega(B, X) = d_\omega(B, X).$$

Similarly:

$$d_\omega(B, X) \leq d_\omega(B, A) + d_\omega(A, X) = d_\omega(A, X).$$

Therefore $d_\omega(A, X) = d_\omega(B, X)$, so $\Delta_{A \rightarrow B}(X) = 0$ for all X .

□

> **No distinguishability $\rightarrow$ no direction.** Direction only exists between correlation-distinguishable classes.

2.5.9 Relation to Continuum Geometry

In a smooth large-scale limit, these relational directions may become ordinary tangent vectors:

$$T_A^{\mathrm{em}} \mathcal{X}_\omega \rightarrow T_x \Sigma.$$

Only in a smooth coarse-grained limit does this tangent cone become an ordinary tangent space. Thus, direction, like distance and position, is an emergent relational property rather than a fundamental structure.

2.5.10 Guardrails for This Subsection

1. **Direction here is reconstructed, not assumed.** No vector spaces, coordinate axes, or background orientations are postulated.
2. **Distance-profile identity is an extra condition.** The drafts support the existence of a localisation refinement target, but the exact condition that guarantees uniqueness of localisation is formalised here explicitly.

3. **Direction equivalence is a formal completion.** The drafts name the direction/localisation refinement item, but do not provide a fully retrievable theorem statement for it. The profile-based direction classes above are a disciplined mathematical completion of that target.

4. **This is still metric-profile geometry, not smooth differential geometry.** Further coarse-graining and continuum-limit work would be needed before claiming tangent spaces or manifold structure.

2.6 Dimensional Closure: Relational Inference Selects Three Spatial Dimensions

2.6.1 Purpose of This Subsection

The framework does not assume a pre-existing 3D space or 4D spacetime manifold. Space must emerge from the zero-preserving relational structure. Before we can ask *why* the universe is three-dimensional, we must first establish *how* an emergent correlation graph can possess any stable dimension at all.

The purpose of this subsection is to prove that the emergent spatial dimension is uniquely selected as $D=3$ by the closure requirements of relational inference.

2.6.2 The Relational Inference Argument

In this framework, local existence, position, direction, distance, and duration are not primitives. They are all inferred only through relations among configurations in the emergent correlation geometry.

Definition 2.11 (Relational Inference Map).

Let $\mathcal{X}_\omega$ be the emergent point set. Define the **relational inference map** at a point $x \in \mathcal{X}_\omega$:

$$\mathcal{I}_x: \mathcal{N}(x) \rightarrow \mathbb{R}^4,$$

where $\mathcal{N}(x)$ is the local correlation neighbourhood. The four output components encode:

1. **Existence**: distinguishability of x from the null configuration.
2. **Position**: separation of x from neighbouring configurations.
3. **Direction**: orientation of x relative to surrounding configurations.
4. **Duration**: causal-depth ordering of x relative to its predecessor and successor chain.

The **inference rank** at x is:

$$r(x) = \dim(\text{Im}(\mathcal{I}_x)).$$

A relational geometry is **inference-nondegenerate** at x if $r(x) = 4$, meaning all four inferred observables are independently distinguishable.

2.6.3 Spatial vs. Temporal Channels

Duration is supplied by the causal layer (Section 1), not the spatial correlation layer. So the spatial correlation geometry must independently supply three channels: existence, position, and direction.

In a D -dimensional correlation geometry, the maximum number of independent spatial relational channels at x is bounded by D :

$$r_{\{\mathrm{spatial}\}}(x) \leq D.$$

The three spatial inference channels—existence, position, and direction—are structurally independent. They cannot be universally reduced to one another:

- **Existence** requires distinguishability from the null configuration (binary relational test).
- **Position** requires pairwise separation measurement $d_{\omega}(x, y)$ (scalar relational comparison).
- **Direction** requires comparison of distance profiles $\Phi_x - \Phi_y$ (vectorial relational comparison).

These three tests are of different algebraic type (binary, scalar, vectorial), so they cannot collapse into fewer than three independent channels without losing distinguishability.

2.6.4 Theorem — Relational Closure Selects Three Spatial Dimensions

Theorem 2.9 (Relational Closure Selects Three Spatial Dimensions).

Let local existence, position, direction, distance, and duration be inferred only through relations among configurations in the emergent correlation geometry. Then the smallest nondegenerate relational network that allows these inferences to close without collapse or redundancy is $D=3$.

Proof.

We prove by contradiction in three cases.

Case 1: $D < 3$ (Under-closure).

Suppose the emergent correlation geometry has effective dimension $D < 3$.

The three spatial inference channels (existence, position, direction) are structurally independent and require three independent spatial relational channels. However, in a D -dimensional geometry:

$$r_{\{\mathrm{spatial}\}}(x) \leq D < 3.$$

Therefore, at least one inference channel must collapse. The network cannot cleanly distinguish existence, position, and direction. Inference becomes degenerate.

This generates a positive under-closure defect:

$$\chi_{\{\mathrm{under}\}}(D) > 0.$$

Hence:

$$\boxed{D < 3 \quad \rightarrow \quad \chi_C(D) > 0 \quad \rightarrow \quad D \notin \frac{D}{\mathrm{adm}}}$$

Case 2: $D > 3$ (Over-closure).

Suppose the emergent correlation geometry has effective dimension $D > 3$.

The minimal inference closure rank is $r_{\min} = 3$. When $D > 3$, the geometry supplies excess independent relational directions:

$$r_{\mathrm{spatial}}(x) = D > 3 = r_{\min}.$$

The excess direction count is:

$$x_D(x) = D - 3 > 0.$$

The overcomplete structure introduces degeneracy in the inverse inference problem: given the observed relational data, the underlying configuration cannot be uniquely recovered because multiple relational paths produce the same inference signature. Position, direction, and distance cease to remain cleanly distinguishable because multiple relational paths encode the same observables, creating redundancy and near-indistinguishability.

Unless all excess directions are pure gauge (which would collapse the effective dimension back to $D=3$), the unmatched excess generates a positive over-closure defect:

$$\chi_{\mathrm{over}}(D) > 0.$$

Hence:

$$\boxed{D > 3 \quad \rightarrow \quad \chi_C(D) > 0 \quad \rightarrow \quad D \notin \frac{D}{\mathrm{adm}}}$$

Case 3: $D = 3$ (Balanced Closure).

Suppose the emergent correlation geometry has effective dimension $D = 3$.

Then:

$$r_{\mathrm{spatial}}(x) = 3 = r_{\min}.$$

The under-closure defect vanishes:

$$\chi_{\mathrm{under}}(3) = 0.$$

The over-closure defect vanishes:

$$\chi_{\mathrm{over}}(3) = 0.$$

If the remaining algebraic, boundary, flux, and record-sector closure defects also vanish in the balanced regime, then:

$$\chi_C(3) = 0.$$

Hence:

$$\boxed{D = 3 \quad \rightarrow \quad \chi_C(3) = 0 \quad \rightarrow \quad 3 \in \frac{D}{\mathrm{adm}}}$$

****Combining all three cases:****

$$\forall C(D) = 0 \iff D = 3.$$

Since the master constraint contains $\forall C(D)$ non-negatively, physical admissibility selects $D=3$. Combined with the emergent proper-time parameter from the causal layer, the selected spacetime phase is $3+1$. $\blacksquare$

2.6.3 Interpretation

The proof does not say "three dimensions because they are convenient." It says three dimensions are the ****unique balance point**** where:

- Existence can be distinguished from non-existence (binary relational test).
- Position can be measured (scalar relational comparison).
- Direction can be oriented (vectorial relational comparison).

without collapse ($D < 3$) and without redundancy-induced indistinguishability ($D > 3$). The causal layer supplies the fourth independent channel—duration—separately, yielding $3+1$ spacetime.

2.6.4 Guardrails for This Subsection

1. ****The closure-defect route is proven.**** The theorem establishes that $D \neq 3$ produces positive closure defect.
2. ****The uniqueness lemma is conditionally closed.**** The proof depends on the structural independence of existence, position, and direction as inference channels, and on the over-closure argument for $D > 3$.
3. ****This is a relational argument, not a geometric counting trick.**** The dimension is not chosen because of ball-volume scaling or aesthetic preference. It is enforced by the closure requirements of relational inference.

2.7 Guardrails and Summary of Section 2

2.7.1 Guardrails for Section 2

1. ****Correlation is pre-geometric.**** The kernel K_ω is not yet a distance, metric, or coordinate chart.
2. ****Localisable does not mean spatially local in advance.**** The term "localisable" is operational and framework-internal.
3. ****Approximate metricity is a formal completion.**** Exact theorem-backed metricity requires the multiplicative consistency condition.
4. ****Perfect correlation is not algebraic identity.**** The relation $A \sim_\omega B$ expresses zero emergent separation, not literal equality in $\mathcal{A}$.
5. ****Direction is reconstructed, not assumed.**** No vector spaces, coordinate axes, or background orientations are postulated.

6. **Dimensional closure** is a relational argument. $D=3$ emerges from inference closure, not from geometric counting.

2.7.2 Summary of Section 2

Section 2 reconstructed spatial structure from correlation geometry:

1. **The correlation kernel** K_ω was defined on localisable zero-preserving observables.
2. **Correlation distance** $d_\omega = -\ell_0 \log K_\omega$ was defined and shown to be non-negative.
3. Under **multiplicative correlation consistency**, d_ω was proven to be a pseudometric.
4. **Emergent points** were defined as equivalence classes of perfectly correlated observables: $\mathcal{X}_\omega = \mathcal{A}_{\{\mathrm{loc}\}} / \sim_\omega$.
5. The quotient was shown to carry a genuine **metric** $\tilde{d}_\omega$.
6. **Approximate metricity** was established under coarse-graining: $d_\omega(A,C) \leq d_\omega(A,B) + d_\omega(B,C) + \epsilon$.
7. **Observable-dependent direction** was reconstructed from distance-profile differences.
8. **Dimensional closure** was proven: $D=3$ is the unique spatial dimension where relational inference of existence, position, and direction closes without degeneracy or redundancy.

The conceptual chain is:

$$\boxed{\text{correlation kernel}} \quad \longrightarrow \quad \text{correlation distance} \quad \longrightarrow \quad \text{pseudometric} \quad \longrightarrow \quad \text{emergent points} \quad \longrightarrow \quad \text{approximate metric space} \quad \longrightarrow \quad \text{direction} \quad \longrightarrow \quad D=3.$$

This completes the reconstruction of proto-space from relational correlation structure.

Section 3 — Emergent Time and Burden-Clock Geometry

3.0 Purpose of Emergent Time and Burden-Clock Geometry

3.0.1 Transition from Space to Time

Section 1 established the causal dependency order $\prec$ and the open extension principle. Section 2 reconstructed emergent spatial structure from the correlation kernel K_ω , proving that correlation distance becomes a pseudometric under multiplicative consistency, and that emergent points arise as equivalence classes of perfectly correlated observables. Section 2 also proved that $D=3$ is the unique spatial dimension where relational inference closes without degeneracy.

However, space is not time. A correlation distance tells us which relational contexts are near or far in an emergent spatial sense. It does not by itself tell us how duration is measured,

why clocks may run at different rates, how temporal ordering becomes quantitative, or how time-dilation-like behaviour could arise.

The causal dependency relation $e_i \prec e_j$ provides order. It says that one admissible event is dependency-prior to another. But a partial order alone does not define proper time. It gives sequence, precedence, or causal depth, but not duration.

Thus, after constructing emergent space from correlation, the next task is to reconstruct emergent time from causal structure together with a local measure of relational difficulty.

That measure is called **burden**.

3.0.2 Why Causal Order Alone Is Not Time

A causal chain has the form:

$$e_0 \prec e_1 \prec \cdots \prec e_n.$$

This chain tells us that each event depends on the previous structure. It gives an order-theoretic notion of depth:

$$L_C(e_0, e_n) = \max\{n \mid e_0 \prec e_1 \prec \cdots \prec e_n\}.$$

But this depth is only a count of dependency steps. It does not yet tell us how much physical time passes along the chain.

For example, two chains may have the same number of dependency steps but differ in relational difficulty. One may pass through a low-burden region, where extension is easy and clock rate is high. Another may pass through a high-burden region, where maintaining admissibility is costly and clock rate is suppressed.

Therefore, causal depth must be weighted.

> **Causal order gives before/after; burden-weighted causal order gives duration.**

This section develops that second structure.

3.0.3 The Role of Burden

Burden measures the local relational load associated with preserving admissibility.

At the foundational level, physicality means constraint compatibility:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \forall \alpha.$$

At the level of extension, admissibility means that a structure may continue only through zero-preserving enlargement:

$$\frac{M}{\mathcal{U}_{s+1}} = 0.$$

However, not all admissible extensions are equally easy. Some regions of relational structure may require more constraint reconciliation, more dependency closure, more correlation adjustment, or more structural coordination in order to remain physical.

This local difficulty is what burden represents.

> **Burden is not constraint failure; it is the cost of maintaining constraint compatibility.**

A high-burden region is not necessarily a region of constraint violation. Rather, it is a region where preserving constraint-zero admissibility is costly. This distinction is crucial. If a structure is physical, its constraints still vanish in the appropriate sense. Burden measures the internal load required to keep them vanishing under extension, interaction, localisation, or deformation.

3.0.4 Burden and Clock Rate

The central claim of the burden-time relation is that higher burden suppresses local clock rate.

A simple functional form is:

$$\alpha(B) = \frac{1}{1 + \lambda B}, \quad \lambda > 0,$$

where B is a non-negative burden measure and $\alpha(B)$ is a local clock factor.

If $B = 0$, then $\alpha(0) = 1$. The clock is unsuppressed relative to the chosen reference scale.

If $B > 0$, then $0 < \alpha(B) < 1$. The local clock contribution is reduced.

If burden increases, the clock factor decreases:

$$B_1 > B_2 \quad \Longleftrightarrow \quad \alpha(B_1) < \alpha(B_2).$$

> **Higher burden means slower internal clock rate.**

This provides the first theorem-safe bridge toward time-dilation-like behaviour.

3.0.5 Proper Time as Weighted Causal Depth

Let γ be a causal chain:

$$\gamma: e_0 \prec e_1 \prec \dots \prec e_n.$$

An unweighted depth count would assign a duration proportional to n . But if each event or step carries burden, then the contribution of each step should be weighted by the local clock factor.

A burden-weighted proper-time functional may be written as:

$$\tau[\gamma] = \sum_{e \in \gamma} \epsilon \cdot \alpha(e) = \sum_{e \in \gamma} \frac{\epsilon}{1 + \lambda B(e)}.$$

This formula expresses proper time as accumulated causal depth, suppressed by local burden.

When burden is low, each causal step contributes nearly ϵ .

When burden is high, each causal step contributes less than ϵ .

> **Proper time is accumulated admissible causal succession, locally slowed by burden.**

3.0.6 Why This Resembles Gravitational Time Dilation

In familiar gravitational physics, clocks deeper in a gravitational potential run more slowly relative to clocks far away. The present framework does not assume spacetime gravity at the foundation. However, the burden-clock relation produces an analogous structure.

If a region carries high burden, then $B(e)$ is large. Therefore $\alpha(e) = 1/(1 + \lambda B(e))$ is small. The accumulated proper time along chains passing through that region is reduced.

This gives a relational explanation of clock suppression:

> **Clocks slow where maintaining relational admissibility is more burdensome.**

This is not yet a full theory of gravity. It is a clock-rate bridge. Later sections may connect burden density, burden flux, and burden stress to effective metric structure. But the first step is simply the monotonic relation between burden and clock rate.

3.0.7 The Causal Reconciliation Principle

The burden-clock relation is grounded in a deeper principle.

> **The effective speed of light and the rate of time are bounded by the degree to which relational structures can reconcile with causal dependency.**

This is the **Causal Reconciliation Principle**. It is not a new primitive axiom, but a deeper grounding of the existing causal, relational, and burden-based structure.

When reconciliation is efficient, causal propagation is fast and time advances normally. When burden or constraint loading increases, reconciliation slows, the causal ceiling tightens, and proper time becomes increasingly suppressed.

In compact form:

$$c_{\mathrm{eff}} \sim \frac{\ell_0}{\tau_{\mathrm{rec}}}, \quad d\tau \sim \frac{d\sigma}{1 + \lambda B}, \quad \tau_{\mathrm{rec}} \uparrow \Rightarrow c_{\mathrm{eff}} \downarrow, \quad d\tau \downarrow.$$

This principle unifies the causal speed limit (Section 1) with the burden-clock suppression (this section). The same mechanism that bounds propagation speed also bounds the rate of temporal flow.

3.0.8 What This Section Establishes

This section establishes the following:

1. **Constraint burden** $B_R(e)$ is defined as the local cost of maintaining zero-closure.
2. A **local clock factor** $\alpha(B) = 1/(1 + \lambda B)$ is introduced.
3. **Burden-weighted proper time** $\tau[\gamma] = \sum_{e \in \gamma} \epsilon \alpha(e)$ is defined.
4. The theorem **"burden suppresses clock rate"** is proven: $B_1 > B_2 \rightarrow \alpha(B_1) < \alpha(B_2)$.
5. **Non-uniform burden** is shown to induce non-uniform clock accumulation, producing temporal geometry.
6. An **effective potential** $\Phi_C = c_{\text{eff}}^2 \log \alpha$ is derived.
7. The **weak-burden limit** $\Phi_C \approx -c_{\text{eff}}^2 \lambda B$ is established.
8. The **scalar burden-to-lapse bridge** $d\tau = \alpha(x) \, d\sigma$ is formalised.
9. The relation between burden-clock geometry and **emergent gravity** is prepared.

The theorem-safe structure is:

$$\boxed{\text{causal order} + \text{constraint burden} \quad \rightarrow \quad \text{burden-weighted proper time} \quad \rightarrow \quad \text{temporal geometry}.}$$

3.0.9 Guardrails for This Section

1. **Time is still emergent.** Even after defining burden-weighted proper time, time is not a primitive background parameter. It is reconstructed along causal chains.
2. **Burden is not violation.** A high-burden region may still be exactly physical: $\frac{M}{\text{volume}} = 0$. Burden measures the cost of preserving physicality, not the failure of physicality.
3. **Clock suppression is not yet a full metric equation.** The formula $\alpha(B) = 1/(1 + \lambda B)$ defines a clock factor. It does not by itself define the full metric tensor.
4. **The burden-clock relation is a model choice unless derived.** The rational form $\alpha(B) = 1/(1 + \lambda B)$ is mathematically simple and monotone. Other monotone decreasing functions could serve similar roles. The theorem-safe statement is: any positive decreasing clock factor yields burden-induced clock suppression.

3.1 Constraint Burden

3.1.1 Purpose of This Subsection

Before defining emergent proper time, the framework needs a measure of how much relational cost is associated with maintaining admissibility at a given event or region. This measure is **constraint burden**.

The purpose of this subsection is to define burden precisely, distinguish it from constraint violation, and establish its role as the local reconciliation cost that will later weight the flow of proper time.

3.1.2 The Concept of Reconciliation Cost

At the foundational level, physicality means constraint compatibility:

$$\omega(\hat{C}_{\alpha}^{\dagger} \hat{C}_{\alpha}) = 0 \quad \forall \alpha.$$

At the level of open extension, admissibility means that a structure may continue only through zero-preserving enlargement:

$$\frac{M}{\mathcal{U}_{s+1}} = 0.$$

However, not all admissible extensions are equally easy. Some regions of relational structure may require more constraint reconciliation, more dependency closure, more correlation adjustment, or more structural coordination in order to remain physical.

This local difficulty is what burden represents.

Definition 3.1 (Constraint Burden).

Let $R \subseteq E$ be a subsystem or coarse-grained region. The **constraint burden** of R is a non-negative quantity:

$$B(R) \geq 0,$$

measuring the total relational reconciliation cost required for R to remain compatible with the master zero constraint.

Interpretation.

Burden measures the local load of preserving admissibility. It is not the amount by which physicality fails. A physical event may have non-zero burden while still satisfying all constraints.

> **$B(R) > 0$** means the region is costly to maintain admissibly, not that it is unphysical.

3.1.3 Abstract Definition

At the abstract level, burden may be defined as:

$$B(R) = \sum_{\alpha} w_{\alpha} b_{\alpha}(R),$$

where $b_{\alpha}(R) \geq 0$ measures how strongly subsystem R participates in, disturbs, or depends on the primitive constraint $\hat{C}_{\alpha}$.

Each $b_{\alpha}(R)$ represents the local contribution of region R to the difficulty of satisfying constraint α .

3.1.4 Operator-Form Candidate

In a represented Hilbert-space setting, if Π_R is a localisation or projection operator associated with region R , a candidate operator expression for burden is:

Definition 3.2 (Operator-Form Burden).

$$B(R) = \sum_{\alpha} w_{\alpha} \left| [\hat{C}_{\alpha}, \Pi_R] \right|_{\omega}^2,$$
 where $|\cdot|_{\omega}$ is a state-dependent norm.

Interpretation.

This measures how strongly the localisation operator Π_R fails to commute with the constraints, evaluated in the physical state ω . If R is in a region where the constraints are trivially satisfied and nothing interferes, then $[\hat{C}_{\alpha}, \Pi_R] \approx 0$ and $B(R) \approx 0$. If R is in a region of intense relational activity—interactions, dense correlations, high structural complexity—then the commutators are non-trivial and $B(R) > 0$.

3.1.5 The Three-Component Decomposition

The framework refines burden from a single scalar into a source-origin decomposition with three distinct channels:

Definition 3.3 (Component Burden Decomposition).

The total burden density decomposes as:

$$\rho_B(x) = \rho_{\text{mode}}(x) + \rho_{\text{int}}(x) + \rho_{\text{rel}}(x),$$

where:

- ρ_{mode} : burden from stable modes (matter-like occupancy),
- ρ_{int} : burden from interactions (non-additive composition cost),
- ρ_{rel} : burden from relational/correlation structure (network binding).

Interpretation.

These three components are not three versions of the same thing. They are three structurally distinct source channels:

- **Mode burden** is generated by the existence and occupation of stable field or particle-like modes. A star, a planet, or an atom contributes mode burden simply because they are realised configurations.
- **Interaction burden** arises from non-additivity in composition. A neutron star carries more interaction burden than diffuse gas. A bound system carries interaction burden given by $B_{\text{int}}(R_1, R_2) = B(R_1 \cup R_2) - B(R_1) - B(R_2)$.
- **Relational burden** is generated by correlations and belongs to the network itself. It measures the burden required to maintain relational consistency across configurations. Unlike the first two, ρ_{rel} does not belong to a single object—it belongs to the correlation web.

This decomposition will become crucial in later sections when the burden tensor sources effective geometry and when the cosmological implications of relational burden are explored.

3.1.6 Burden Is Not Constraint Violation

This distinction must remain absolutely clear throughout the manuscript.

A physical state satisfies:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0.$$

A represented physical vector satisfies:

$$\hat{\mathfrak{M}}_\omega \psi = 0.$$

An effective physical configuration satisfies:

$$\mathfrak{M}(x) = 0.$$

These are zero-violation conditions. Burden, by contrast, may be non-zero even when the structure is physical. It measures the load of preserving physicality, not the failure of physicality.

In a physical but burdened region, one may have:

$$\mathfrak{M} = 0 \quad \text{while} \quad B > 0.$$

> $\mathfrak{M}$ measures constraint violation; B measures constraint-maintenance load.

3.1.7 Guardrails for This Subsection

1. Burden is not energy. At this stage, burden is a relational cost measure. Energy requires additional structure (time translation, Hamiltonian, conservation). Burden may become energy-like under additional assumptions, but that identification is not automatic.
2. Burden is not violation. $B > 0 \not\Rightarrow \mathfrak{M} > 0$. A region may be exactly physical and still carry high burden.
3. The three-component decomposition is structural. Mode, interaction, and relational burden are not interchangeable. They represent distinct source channels that may activate differently in different physical regimes.

3.2 Local Clock Factor and Burden Suppression

3.2.1 Purpose of This Subsection

The previous subsection defined constraint burden $B(R)$ as the local cost of maintaining admissibility. The next step is to connect burden to the rate at which proper time accumulates.

The central claim is that higher burden suppresses local clock rate. This subsection formalises that claim through the definition of a **local clock factor** and proves the monotonic suppression theorem.

3.2.2 Definition — Local Clock Factor

Definition 3.4 (Local Clock Factor).

Let $B \geq 0$ be a burden measure and $\lambda > 0$ a coupling constant. The **local clock factor** is:

$$\boxed{\alpha(B) = \frac{1}{1 + \lambda B}}.$$

Since $B \geq 0$ and $\lambda > 0$, we have:

$$1 + \lambda B \geq 1,$$

and therefore:

$$0 < \alpha(B) \leq 1.$$

Interpretation.

- If $B = 0$, then $\alpha(0) = 1$. The clock is unsuppressed.
- If $B > 0$, then $0 < \alpha(B) < 1$. The local clock contribution is reduced.
- As $B \rightarrow \infty$, $\alpha(B) \rightarrow 0$. The clock factor approaches zero.

The clock factor $\alpha(B)$ represents the fraction of the elementary causal-step time that accumulates at a given burden level.

> **$\alpha(B)$ is the local reconciliation rate: the fraction of the elementary tick that survives under burden loading.**

3.2.3 Why a Rational Form?

The rational form $\alpha(B) = 1/(1 + \lambda B)$ is chosen for four reasons:

1. **Positivity:** $\alpha(B) > 0$ for all finite $B \geq 0$. The clock never literally stops for finite burden.
2. **Boundedness:** $\alpha(B) \leq 1$. The clock never runs faster than the unsuppressed rate.
3. **Monotonicity:** $\alpha'(B) < 0$. Higher burden always suppresses the clock.
4. **Simplicity:** It is the simplest non-trivial function satisfying all three conditions above.

However, the essential structure only requires a positive decreasing function. The theorem-safe general statement is:

> **Any positive decreasing clock factor $\alpha(B)$ with $\alpha(0) = 1$ and $\alpha'(B) < 0$ yields burden-induced clock suppression.**

The rational form is one canonical ansatz. Other forms (e.g., $\alpha(B) = e^{-\lambda B}$) could serve similar roles. The manuscript uses the rational form because it gives simple algebraic expressions and clear monotonic behaviour.

3.2.4 Theorem — Burden Suppresses Clock Rate

Theorem 3.1 (Burden Suppresses Clock Rate).

Let $\alpha(B) = 1/(1 + \lambda B)$ with $\lambda > 0$. If $B_1 > B_2 \geq 0$, then:

$$\boxed{\alpha(B_1) < \alpha(B_2)}.$$

Proof.

Since $B_1 > B_2$ and $\lambda > 0$:

$$\lambda B_1 > \lambda B_2.$$

Adding 1 to both sides:

$$1 + \lambda B_1 > 1 + \lambda B_2.$$

Both sides are positive. Taking reciprocals reverses the inequality:

$$\frac{1}{1 + \lambda B_1} < \frac{1}{1 + \lambda B_2}.$$

Therefore:

$$\alpha(B_1) < \alpha(B_2). \quad \blacksquare$$

3.2.5 Corollary — Limiting Behaviour

Corollary 3.1.

$$\lim_{B \rightarrow \infty} \alpha(B) = 0.$$

Proof.

As $B \rightarrow \infty$, $1 + \lambda B \rightarrow \infty$, so $1/(1 + \lambda B) \rightarrow 0$. $\blacksquare$

Interpretation.

In the limit of infinite burden, the clock factor approaches zero. This means proper time accumulation flattens toward zero. However, for any finite burden $B < \infty$, the clock factor remains strictly positive: $\alpha(B) > 0$.

> **For finite burden, $\alpha > 0$; the clock slows but does not literally stop.**

This is the guardrail against interpreting black-hole-like regimes as places where "time ends." Time does not end; it is suppressed toward zero in the exterior projection. The interior relational structure continues, but its reconciliation rate becomes extremely slow.

3.2.6 Lemma — Clock Factor Bounds

Lemma 3.1.

For all $B \geq 0$ and $\lambda > 0$:

$$0 < \alpha(B) \leq 1.$$

Moreover:

$$\alpha(B) = 1 \iff B = 0.$$

Proof.

Since $B \geq 0$ and $\lambda > 0$, we have $1 + \lambda B \geq 1 > 0$. Therefore $\alpha(B) = 1/(1 + \lambda B) > 0$ and $\alpha(B) \leq 1$.

Equality $\alpha(B) = 1$ holds if and only if $1 + \lambda B = 1$, which requires $\lambda B = 0$. Since $\lambda > 0$, this occurs exactly when $B = 0$. $\blacksquare$

3.2.7 Guardrails for This Subsection

1. **The clock factor is not a full metric.** $\alpha(B)$ is a scalar suppression factor. It does not by itself define the full metric tensor $g_{\{\mu\nu\}}$.
2. **The rational form is a model choice.** The specific function $\alpha(B) = 1/(1+\lambda B)$ is used for simplicity. The theorem-safe content is: any positive, bounded, strictly decreasing $\alpha(B)$ with $\alpha(0) = 1$ produces clock suppression.
3. **$\alpha \rightarrow 0$ does not mean $\alpha = 0$.** For finite burden, the clock factor is strictly positive. The approach to zero is asymptotic, not literal.

3.3 Burden-Weighted Proper Time

3.3.1 Purpose of This Subsection

The previous subsection defined the local clock factor $\alpha(B)$ and proved that burden suppresses it monotonically. The next step is to use this clock factor to weight causal depth, producing a proper-time functional along causal chains.

The purpose of this subsection is to define **burden-weighted proper time** rigorously and establish its basic properties.

3.3.2 Definition — Burden-Weighted Proper Time

Definition 3.5 (Burden-Weighted Proper Time).

Let $\gamma = (e_0, e_1, \dots, e_n)$ be a finite causal chain with $e_0 \prec e_1 \prec \dots \prec e_n$. Let $\epsilon > 0$ be the elementary reconciliation interval. Let $B_R(e_k)$ be the constraint burden of subsystem R at event e_k . Let $\lambda > 0$.

The **burden-weighted proper time** along γ is:

$$\boxed{\tau_R[\gamma] = \sum_{k=0}^n \frac{\epsilon}{1 + \lambda B_R(e_k)}} \quad \square$$

Equivalently, using $\alpha(B) = 1/(1 + \lambda B)$:

$$\boxed{\tau_R[\gamma] = \sum_{k=0}^n \epsilon \alpha(B_R(e_k))} \quad \square$$

Interpretation.

Each causal step e_k contributes an amount $\epsilon \cdot \alpha(B_R(e_k))$ to the accumulated proper time. When burden is low, the contribution is close to ϵ . When burden is high, the contribution is reduced.

> **Proper time is accumulated causal depth, locally suppressed by burden.**

3.3.3 Basic Bounds

Theorem 3.2 (Basic Bounds).

For any causal chain $\gamma = (e_0, \dots, e_n)$ with $n+1$ events:

$$0 < \tau_R[\gamma] \leq (n+1)\epsilon. \quad \square$$

Proof.

Since $0 < \alpha(B_R(e_k)) \leq 1$ for all k (Lemma 3.1), each term satisfies:
 $0 < \epsilon \alpha(B_R(e_k)) \leq \epsilon.$

Summing over $k = 0, \dots, n$:

$$0 < \sum_{k=0}^n \epsilon \alpha(B_R(e_k)) \leq \sum_{k=0}^n \epsilon = (n+1)\epsilon.$$

Therefore:

$$0 < \tau_R[\gamma] \leq (n+1)\epsilon. \quad \square$$

Interpretation.

Burden-weighted proper time is always strictly positive (for finite burden) and never exceeds the unburdened causal-step time. Burden can only reduce clock accumulation relative to the unburdened count.

> ***Burden can only reduce clock accumulation relative to the unburdened causal count.***

3.3.4 Constant-Burden Case

If the burden is constant along the chain, $B_R(e_k) = B_0$ for all k , then:

$$\tau_R[\gamma] = \sum_{k=0}^n \frac{\epsilon}{1 + \lambda B_0} = \frac{(n+1)\epsilon}{1 + \lambda B_0}.$$

Thus, constant burden rescales causal depth by a uniform suppression factor:

$$\boxed{\tau = \alpha(B_0) \times \text{unweighted causal-step time}.}$$

This is the simplest burden-time relation. The clock runs at a uniformly reduced rate.

3.3.5 Variable-Burden Case

If burden varies along the chain, then:

$$\tau_R[\gamma] = \sum_{k=0}^n \frac{\epsilon}{1 + \lambda B_R(e_k)}.$$

There is no single suppression factor. An effective average clock factor may be defined as:

$$\bar{\alpha}_\gamma = \frac{1}{n+1} \sum_{k=0}^n \alpha(B_R(e_k)),$$

so that:

$$\tau_R[\gamma] = (n+1)\epsilon \bar{\alpha}_\gamma.$$

However, because $\alpha(B) = 1/(1+\lambda B)$ is nonlinear, the effective average burden is not generally the arithmetic average of $B_R(e_k)$.

> ***Variable burden produces genuinely path-dependent proper time.***

This is the origin of path-dependent temporal geometry in the framework.

3.3.6 Theorem — Higher Burden Reduces Proper Time for Equal Depth

Theorem 3.3 (Higher Burden Reduces Proper Time for Equal Depth).

Let $\gamma_1 = (e_0, \dots, e_n)$ and $\gamma_2 = (e'_0, \dots, e'_n)$ be two causal chains of equal depth n . If $B_R(e_k) > B_R(e'_k)$ for every k , then:

$$\boxed{\tau_R[\gamma_1] < \tau_R[\gamma_2].}$$

Proof.

By Theorem 3.1, $B_R(e_k) > B_R(e'_k)$ implies $\alpha(B_R(e_k)) < \alpha(B_R(e'_k))$ for every k . Therefore:

$$\epsilon, \alpha(B_R(e_k)) < \epsilon, \alpha(B_R(e'_k)) \quad \forall k.$$

Summing:

$$\sum_{k=0}^n \epsilon, \alpha(B_R(e_k)) < \sum_{k=0}^n \epsilon, \alpha(B_R(e'_k)).$$

Therefore:

$$\tau_R[\gamma_1] < \tau_R[\gamma_2]. \quad \blacksquare$$

Interpretation.

For equal causal depth, the chain passing through higher burden accumulates less proper time. This is the first time-dilation-like result:

> **For equal causal depth, uniformly higher burden implies lower accumulated proper time.**

3.3.7 Relation to Unburdened Causal Time

The unburdened causal time is:

$$\tau_0[\gamma] = (n+1)\epsilon.$$

The burden-weighted proper time satisfies:

$$\tau_R[\gamma] = \sum_{k=0}^n \epsilon, \alpha(B_R(e_k)) \leq \sum_{k=0}^n \epsilon = \tau_0[\gamma].$$

Thus:

$$\tau_R[\gamma] \leq \tau_0[\gamma].$$

The ratio:

$$\frac{\tau_R[\gamma]}{\tau_0[\gamma]} = \frac{1}{n+1} \sum_{k=0}^n \alpha(B_R(e_k)) = \bar{\alpha}_\gamma$$

gives the average clock suppression along the chain.

3.3.8 Continuum Approximation

If causal chains become dense in an emergent continuum limit, the sum may be approximated by an integral.

Let s be a parameter along a causal path γ , and let $B(s)$ be the burden profile along the path. Then:

$$\tau[\gamma] \approx \int_{\gamma} \alpha(B(s)) \, d\sigma = \int_{\gamma} \frac{d\sigma}{1 + \lambda B(s)},$$

where $d\sigma$ is an elementary causal-depth measure.

Using the rational clock factor:

$$\tau[\gamma] \approx \int_{\gamma} \frac{d\sigma}{1 + \lambda B(s)}.$$

This continuum expression is an approximation. The discrete causal-chain definition is more primitive in the present framework.

> **The integral form is an effective continuum limit of a causal-chain sum.**

3.3.9 Guardrails for This Subsection

1. **Proper time is chain-relative.** $\tau[\gamma]$ is assigned to a causal chain γ . It is not a universal external time coordinate.
2. **The step scale ϵ is not automatically Planck time.** ϵ is a model parameter. It should not be identified with a known fundamental constant unless a later calibration is provided.
3. **Burden-weighted proper time is not yet coordinate time.** $\tau[\gamma]$ is local/path proper time, not a global t .
4. **Clock suppression is not yet full gravity.** The result $B_1 > B_2 \Rightarrow \alpha_1 < \alpha_2$ shows burden suppresses clocks. This resembles gravitational time dilation, but a full gravitational theory requires spatial metric response, source structure, conservation laws, and field equations.

3.4 Non-Uniform Burden as Temporal Geometry

3.4.1 Purpose of This Subsection

The previous subsection defined burden-weighted proper time along a causal chain and proved that higher burden reduces clock accumulation. The present subsection develops the next consequence: when burden varies from region to region, clock rates become non-uniform. This non-uniformity is the first form of **temporal geometry** in the framework.

> **Non-uniform burden induces non-uniform time flow.**

This does not yet require a full spacetime metric. It requires only causal chains, emergent locality, and a burden-dependent clock factor. The result is a relational version of time-rate variation: different regions accumulate different amounts of proper time over comparable causal depth.

3.4.2 From Local Burden to Local Clock Rate

Let burden be assigned to emergent locations. If $x \in \text{mathcal{X}}_\omega$ is an emergent point in the correlation geometry, let $B(x) \geq 0$ denote the local burden at x .

Define the local clock factor:

$$\boxed{\alpha(x) = \frac{1}{1 + \lambda B(x)}, \quad \lambda > 0.}$$

This is the spatially local version of the event-based formula $\alpha(e) = 1/(1 + \lambda B(e))$.

If burden is uniform, $B(x) = B_0$ for all x , and the clock factor is also uniform:

$$\alpha(x) = \frac{1}{1 + \lambda B_0} \quad \text{for all } x.$$

If burden is non-uniform, $B(x_1) \neq B(x_2)$ generally implies $\alpha(x_1) \neq \alpha(x_2)$.

Thus, local burden variation produces local clock-rate variation.

The core relation is:

$$\boxed{B(x_1) > B(x_2) \quad \Longleftrightarrow \quad \alpha(x_1) < \alpha(x_2).}$$

3.4.3 Definition — Temporal Geometry

Definition 3.6 (Temporal Geometry).

The **temporal geometry** of an emergent region is the assignment:

$$x \mapsto \alpha(x),$$

where $\alpha(x) = 1/(1 + \lambda B(x))$ is the local clock factor induced by the burden field $B(x)$.

Interpretation.

Temporal geometry is the spatial pattern of local clock factors. It is not yet full spacetime curvature. It is the first layer of clock-rate structure.

> **Temporal geometry is the spatial pattern of clock-rate suppression induced by burden.**

3.4.4 Region-Wise Clock Comparison

Let R_1 and R_2 be two emergent regions in the correlation geometry. Suppose each region has approximately constant burden:

$$B(x) \approx B_1 \quad \text{for } x \in R_1,$$

$$B(x) \approx B_2 \quad \text{for } x \in R_2.$$

Then the corresponding regional clock factors are:

$$\alpha_1 = \frac{1}{1 + \lambda B_1}, \quad \alpha_2 = \frac{1}{1 + \lambda B_2}.$$

If $B_1 > B_2$, then $\alpha_1 < \alpha_2$.

Now consider two causal chains γ_1 and γ_2 of equal causal-step count n , one remaining inside R_1 and the other remaining inside R_2 . Their proper times are approximately:

$$\tau_1 \approx n \epsilon \alpha_1, \quad \tau_2 \approx n \epsilon \alpha_2.$$

Therefore:

$$\tau_1 < \tau_2.$$

> **Over equal causal depth, higher-burden regions accumulate less proper time.**

This is the relational-clock analogue of gravitational time dilation.

3.4.5 Theorem — Non-Uniform Burden Induces Non-Uniform Proper Time

Theorem 3.4 (Non-Uniform Burden Induces Non-Uniform Proper Time).

Let R_1 and R_2 be two regions with constant burdens B_1 and B_2 respectively. Let $\alpha_i = 1/(1 + \lambda B_i)$ for $i = 1, 2$. Consider two causal chains γ_1 and γ_2 , each with n links, where γ_i lies entirely in R_i . If $B_1 > B_2$, then:

$$\boxed{\tau[\gamma_1] < \tau[\gamma_2].}$$

Proof.

Since $B_1 > B_2$ and $\lambda > 0$, Theorem 3.1 gives $\alpha_1 < \alpha_2$. For equal link-count n , the proper time along each chain is:

$$\tau[\gamma_i] = n \epsilon \alpha_i.$$

Therefore:

$$\tau[\gamma_1] = n \epsilon \alpha_1 < n \epsilon \alpha_2 = \tau[\gamma_2],$$

because $n \epsilon > 0$. $\square$

3.4.6 Burden Gradients

If burden varies continuously across emergent space, one may write:

$$B = B(x), \quad \alpha = \alpha(x) = \frac{1}{1 + \lambda B(x)}.$$

A burden gradient $\nabla B(x)$ then induces a clock-factor gradient $\nabla \alpha(x)$.

Using the rational clock factor:

$$\alpha(x) = (1 + \lambda B(x))^{-1}.$$

Differentiating formally:

$$\nabla \alpha(x) = -\frac{\lambda \nabla B(x)}{(1 + \lambda B(x))^2}.$$

Thus, where burden increases, the clock factor decreases.

> **Where burden increases, clock rate decreases.**

The continuum gradient expression is useful later, but it should not be treated as primitive. The foundational statement is the discrete comparison:

$$B(x_1) > B(x_2) \quad \Longleftrightarrow \quad \alpha(x_1) < \alpha(x_2).$$

3.4.7 Effective Potential from Clock Rate

It is useful to express the clock factor in terms of an effective potential.

Definition 3.7 (Burden-Clock Potential).

Define the **burden-clock potential**:

$$\Phi_C(x) = c_{\mathrm{eff}}^2 \log \alpha(x).$$

Since $\alpha(x) = 1/(1 + \lambda B(x))$:

$$\Phi_C(x) = c_{\mathrm{eff}}^2 \log \left(\frac{1}{1 + \lambda B(x)} \right) = -c_{\mathrm{eff}}^2 \log(1 + \lambda B(x)).$$

Since $1 + \lambda B(x) \geq 1$:

$$\log(1 + \lambda B(x)) \geq 0,$$

and therefore:

$$\Phi_C(x) \leq 0.$$

Interpretation.

Higher burden makes $\Phi_C(x)$ more negative. Thus:

> **Higher burden corresponds to a deeper effective temporal potential.**

3.4.8 Weak-Burden Expansion

If burden is small in the sense that $\lambda B(x) \ll 1$, then one may expand:

$$\log(1 + \lambda B(x)) \approx \lambda B(x) - \frac{\lambda^2 B(x)^2}{2} + \dots$$

To first order:

$$\log(1 + \lambda B(x)) \approx \lambda B(x).$$

Therefore:

$$\Phi_C(x) \approx -c_{\mathrm{eff}}^2 \lambda B(x).$$

Similarly:

$$\alpha(x) = \frac{1}{1 + \lambda B(x)} \approx 1 - \lambda B(x).$$

> **In the weak-burden regime, the temporal potential is approximately proportional to negative burden: $\Phi_C(x) \approx -c_{\mathrm{eff}}^2 \lambda B(x)$.**

This gives a simple linear bridge between burden and clock potential in the weak regime.

3.4.9 Redshift-Like Ratio

Consider two regions R_1 and R_2 with clock factors α_1 and α_2 . The ratio of clock rates is:

$$\frac{\alpha_1}{\alpha_2} = \frac{1 + \lambda B_2}{1 + \lambda B_1}.$$

If $B_1 > B_2$, then:

$$\frac{\alpha_1}{\alpha_2} < 1.$$

Thus, the high-burden region has a lower clock rate relative to the low-burden region.

One may define a redshift-like quantity:

$$1 + z_{12} = \frac{\alpha_2}{\alpha_1} = \frac{1 + \lambda B_1}{1 + \lambda B_2}.$$

Then $z_{12} > 0$ when $B_1 > B_2$.

> **Signals from higher-burden regions appear slower relative to lower-burden references.**

This remains an analogy until signal propagation and observational comparison are defined. It is safer to say "redshift-like" rather than "redshift" at this stage.

3.4.10 Guardrails for This Subsection

1. **Non-uniform burden gives lapse-like temporal geometry, not full curvature by itself.** Full curvature requires a full metric structure.
2. **The burden potential is not yet the Newtonian potential.** Φ_C is an effective burden-clock potential. Identification with Newtonian gravity requires additional calibration and weak-field matching.
3. **Redshift-like ratios require signal interpretation.** The ratio α_1/α_2 gives a clock-rate comparison. Calling this a redshift requires signal propagation, frequencies, transmission, and observation.
4. **Continuum gradients are effective.** Expressions such as $\nabla B(x)$ and $\nabla \alpha(x)$ assume continuum differentiability. They are not available at the most primitive relational level. The foundational statement is finite and order-based: $B(x_1) > B(x_2) \rightarrow \alpha(x_1) < \alpha(x_2)$.

3.5 Scalar Burden-to-Lapse Bridge

3.5.1 Purpose of This Subsection

The previous subsections established that non-uniform burden produces non-uniform clock rates and an effective potential. The present subsection formalises the connection between the burden field and an effective **"lapse function"**, which is the scalar temporal component of an effective metric.

> **Burden becomes temporal geometry through a lapse-like scalar field.**

This is still not the full burden-to-metric dictionary. It concerns only the scalar time-rate part of the effective geometry. Spatial metric response, shift terms, burden flux, burden stress, and tensorial source structure belong to later sections.

3.5.2 From Clock Factor to Lapse

In ordinary geometric language, a lapse function controls the relation between a coordinate-like ordering parameter and local proper time. The present framework does not assume such a coordinate time at the foundation. Nevertheless, once causal order and extension order have been introduced, one may define a relational ordering parameter σ that labels causal depth or admissible extension increments.

This parameter is not physical time by itself. It is an order label.

The local proper-time increment is then defined by:

$$\boxed{d\tau = \alpha(x) \, d\sigma.}$$

Here, $\alpha(x)$ acts as a **lapse-like conversion factor**. It tells us how much local proper time accumulates per unit of relational order.

Using the burden-clock relation:

$$\boxed{\alpha(x) = \frac{1}{1 + \lambda B(x)}.}$$

Thus:

$$\boxed{d\tau = \frac{d\sigma}{1 + \lambda B(x)}.}$$

This equation is the **scalar burden-to-lapse bridge** in its simplest form.

> **Local burden suppresses the conversion of causal or extension order into proper time.**

3.5.3 Definition — Burden Lapse

Definition 3.8 (Burden Lapse).

Let $B(x) \geq 0$ be a scalar burden field over an emergent correlation-geometric region, where $x \in \mathcal{X}_\Omega$. Let $\lambda > 0$. The **burden lapse** is the scalar field:

$$\boxed{\alpha_B(x) = \frac{1}{1 + \lambda B(x)}.}$$

When no ambiguity is likely, we write $\alpha(x)$ instead of $\alpha_B(x)$.

The burden lapse satisfies:

$$0 < \alpha_B(x) \leq 1.$$

It equals one precisely where burden vanishes:

$$\alpha_B(x) = 1 \iff B(x) = 0.$$

It decreases monotonically as burden increases.

3.5.4 Effective Temporal Line Element

In a continuum approximation, one may encode the lapse relation in a temporal line element.

Let $d\sigma$ represent an infinitesimal ordering increment, and let c_{eff} be an effective conversion speed or signal scale. Then the time-like part of an effective line element may be written as:

$$\boxed{ds^2_{\mathrm{time}} = -\alpha_B(x)^2 \, , \, c_{\mathrm{eff}}^2 \, , \, d\sigma^2.}$$

Substituting the burden lapse:

$$\boxed{ds^2_{\mathrm{time}} = -\frac{c_{\mathrm{eff}}^2}{(1 + \lambda B(x))^2} \, , \, d\sigma^2.}$$

This expression should be interpreted carefully. It is not yet a complete spacetime metric. It contains only the lapse-like scalar temporal factor. A full metric would require spatial components, possible shift terms, compatibility with causal structure, and continuum regularity.

> **The scalar burden lapse supplies the time-time part of an effective metric ansatz.**

3.5.5 Relation to Proper Time Along Chains

The scalar lapse field connects directly to burden-weighted proper time.

For a causal chain passing through locations $x_0, x_1, \dots, x_n$, the proper time is:

$$\tau[\gamma] = \sum_{k=0}^n \epsilon \, , \, \alpha_B(x_k).$$

Using the lapse formula:

$$\tau[\gamma] = \sum_{k=0}^n \frac{\epsilon}{1 + \lambda B(x_k)}.$$

Thus, the scalar lapse is the local factor appearing in the chain-sum definition of proper time.

In the continuum approximation:

$$\tau[\gamma] \approx \int_{\gamma} \alpha_B(x) \, , \, d\sigma = \int_{\gamma} \frac{d\sigma}{1 + \lambda B(x)}.$$

So the burden lapse is not an additional object unrelated to the previous subsection. It is the local field version of the same clock-suppression rule.

3.5.6 The Full Effective Metric Ansatz

Combining the temporal lapse with the spatial correlation metric from Section 2, one may write a first weak-field effective metric ansatz:

Definition 3.9 (Effective Burden Metric Ansatz).

$$\boxed{ds^2_{\mathrm{eff}} = -\alpha_B(x)^2 \, , \, c_{\mathrm{eff}}^2 \, , \, d\sigma^2 + h_{ij}(x) \, , \, dx^i \, dx^j,}$$

where:

- $\alpha_B(x) = 1/(1 + \lambda B(x))$ is the burden lapse,
- $h_{ij}(x)$ is the emergent spatial metric from correlation geometry,

- $d\sigma$ is the causal-depth or extension-order parameter.

This is the first effective metric ansatz in the framework. It is not yet derived from a full field equation. It is a constitutive relation: burden density controls the temporal component, while correlation geometry controls the spatial component.

3.5.7 Guardrails for This Subsection

1. **The lapse is not primitive time.** $\alpha_B(x)$ is a conversion factor from relational order to proper time, not a background clock.
2. **Scalar lapse is not full metric.** The expression $ds^2_{\mathrm{time}} = -\alpha_B^2 c_{\mathrm{eff}}^2 d\sigma^2$ contains only the time-like scalar part. A full metric requires spatial components, shift terms, and curvature.
3. **The potential is not automatically Newtonian gravity.** $\Phi_C = c_{\mathrm{eff}}^2 \log \alpha_B$ is a burden-clock potential. Identification with Newtonian gravity requires weak-field matching, probe dynamics, and empirical calibration.
4. **Gradient equations are effective.** Expressions such as $\nabla \Phi_C$ and ∇B require continuum differentiability. The foundational statement is discrete and comparative.

3.6 Guardrails and Summary of Section 3

3.6.1 Guardrails for Section 3

1. **Time is still emergent.** Even after defining burden-weighted proper time, time is not a primitive background parameter. It is reconstructed along causal chains.
2. **Burden is not violation.** $B > 0 \not\Rightarrow \frac{M}{V} > 0$. A region may be exactly physical and still carry high burden.
3. **Clock suppression is not yet a full metric equation.** $\alpha(B) = 1/(1 + \lambda B)$ defines a scalar clock factor, not the full metric tensor.
4. **The burden-clock relation is a model choice unless derived.** The rational form is one explicit ansatz. The theorem-safe content is: any positive decreasing clock factor yields burden-induced clock suppression.
5. **Non-uniform burden gives lapse-like temporal geometry, not full curvature.**
6. **The burden potential is not yet the Newtonian potential.**
7. **Redshift-like ratios require signal interpretation.**
8. **Continuum gradients are effective, not foundational.**

3.6.2 Summary of Section 3

Section 3 reconstructed emergent time from causal order weighted by constraint burden.

1. **Constraint burden** $B(R)$ was defined as the local cost of maintaining zero-closure. It decomposes into mode, interaction, and relational channels: $\rho_B = \rho_{\mathrm{mode}} + \rho_{\mathrm{int}} + \rho_{\mathrm{rel}}$.
2. The **local clock factor** $\alpha(B) = 1/(1 + \lambda B)$ was introduced. It satisfies $0 < \alpha \leq 1$ and decreases monotonically with burden.

3. **Burden-weighted proper time** was defined: $\tau[\gamma] = \sum_{e \in \gamma} \epsilon \alpha(B(e))$.
4. The theorem **"burden suppresses clock rate"** was proven: $B_1 > B_2 \rightarrow \alpha(B_1) < \alpha(B_2)$.
5. **Non-uniform burden** was shown to induce non-uniform clock accumulation, producing temporal geometry.
6. An **effective potential** $\Phi_C = c_{\text{eff}}^2 \log \alpha$ was derived.
7. The **weak-burden limit** $\Phi_C \approx -c_{\text{eff}}^2 \lambda B$ was established.
8. The **scalar burden-to-lapse bridge** $d\tau = \alpha(x) d\sigma$ was formalised, yielding the effective metric ansatz $ds^2 = -\alpha_B^2 c_{\text{eff}}^2 d\sigma^2 + h_{ij} dx^i dx^j$.
9. The **Causal Reconciliation Principle** was stated: causal speed and time rate are limited by the reconciliation of relational structure with causal dependency.

The conceptual chain is:

$\boxed{\text{causal order} + \text{constraint burden} \quad \rightarrow \quad \text{burden-weighted proper time} \quad \rightarrow \quad \text{temporal geometry} \quad \rightarrow \quad \text{effective lapse}.}$

This completes the reconstruction of emergent temporal structure from causal depth and burden.

Section 4 — Fields, Particles, and Interactions

4.0 Purpose of Fields, Particles, and Interactions

4.0.1 Transition from Geometry to Matter

Section 2 reconstructed emergent spatial structure from the correlation kernel K_ω , proving that correlation distance becomes a pseudometric under multiplicative consistency, and that emergent points arise as equivalence classes of perfectly correlated observables. Section 3 reconstructed emergent time from causal depth weighted by constraint burden, deriving the burden-clock suppression theorem and the effective temporal geometry.

The framework now possesses:

1. A primitive relational algebra $\mathcal{A}$ with constraint ideal $\mathcal{I}_C$.
2. A physical sector defined by $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$.
3. An emergent spatial metric space $(\mathcal{X}_\omega, \tilde{d}_\omega)$.
4. An emergent temporal structure $\tau[\gamma] = \sum_{e \in \gamma} \epsilon / (1 + \lambda B(e))$.
5. An effective metric ansatz $ds^2 = -\alpha_B^2 c_{\text{eff}}^2 d\sigma^2 + h_{ij} dx^i dx^j$.

However, the framework does not yet contain fields, particles, or interactions. The emergent geometry is a relational substrate, but it is not yet populated by matter-like excitations.

The purpose of this section is to reconstruct these structures from the relational constraint framework.

> **Fields and particles are not primitive occupants of spacetime; they are stable zero-preserving modes of relational structure.**

In ordinary field theory, one begins with a spacetime manifold and defines fields on it. In this framework, that order is reversed. Emergent space is first reconstructed from correlation geometry. Emergent time is reconstructed from causal order and burden. Only after these structures are available can one define field-like and particle-like behaviour.

4.0.2 What This Section Introduces

Section 4 introduces the following:

1. **Fields** — stable, zero-preserving patterns of values over the emergent correlation geometry.
2. **Field modes** — eigenmodes of the correlation Laplacian that satisfy the field-sector zero condition.
3. **Particles** — localised, stable, identifiable, zero-preserving field modes.
4. **Particle identity** — persistence of a localised mode through admissible extension.
5. **Interactions** — non-additive constraint burden or constraint-response effects among modes.
6. **Interaction vertices** — minimal events where multiple stable modes jointly preserve zero.

These structures are still pre-geometric in the sense that they do not assume a background spacetime manifold. They describe how matter-like structure can emerge from the relational constraint framework.

4.0.3 Why Fields Come After Space and Direction

A field normally means something like $\phi(x)$, where x is a spacetime point. But in this theory, x is not primitive. We only obtained emergent points at Layer 5 as correlation-equivalence classes:

$$\mathcal{X}_\omega = \mathcal{A}_{\{\mathrm{loc}\}} / \sim_\omega.$$

So we cannot define fields before defining emergent points. Likewise, field variation requires directions. But direction is not primitive either. We only obtained emergent directions at the end of Section 2 as profile-difference equivalence classes.

Only now can we define:

- field value,
- field variation,
- field equations.

This ordering is important.

> ****Not: space first, then fields; but: relational correlations first, emergent space second, field modes third.****

4.0.4 Fields as Relational Mode Assignments

Once an emergent point set $\mathcal{X}_\omega$ exists, one may define a field-like object as an assignment of relational data to emergent locations or regions.

If $\mathcal{X}_\omega$ is the emergent point set, a scalar-like field may be represented schematically as:

$\phi: \mathcal{X}_\omega \rightarrow V$,
where V is a value space.

But this notation is effective. It should not be read as a primitive definition. The deeper interpretation is that ϕ summarises a family of relational modes supported across correlation-equivalence classes.

> ****A field is a stable assignment of zero-preserving mode data over emergent correlation geometry.****

This allows field-like behaviour to arise without assuming fields as fundamental background objects.

4.0.5 Particles as Stable Localised Modes

Particles are not introduced as primitive point objects. They arise as special stable field configurations with localisation, persistence, and identity.

A particle is a field excitation that is:

1. localised in correlation geometry,
2. stable under admissible extension,
3. zero-preserving,
4. identifiable across causal progression.

> ****A particle is a stable, localised, zero-preserving mode.****

This definition preserves the framework's hierarchy. The theory first defines physical admissibility, then zero-preserving observables, then causal dependency, then emergent space, then burden-weighted time, then field modes. Particle identity appears only after all of these structures are available.

4.0.6 Interactions as Non-Additive Burden

With particles defined, the next question is how they interact.

In this framework, interactions are not introduced as primitive forces. They arise when the joint constraint cost of multiple modes is not equal to the sum of their individual costs.

> **Interaction is non-additivity of constraint burden.**

If two modes ϕ and χ combine, and the burden of the combined configuration differs from the sum of separate burdens, then the modes interact.

This gives a general relational interpretation of interaction without importing a specific force law.

4.0.7 Structural Sequence of Section 4

The section proceeds in the following order:

1. **Fields** are defined as stable zero-preserving mode assignments over $\mathcal{X}_\omega$.
2. **Field modes** are constructed as eigenmodes of the correlation Laplacian.
3. **Particles** are defined as localised, stable, identifiable, zero-preserving field modes.
4. **Particle identity** is formalised through coherent tracking across causal antichains.
5. **Interactions** are defined as non-additive constraint burden.
6. **Interaction vertices** are introduced as multi-mode zero-closure events.

The conceptual sequence is:

$$\boxed{\text{correlation geometry}} \quad \longrightarrow \quad \text{field modes} \quad \longrightarrow \quad \text{localised stable modes} \quad \longrightarrow \quad \text{particles} \quad \longrightarrow \quad \text{interactions}.$$

4.1 Fields as Stable Zero-Preserving Modes

4.1.1 Purpose of This Subsection

The previous subsection introduced the role of fields, particles, and interactions within the framework. It argued that fields should not be treated as primitive entities defined on a pre-existing spacetime manifold. Instead, field-like structures must be reconstructed after the framework has already produced physical admissibility, causal dependency, emergent space, burden-weighted time, and local clock structure.

The purpose of this subsection is to define the first of these matter-like structures: field modes.

> **A field mode is a zero-preserving relational pattern distributed over emergent space.**

This subsection establishes how field-like behaviour can be described once the correlation geometry $(\mathcal{X}_\omega, D_\omega)$ has been constructed.

4.1.2 The Emergent Spatial Arena

From Section 2, localisable zero-preserving observables $A, B \in \mathcal{A}_{\mathrm{loc}}$ were equipped with a state-dependent correlation kernel $K_\omega(A, B)$. The corresponding logarithmic distance was:

$$d_\omega(A, B) = -\ell_0 \log K_\omega(A, B).$$

Under suitable assumptions, d_ω becomes a pseudometric. Quotienting by zero-distance equivalence gives an emergent point set:

$$\mathcal{X}_\omega = \mathcal{A}_{\mathrm{loc}} / \sim_\omega.$$

The induced metric on equivalence classes is:

$$D_\omega([A]_\omega, [B]_\omega) = d_\omega(A, B).$$

This metric space, or approximate metric space in the coarse-grained case, is the proto-spatial arena on which field modes may be described.

> **Field modes live on emergent correlation geometry, not on primitive background space.**

4.1.3 Local Mode Data

Let $x \in \mathcal{X}_\omega$ be an emergent point or coarse-grained region. A field mode assigns to x some local relational data.

In an effective notation, a scalar-like mode may be written as:

$$\phi(x) \in V,$$

where V is a value space.

But this notation is shorthand. The deeper meaning is that $\phi(x)$ represents mode data associated with the equivalence class of localisable observables corresponding to x .

If $x = [A]_\omega$, then $\phi(x)$ should be understood as relational data supported by, or associated with, the class $[A]_\omega$.

> **$\phi([A]_\omega)$ is effective notation for mode data attached to a correlation-equivalence class.**

The field notation is useful, but it should not obscure the relational origin of the construction.

4.1.4 Definition — Field on Emergent Points

Definition 4.1 (Field).

Let $(\mathcal{X}_\omega, D_\omega)$ be an emergent correlation-geometric space, exact or approximate. Let V be a value space representing possible local mode data. A **field** is an assignment:

$$\boxed{\phi: \mathcal{X}_\omega \rightarrow V}$$

or, more generally, an assignment of compatible local data to regions of $\mathcal{X}_\omega$, such that the associated relational operation or configuration is zero-preserving.

4.1.5 The Zero-Preservation Condition

A field is physical only if it preserves the master zero constraint. In represented Hilbert-space language, if the mode is represented by an operator or family of operators $\hat{\phi}(f)$, smeared over test data f , then physical admissibility requires:

$$\boxed{\hat{\phi}(f) \in \mathcal{H}_{\mathrm{phys}}^{\omega} \subseteq \mathcal{H}_{\mathrm{phys}}^{\omega}.}$$

Equivalently:

$$\hat{\phi}(f) \in \ker \hat{\mathcal{M}}_{\omega} \subseteq \ker \hat{\mathcal{M}}_{\omega}.$$

In effective classical language, if the field mode deforms a physical configuration x into x_{ϕ} , then:

$$\boxed{\mathcal{M}(x) = 0 \quad \Longleftrightarrow \quad \mathcal{M}(x_{\phi}) = 0.}$$

Thus, field modes are physical only if they preserve constraint-zero admissibility.

> **Field admissibility first; field dynamics second.**

This preserves the logic of the framework: constraint compatibility is prior to dynamics.

4.1.6 The Correlation Laplacian

To define field dynamics, we need an operator that measures spatial variation. The natural candidate is the **correlation Laplacian**, built from the emergent relational weights.

Definition 4.2 (Relational Weights).

For emergent points $x_i, x_j \in \mathcal{X}_{\omega}$, define relational weights:

$$\boxed{W_{ij} = \exp\left(-\frac{D_{\omega}(x_i, x_j)}{\ell_0}\right).}$$

Since $D_{\omega}(x_i, x_j) = -\ell_0 \log K_{\omega}(i, j)$, this is equivalently:

$$W_{ij} = K_{\omega}(i, j).$$

Definition 4.3 (Correlation Laplacian).

The **correlation Laplacian** is the operator:

$$\boxed{(\Delta_{\omega} \phi)(x_i) = \sum_j W_{ij} [\phi(x_j) - \phi(x_i)].}$$

This operator measures the deviation of $\phi(x_i)$ from its correlation-neighbourhood average. It generalises the usual graph Laplacian without presupposing a metric manifold.

4.1.7 Field Modes as Eigenmodes

A natural class of modes satisfies the eigenvalue relation:

$$\boxed{\Delta_{\omega} \phi_n = \lambda_n \phi_n.}$$

Here λ_n is the relational spectral value of the mode.

When combined with emergent time, field modes satisfy a relational wave equation of schematic form:

$$\partial_{\tau}^2 \phi + c_{\mathrm{eff}}^2 L_{\omega} \phi + m_{\mathrm{eff}}^2 \phi = 0,$$
 where $L_{\omega} = -\Delta_{\omega}$ is the positive correlation Laplacian.

4.1.8 Stable Field Modes

A field mode is **stable** if small perturbations remain bounded under emergent-time flow.

Definition 4.4 (Stable Field Mode).

A field mode ϕ_n is stable if there exists a finite constant $C > 0$ such that:

$$\boxed{|\phi_n(\tau)| \leq C |\phi_n(0)|}$$
 over the relevant emergent-time interval.

Stability means the mode does not blow up under evolution.

4.1.9 Guardrails for This Subsection

1. **Fields are not fundamental.** They are derived from the emergent correlation geometry.
2. **The correlation Laplacian is a toy operator.** It is the natural field operator on the emergent geometry, but it is not yet a derived continuum operator.
3. **$\Box_{\mathrm{RCF}} \phi = 0$ is a candidate equation.** It needs a continuum proof.
4. **Field support is emergent.** Support statements refer to emergent regions $R \subseteq \mathcal{X}_{\omega}$, not subsets of a primitive spacetime manifold.

4.2 Particles as Localised Stable Field Modes

4.2.1 Purpose of This Subsection

The previous subsection defined field modes as stable zero-preserving patterns over the emergent correlation geometry. A field mode alone is not yet a particle. To behave like a particle, a mode must satisfy additional conditions: localisation, stability, identity persistence, and zero-preservation.

The purpose of this subsection is to define when a mode becomes particle-like.

> **A particle is a stable, localised, identifiable, zero-preserving field mode.**

4.2.2 Why Particles Come After Fields

A particle usually seems like a little object moving through space. But in this framework:

- Space is emergent.
- Time is emergent.
- Fields are emergent.
- Therefore particles cannot be fundamental.

The correct order is:

\$\$ \text{\text{space}} \text{\text{fields}} \text{\text{stable modes}} \text{\text{localised stable modes}} \text{\text{particles}}. \$\$

A particle is not "a little object inside space." It is a persistent relational pattern that becomes object-like only after emergent space and time have been reconstructed.

4.2.3 Definition — Particle-Like Excitation

Definition 4.5 (Particle-Like Excitation).

Let $\delta\phi$ be a field excitation over a background configuration ϕ_0 . A

particle-like excitation is an excitation satisfying four conditions:

1. **Zero-Preservation:** $\delta\phi \in \Psi \cap T_\Psi \ker \hat{\mathcal{M}}$ (or $\mathcal{M}[\phi_0 + \delta\phi] \approx 0$).
2. **Localisation:** The excitation has finite spread in the correlation geometry.
3. **Stability:** The amplitude remains bounded over proper time.
4. **Identifiability:** The excitation can be coherently tracked across successive causal antichains.

4.2.4 The Localisation Functional

Definition 4.6 (Localisation Functional).

Let $\delta\phi_\Sigma$ be a field excitation on an antichain Σ . Let x_0 be a candidate localisation centre. Define:

$$\boxed{L[\delta\phi_\Sigma; x_0] = \frac{\sum_{x \in \Sigma} D_\omega(x, x_0)^2}{\sum_{x \in \Sigma} |\delta\phi_\Sigma(x)|^2}}.$$

The excitation is **localised** if there exists some x_0 such that $L[\delta\phi_\Sigma; x_0] < \infty$.

4.2.5 Particle Identity Across Antichains

Let $\Sigma_s, \Sigma_{s+1}, \dots, \Sigma_{s+k}$ be a sequence of causally compatible antichains. A localised excitation has **persistent identity** if there exists a sequence of excitations $\delta\phi_{\Sigma_s}, \delta\phi_{\Sigma_{s+1}}, \dots$ such that:

1. Each excitation is localised.
2. Each excitation is stable.
3. Their centres form a coherent track.
4. Internal mode data are preserved up to allowed phase/gauge equivalence.

4.2.6 Theorem — Particle Identity

Theorem 4.1 (Particle Identity).

Let $\delta\phi$ be a stable physical field mode on $(\mathcal{X}_\omega, D_\omega)$.

Suppose:

1. $\delta\phi$ is zero-preserving.

2. $\delta\phi$ is localised on each antichain Σ_t : $L[\delta\phi_{\Sigma_t}; X_t] < \infty$.
3. The localisation centres X_t form a coherent finite-speed track.
4. The internal mode class is preserved: $\delta\phi_{\Sigma_{t+1}} \sim_{\text{mode}} \delta\phi_{\Sigma_t}$.

Then $\delta\phi$ defines a particle-like excitation.

Proof.

Condition 1 ensures the excitation remains physical. Conditions 2 and 3 imply persistent localisation. Condition 4 ensures identifiability across emergent time. Together, these satisfy the definition of a particle-like excitation. $\blacksquare$

4.2.7 Interpretation

This theorem says that particles are not primitive points. They are:

$\boxed{\text{particle} = \text{trackable localised stable field mode}.}$

A particle trajectory is not a continuous line in background space. It is a discrete sequence of localised mode continuations through emergent correlation geometry.

> ***Particle identity is relational continuity, not primitive objecthood.***

4.3 Interactions as Non-Additive Constraint Burden

4.3.1 Purpose of This Subsection

With particles defined as stable localised modes, the next question is how they interact.

In this framework, interactions are not introduced as primitive forces. They arise when the joint constraint cost of multiple modes is not equal to the sum of their individual costs.

> ***Interaction is non-additivity of constraint burden.***

4.3.2 Definition — Interaction Functional

Definition 4.7 (Interaction Functional).

Let $\mathcal{M}[\phi]$ be a non-negative constraint cost functional. For two field modes ϕ and χ , define:

$\boxed{\mathcal{M}[\phi, \chi] := \mathcal{M}[\phi + \chi] - \mathcal{M}[\phi] - \mathcal{M}[\chi].}$

If $\mathcal{M}[\phi, \chi] = 0$, the modes are non-interacting. If $\mathcal{M}[\phi, \chi] \neq 0$, the modes interact.

4.3.3 Theorem — Interaction Non-Additivity

****Theorem 4.2 (Interaction Non-Additivity).****

$\mathcal{I}[\phi, \chi] = 0$ if and only if the joint constraint cost is additive. $\mathcal{I}[\phi, \chi] \neq 0$ if and only if the joint configuration carries additional or reduced mutual constraint cost.

Proof.

By definition, $\mathcal{I}[\phi, \chi] = \mathcal{M}[\phi + \chi] - \mathcal{M}[\phi] - \mathcal{M}[\chi]$. If $\mathcal{I} = 0$, then $\mathcal{M}[\phi + \chi] = \mathcal{M}[\phi] + \mathcal{M}[\chi]$, so the cost is additive. If $\mathcal{I} \neq 0$, then $\mathcal{M}[\phi + \chi] \neq \mathcal{M}[\phi] + \mathcal{M}[\chi]$, so the joint configuration carries a mutual contribution not present in either field alone. $\blacksquare$

4.3.4 Sign of Interaction

- If $\mathcal{I}[\phi, \chi] > 0$, the joint configuration carries ****additional**** reconciliation burden. This may correspond to repulsion or incompatibility.
- If $\mathcal{I}[\phi, \chi] < 0$, the joint configuration carries ****reduced**** burden. This may correspond to binding or constructive compatibility.

> ****Binding is burden reduction; repulsion is burden increase.****

4.3.5 Many-Body Generalisation

For n modes $\phi_1, \dots, \phi_n$:

$$\boxed{\mathcal{I}_n[\phi_1, \dots, \phi_n] := \mathcal{M}[\sum_{a=1}^n \phi_a] - \sum_{a=1}^n \mathcal{M}[\phi_a].}$$

If $\mathcal{I}_n \neq 0$, the collection has many-body interaction.

4.3.6 Locality of Interactions

Interactions should be local relative to the emergent correlation geometry. Let $W_{ij} = \exp(-D_{\omega(i,j)} / \ell_0)$. A local interaction cost may be written:

$$\boxed{\mathcal{M}_{\{\text{int}\}}[\phi, \chi] = g^2 \sum_{i,j} W_{ij} |\phi_i|^2 |\chi_j|^2.}$$

This formalises the statement: interaction strength is controlled by proximity in correlation geometry.

> ****Interaction strength decays with relational distance.****

4.3.7 Interaction Vertices

An ****interaction vertex**** is a minimal event where multiple stable modes jointly preserve zero. It is represented abstractly by a multi-mode compatibility condition $\mathcal{C}_{\{\text{vertex}\}}(\phi_1, \phi_2, \dots) = 0$.

An interaction event is physical precisely when the multi-mode compatibility constraint vanishes.

> **An interaction vertex is a multi-mode zero-closure event.**

4.3.8 Guardrails for This Subsection

1. **Interaction is not automatically force.** Non-additive burden is an interaction source, not yet a force equation.
2. **Non-additivity must remain admissible.** Physical interaction means non-additivity inside the zero sector.
3. **Negative burden interaction is not automatically attraction.** Binding-like behaviour requires an added stability or dynamics criterion.
4. **Positive burden interaction is not automatically repulsion.** Repulsion-like behaviour requires an added motion or selection rule.
5. **Interaction locality depends on emergent geometry.** If the geometry is approximate or state-dependent, interaction locality is also approximate or state-dependent.

4.4 Guardrails and Summary of Section 4

4.4.1 Guardrails for Section 4

1. **Fields are not fundamental.** They are derived from the emergent correlation geometry.
2. **Particles are not primitive objects.** They are stable, localised field modes.
3. **Interactions do not require a background force field.** They arise from constraint nonlinearity and burden redistribution.
4. **No spacetime manifold is assumed at the outset.** Fields live on $\mathcal{X}_\omega$, the emergent point space.
5. **A mode is not automatically a particle.** Particle-like modes are a special class of stable localised modes.
6. **Particle identity is relational, not absolute.** It is structural continuity, not primitive individuality.

4.4.2 Summary of Section 4

Section 4 established the matter layer of the framework:

1. **Fields** were defined as stable zero-preserving mode assignments $\phi: \mathcal{X}_\omega \rightarrow V$.
2. The **correlation Laplacian** Δ_ω was introduced as the natural field operator.
3. **Stable field modes** were defined as bounded eigenmodes satisfying $\Delta_\omega \phi_n = \lambda_n \phi_n$.
4. **Particles** were defined as localised, stable, identifiable, zero-preserving field modes.
5. **Particle identity** was formalised through coherent tracking across causal antichains.

6. **Interactions** were defined as non-additive constraint burden: $\mathcal{I}[\phi, \chi] = \mathcal{M}[\phi + \chi] - \mathcal{M}[\phi] - \mathcal{M}[\chi]$.
7. **Interaction vertices** were introduced as multi-mode zero-closure events.

The conceptual chain is:

$\boxed{\text{correlation geometry} \rightarrow \text{field modes} \rightarrow \text{localised stable modes} \rightarrow \text{particles} \rightarrow \text{interactions}.}$

This gives the framework a matter layer built entirely from relational structure, with no background spacetime imported.

Section 5 — Gravity as Geometry of Constraint Burden

5.0 Purpose of the Burden Tensor and Gravity Source Structure

5.0.1 Transition from Matter to Gravitational Source

Section 4 reconstructed fields and particles as stable zero-preserving modes on the emergent correlation geometry. Interactions were defined as the non-additive burden generated when these modes compose.

However, the burden-clock bridge established in Section 3 only utilized a *scalar* burden density $\rho_B(x)$ to suppress the local clock rate (the lapse function α_B). While this provided the first temporal geometry, a scalar field alone is insufficient to generate a full, dynamically consistent gravitational theory. General relativity requires a tensorial source—the stress-energy tensor $T_{\mu\nu}$ —to source spatial curvature, frame-dragging, and anisotropic stresses.

The purpose of this section is to promote scalar constraint burden into a tensorial effective source, map it to an effective spacetime metric, and establish the conditions under which the framework recovers an Einstein-like field equation.

> **Gravity is not introduced as a fundamental force. It emerges when variations in constraint burden reshape the emergent time and space.**

5.0.2 The Causal Reconciliation Principle and Gravity

The underlying physical mechanism for this geometric response is the **Causal Reconciliation Principle** (Section 1.6). The rate of time and the speed of causal propagation are bounded by how quickly relational structures can reconcile with causal dependency.

When a region contains a high concentration of stable modes (matter) or complex interactions, the local reconciliation cost (burden) increases. This slows the local clock rate (temporal curvature) and alters the relational correlation distances (spatial curvature).

Gravity, in this framework, is the macroscopic geometric response of the emergent spacetime to the microscopic distribution of constraint-reconciliation load.

5.0.3 What This Section Establishes

This section establishes the following:

1. The **component-selective burden tensor** $\Theta^{(B)}_{\mu\nu}$ is constructed from mode, interaction, and relational sources.
2. The **active-source conservation theorem** proves that the total active burden must be covariantly divergence-free.
3. The **burden-to-metric dictionary** maps the burden tensor components to an effective ADM-like metric.
4. The **Einstein-like closure theorem** proves that, under smoothness and Lorentz-compatibility assumptions, the minimal geometric response to a conserved burden source takes the form of Einstein's field equations.

5.1 Component-Selective Burden Sources

5.1.1 From Scalar Density to Tensorial Source

In Section 3, the burden density was treated as a scalar $\rho_B(x)$ that dictated the lapse function $\alpha_B(x)$. To source a full spacetime geometry, we must distinguish not only how much burden is present, but how it flows and how it exerts directional stress on the relational network.

Definition 5.1 (Burden Tensor).

Let the coarse-grained emergent domain admit local coordinates $x^\mu = (\tau, x^1, \dots, x^d)$. The **burden tensor** is the effective source object:

$$\boxed{\Theta^{(B)}_{\mu\nu}} = \begin{pmatrix} \rho_B & J^{(B)}_j \\ \Pi^{(B)}_{ij} \end{pmatrix},$$

where:

- $\Theta^{(B)}_{00} = \rho_B$ is the burden density,
- $\Theta^{(B)}_{0i} = \Theta^{(B)}_{i0} = J^{(B)}_i$ is the burden flux,
- $\Theta^{(B)}_{ij} = \Pi^{(B)}_{ij}$ is the relational burden stress.

5.1.2 The Three Source Channels

The framework refines the burden tensor from a single homogeneous block into a source-origin decomposition. The total burden density decomposes as:

$$\rho_B(x) = \rho_{\text{mode}}(x) + \rho_{\text{int}}(x) + \rho_{\text{rel}}(x).$$

Consequently, the full burden tensor decomposes into three distinct structural channels:

$$\boxed{\Theta^{(B)}_{\mu\nu}} = a_{\text{mode}} \Theta^{(\text{mode})}_{\mu\nu} + a_{\text{int}} \Theta^{(\text{int})}_{\mu\nu} + a_{\text{rel}} \Theta^{(\text{rel})}_{\mu\nu},$$

where a_{mode} , a_{int} , $a_{\mathrm{rel}} \in [0, 1]$ are activation weights.

Interpretation of the Channels:

1. **Mode burden (ρ_{mode}):** Generated by the existence and occupation of stable field or particle-like modes. This is the analogue of ordinary visible matter.
2. **Interaction burden (ρ_{int}):** Arises from non-additivity in composition (binding energy, structural complexity). A bound system carries interaction burden given by $B_{\mathrm{int}}(R_1, R_2) = B(R_1 \cup R_2) - B(R_1) - B(R_2)$.
3. **Relational burden (ρ_{rel}):** Generated by the correlation network itself. It measures the burden required to maintain relational consistency across configurations. Unlike the first two, it does not belong to a single object, but to the network topology.

> **RCF burden sourcing is component-selective: mode, interaction, and relational burden may contribute separately or together, while the total active source must remain conserved.**

5.1.3 Symmetry of the Burden Tensor

Theorem 5.1 (Symmetry of the Burden Tensor).

If the relational burden stress is symmetric, $\Pi_{ij}^{(B)} = \Pi_{ji}^{(B)}$, and the burden flux is represented symmetrically, $\Theta_{0i}^{(B)} = \Theta_{i0}^{(B)} = J_i^{(B)}$, then the burden tensor is symmetric:

$$\boxed{\Theta_{\mu\nu}^{(B)} = \Theta_{\nu\mu}^{(B)}} \quad \square$$

Proof. Trivial from the definition: $\Theta_{00}^{(B)} = \rho_B$ is trivially symmetric; $\Theta_{0i}^{(B)} = \Theta_{i0}^{(B)}$ by definition; and $\Theta_{ij}^{(B)} = \Pi_{ij}^{(B)} = \Pi_{ji}^{(B)} = \Theta_{ji}^{(B)}$ by assumption. $\square$

5.2 Active-Source Conservation

5.2.1 Purpose of Conservation

For the burden tensor to act as a valid source for an effective geometric field equation, it must satisfy a conservation law. In standard General Relativity, the Bianchi identities force the Einstein tensor to be divergence-free, which in turn forces the stress-energy tensor to be divergence-free ($\nabla_\mu T^{\mu\nu} = 0$).

The RCF must satisfy an analogous condition: the total active burden source must be covariantly conserved.

5.2.2 Theorem — Active-Source Conservation

Theorem 5.2 (Active-Source Conservation).

Let the total active burden source be:

$$\Theta^{(B)}_{\mu\nu} = a_{\mathrm{mode}} \Theta^{(\mathrm{mode})}_{\mu\nu} + a_{\mathrm{int}} \Theta^{(\mathrm{int})}_{\mu\nu} + a_{\mathrm{rel}} \Theta^{(\mathrm{rel})}_{\mu\nu}.$$

If the emergent geometry is smooth and Lorentz-compatible, and if the geometric side of the field equation is divergence-free by the Bianchi identity, then the total active source must satisfy:

$$\boxed{\nabla_B{}^\mu \Theta^{(B)}_{\mu\nu} = 0.}$$

Proof.

Assume the effective metric g_B satisfies a candidate geometric equation of the form $G_{\mu\nu}[g_B] + \Lambda_B g^{(B)}_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$.

By the contracted Bianchi identities of standard differential geometry, the Einstein tensor satisfies $\nabla_B{}^\mu G_{\mu\nu} = 0$. By metric compatibility, $\nabla_B{}^\mu g^{(B)}_{\mu\nu} = 0$.

Taking the covariant divergence of the field equation yields:

$$\nabla_B{}^\mu \left(G_{\mu\nu} + \Lambda_B g^{(B)}_{\mu\nu} \right) = \nabla_B{}^\mu \left(\kappa_B \Theta^{(B)}_{\mu\nu} \right).$$

Since the left-hand side vanishes identically, and κ_B is constant, we obtain:

$$0 = \kappa_B \nabla_B{}^\mu \Theta^{(B)}_{\mu\nu}.$$

Assuming $\kappa_B \neq 0$, it follows that $\nabla_B{}^\mu \Theta^{(B)}_{\mu\nu} = 0$. $\blacksquare$

5.2.3 Component Exchange

The conservation law applies to the *total* active source. It does not require each component to be individually conserved. The framework allows for exchange between channels:

$$\nabla_B{}^\mu \Theta^{(\mathrm{int})}_{\mu\nu} = - \nabla_B{}^\mu \Theta^{(\mathrm{mode})}_{\mu\nu} - \nabla_B{}^\mu \Theta^{(\mathrm{rel})}_{\mu\nu}.$$

This means interaction burden can be converted into mode burden (e.g., binding energy releasing particles) or absorbed by relational network adjustments, provided the total sum closes.

> **Zero-preserving geometry structurally demands zero-preserving burden flow.**

5.3 Burden Tensor to Effective ADM Metric Mapping

5.3.1 The ADM Dictionary

To explicitly connect the burden tensor to spacetime geometry, we use the Arnowitt-Deser-Misner (ADM) decomposition. The ADM formalism foliates spacetime into spatial slices and uses the spatial metric, lapse, and shift to encode the geometry.

In RCF, ADM is not primitive; it is a continuum bookkeeping layer applied after the emergent geometry has stabilized.

Definition 5.2 (Burden-Metric Dictionary).

Let ρ_B , J_B^i , and Π_B^{ij} be the projections of $\Theta^{(B)}_{\mu\nu}$. Define the burden lapse, shift, and spatial metric as:

- Density to Lapse:** $\rho_B \mapsto N_B$, where $N_B = \frac{1}{1 + \lambda \rho_B}$.
- Flux to Shift:** $J_B^i \mapsto N_B^i$, where $N_B^i = \sigma J_B^i$.
- Stress to Spatial Metric:** $\Pi_B^{ij} \mapsto h^{(B)}_{ij}$, where $h^{(B)}_{ij} = h^{(0)}_{ij} + \eta \Pi_B^{ij} + \zeta \rho_B h^{(0)}_{ij}$.

Here, σ , η , ζ are model-dependent effective coupling coefficients, and $h^{(0)}_{ij}$ is the baseline correlation metric from Section 2.

5.3.2 Theorem — Effective Metric Ansatz

Theorem 5.3 (Effective ADM Metric).

Given the burden-metric dictionary, the effective ADM-like line element is:

$$ds_B^2 = -N_B^2 d\tau^2 + h^{(B)}_{ij} \left(dx^i + N_B^i d\tau \right) \left(dx^j + N_B^j d\tau \right).$$

Proof. Expanding the ADM line element $ds^2 = -N^2 d\tau^2 + h_{ij}(dx^i + N^i d\tau)(dx^j + N^j d\tau)$ and substituting the burden definitions for N_B , N_B^i , and $h^{(B)}_{ij}$ yields the effective burden metric. $\square$

5.3.3 Component Mapping Summary

RCF Burden Object	Effective Metric Role	Interpretation
ρ_B	$N_B \mapsto g_{00}$	Clock suppression / temporal curvature
J_B^i	$N_B^i \mapsto g_{0i}$	Burden flow / frame-dragging shift
Π_B^{ij}	$h^{(B)}_{ij} \mapsto g_{ij}$	Anisotropic relational stress / spatial curvature

This dictionary establishes that scalar burden density controls the rate of time, burden flux controls spatial displacement between slices, and burden stress controls the spatial metric deformation.

5.4 Einstein-Like Burden Geometry Consistency

5.4.1 The Closure Target

With a symmetric, conserved burden tensor $\Theta^{(B)}_{\mu\nu}$ and a smooth, Lorentz-compatible effective metric g_B , the framework is positioned to state its gravitational field equation.

Theorem 5.4 (Einstein-Like Closure, Conditional Uniqueness Form).

Assume the emergent geometry is smooth, Lorentz-compatible, and governed by a symmetric conserved source $\Theta^{(B)}_{\mu\nu}$. Then the minimal geometric field equation must take the form:

$$\boxed{G_{\mu\nu}[g_B] + \Lambda_B g^{(B)}_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}},$$

where $G_{\mu\nu}$ is the Einstein tensor, Λ_B is an effective cosmological constant, and κ_B is an effective coupling constant.

Proof Sketch.

In a smooth 4-dimensional metric geometry, the left-hand side of the field equation must be a symmetric rank-2 tensor built from the metric and its derivatives, and it must be divergence-free.

By Lovelock's theorem, in four dimensions, the only symmetric, divergence-free, natural $(0,2)$ -tensors that are dependent on the metric and its first two derivatives are linear combinations of the Einstein tensor $G_{\mu\nu}$ and the metric tensor $g_{\mu\nu}$.

Therefore, once conservation ($\nabla_B^\mu \Theta^{(B)}_{\mu\nu} = 0$) and smoothness are fixed, the Einstein-like equation is the unique natural closure of the geometric side of the theory. $\blacksquare$

5.4.2 Status of the Equation

It is critical to maintain theorem-safe status language regarding this equation.

1. ***The equation is a conditional reduction theorem.*** It states that *if* the framework recovers a smooth, second-order metric theory in the continuum limit, that theory must be Einstein-like.
2. ***It is not yet a dynamical derivation from the master constraint.*** The full derivation of κ_B and the exact mapping of the algebraic master constraint to the differential Einstein tensor remains an open target.
3. ***Lovelock compatibility protects the left-hand side.*** The framework does not need to arbitrarily postulate the Einstein tensor; it is forced by the uniqueness theorem under the stated assumptions.

> ***Gravity is the equation of state of zero-reconciliation.***

5.5 Guardrails and Summary of Section 5

5.5.1 Guardrails for Section 5

1. ***Gravity is not a fundamental force.*** It is the macroscopic geometric response to constraint burden.
2. ***The burden tensor is not automatically stress-energy.*** $\Theta^{(B)}_{\mu\nu}$ is stress-energy-like, but identification with physical $T_{\mu\nu}$ requires proof that matter excitations generate proportional constraint burden.

3. **The ADM mapping is an ansatz.** The constitutive relation mapping burden to lapse/shift/metric is a model-dependent dictionary, not a primitive law.
4. **Einstein-like closure is an open target.** The equation $G_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$ is a conditional reduction theorem, not a completed derivation from the algebra.
5. **Component activation is model-dependent.** The weights a_{mode} , a_{int} , a_{rel} must be derived or specified in a concrete model.

5.5.2 Summary of Section 5

Section 5 established the gravitational layer of the framework:

1. The **burden tensor** $\Theta^{(B)}_{\mu\nu}$ was defined, decomposing into mode, interaction, and relational channels.
2. The **active-source conservation theorem** proved that the total active burden must be covariantly divergence-free.
3. The **burden-to-metric dictionary** mapped burden density to lapse, flux to shift, and stress to spatial metric, yielding an effective ADM metric.
4. The **Einstein-like closure theorem** proved that, under smoothness and Lorentz-compatibility, the minimal geometric response to conserved burden takes the form of Einstein's field equations.

The conceptual chain is:

$$\boxed{\text{constraint burden}} \quad \longrightarrow \quad \text{burden tensor} \quad \longrightarrow \quad \text{effective metric} \quad \longrightarrow \quad \text{Einstein-like closure}.$$

This completes the reconstruction of gravitational geometry from constraint burden.

Section 6 — Black Holes as Unreconciled Relational Sectors

6.0 Purpose of Black-Hole-Like Domains in RCF

6.0.1 Reframing the Black Hole

In standard general relativity, a black hole is defined as a region of spacetime from which nothing, not even light, can escape to infinity. It is bounded by an event horizon, and its interior typically contains a curvature singularity where the continuum description of spacetime breaks down.

The Relational Constraint Framework does not begin with a pre-existing spacetime manifold, nor does it assume a background metric. Therefore, it cannot define a black hole primitive as a causal boundary in a pre-given geometry, nor can it accept a literal infinite-density singularity as a physical object, since points and metrics are emergent.

Instead, the framework must reconstruct black-hole-like behaviour from the fundamental dynamics of the constraint algebra, burden, and causal dependency.

The purpose of this section is to demonstrate that extreme concentrations of constraint burden naturally produce domains where relational structures cannot reconcile with causal dependency. In these domains, causal propagation is suppressed, local time flattens, and exterior accessibility collapses.

> **A black hole is not a hole in spacetime, nor a literal singularity. It is a strongly burden-loaded, unreconciled relational sector where exterior reconstruction fails.**

6.0.2 What This Section Establishes

This section establishes the following:

1. **The Unreconciled Region Principle**: A black-hole-like region is defined by the failure of relational structures to reconcile with causal dependency, leading to extreme flux suppression.
2. **Projection as Flux Suppression**: The operational observable shadow of the region is a projection of the full relational structure, attenuated by the burden lapse factor.
3. **Boundary Recovery**: The theorem stating that if the boundary record map is injective, the lowest-burden admissible sector of the interior is recoverable.
4. **Entropy-Area Scaling**: The entropy of the region is a coarse-grained count of admissible boundary records, scaling with effective boundary area.
5. **Lowest-Burden Emission**: Relaxation occurs by emitting the simplest admissible carrier sector, not the original infallen matter class.
6. **Hawking-like Appearance**: Under coarse-grained observation, lowest-burden emission appears approximately thermal.

6.1 The Unreconciled Region Principle

6.1.1 Failure of Reconciliation

In Section 1, we established the **Causal Reconciliation Principle**: the effective speed of causal propagation and the rate of proper time are bounded by the degree to which relational structures can reconcile with causal dependency. When reconciliation is efficient, causal propagation is fast and time advances normally. When burden increases, reconciliation slows, the causal ceiling tightens, and proper time is suppressed.

A black-hole-like domain represents the extreme limit of this process. It is a sector where the local burden $B_{\mathrm{active}}(R)$ becomes so large that the reconciliation rate approaches zero.

Definition 6.1 (Unreconciled Region).

Let R be a coarse-grained region of the emergent causal-relational structure. R is an **unreconciled region** if its active burden loading $B_{\mathrm{active}}(R)$ is sufficiently

extreme that relational structures within R cannot reconcile with the causal dependency structure at a rate compatible with exterior propagation.

In this regime:

$$\alpha_R = \frac{1}{1 + \lambda B_{\mathrm{active}}(R)} \ll 1.$$

6.1.2 Black-Hole Relaxation as Lagging Re-equilibration

Because reconciliation fails, the region cannot evolve synchronously with the surrounding open extension. The relaxation or "evaporation" of this region is therefore not a synchronized decay process.

Lemma 6.1 (Lagging Reconciliation).

A black-hole-like region is a lagging relational sector in which relational structures cannot reconcile with causal dependency. This incompatibility suppresses the effective causal speed, flattens the local temporal rate, and forces relaxation to proceed through highly suppressed flux channels.

> **Where reconciliation with causal dependency fails, causal speed and proper time are suppressed.**

The observable slow relaxation arises because the boundary-projected sector re-equilibrates only through highly restricted, weakly accessible flux channels, rather than through direct, unsuppressed causal interaction.

6.1.3 The Relational Relaxation Principle

To make the relaxation rate precise, we introduce the Relational Relaxation Principle.

Theorem 6.1 (Relational Relaxation Principle).

Let R be a strongly burden-loaded unreconciled region. The characteristic relaxation time τ_R satisfies:

$$\tau_R \propto \frac{B_{\mathrm{active}}(R)}{\Gamma_{\mathrm{rel}}(R)}$$

where $\Gamma_{\mathrm{rel}}(R)$ measures the admissible relational coupling between the region and its exterior.

Consequently:

$$\Gamma_{\mathrm{rel}}(R) \rightarrow 0 \quad \rightarrow \quad \tau_R \rightarrow \infty,$$

$$\Gamma_{\mathrm{rel}}(R) > 0 \quad \rightarrow \quad \tau_R < \infty.$$

Proof Sketch.

The relaxation rate is directly proportional to the external coupling strength Γ_{rel} and inversely proportional to the structural inertia (burden) B_{active} that must be unloaded. If the external coupling vanishes, the region is perfectly isolated and cannot relax. If the coupling is non-zero, relaxation is inevitable but scales with the burden load. $\square$

Interpretation.

The black-hole-like region does not evaporate simply because it "contains energy." It relaxes because it remains weakly connected to the surrounding relational network through Γ_{rel} . The emitted sector is not the original matter class, but the lowest-burden excitation sector compatible with the boundary projection.

6.2 Projection as Flux Suppression

6.2.1 The Projection Mechanism

To understand what an exterior observer can reconstruct, we must formalize how the interior is hidden. The framework achieves this through the **Projection-Flux Principle**.

When relational burden becomes extreme, the operational description becomes a projection of the full relational structure. The underlying degrees of freedom are not destroyed, but their visibility is reduced by burden loading and boundary filtering.

Definition 6.2 (Projection as Flux Suppression).

Let Φ denote the admissible relational flux through a region R , and let $\Pi_{\partial R}$ denote projection onto the operationally accessible boundary sector. Under extreme burden loading, the observable flux is:

$$\boxed{\mathcal{P}_{\partial R}(\Phi) = \Pi_{\partial R} \left[\alpha_R \mathcal{C}_{\mathrm{cg}}(\Phi) \right]}$$

where $\alpha_R = \frac{1}{1 + \lambda B_{\mathrm{active}}(R)}$ is the lapse suppression factor, and $\mathcal{C}_{\mathrm{cg}}$ is the coarse-graining map defined by the approximate metric.

6.2.2 The Dual Horizon Ratios

The strength of the boundary projection is governed by the relational isolation of the region. The framework utilizes a dual-ratio structure to classify the regime.

Definition 6.3 (Dual Horizon Ratios).

Let $W_{\mathrm{in}}(R)$ be the total internal relational cohesion, and $W_{\mathrm{out}}(R)$ be the total exterior relational coupling. Define:

- Internal Isolation Ratio:** $\chi_R = \frac{W_{\mathrm{in}}}{W_{\mathrm{out}}}$
- Exterior Accessibility Ratio:** $\varepsilon_R = \frac{W_{\mathrm{out}}}{W_{\mathrm{in}}}$

Theorem 6.2 (Dual Horizon Limits).

The relational horizon limit is characterized by:

$$\varepsilon_R \rightarrow 0 \quad \Longleftrightarrow \quad \chi_R \rightarrow \infty.$$

In this regime, exterior accessibility fails. However, this failure of reconstruction does not imply internal causal annihilation:

$$\boxed{\varepsilon_R \rightarrow 0 \quad \not\Rightarrow \quad \prec_R = \varnothing.}$$

Proof.

The causal dependency order $\prec_R$ is an internal structural property of the event set $\mathcal{E}_R$, generated by constraint closure inclusion ($\Gamma(e_i) \subseteq \Gamma(e_j)$). The exterior accessibility ratio ε_R depends on the state-dependent correlation kernel K_ω evaluated across the boundary. By the Two-Link Principle (Section 1.4), causal links and relational correlation links are logically independent. Suppressing the correlation link (exterior access) does not logically entail the destruction of the internal causal links. $\blacksquare$

> **Black-hole-like isolation is best described by a dual-ratio structure: internal closure dominance and exterior accessibility failure.**

6.2.3 The Black-Hole-Like Condition

Combining the unreconciled region principle with the dual-ratio structure, we define the full black-hole-like condition.

Definition 6.4 (Black-Hole-Like Domain).

A region R is black-hole-like if it satisfies:

$$\boxed{\mathcal{S}_R \geq 1, \quad \chi_R \leq 1, \quad \varepsilon_R \leq 1, \quad \alpha_R \leq 1,}$$

where $\mathcal{S}_R = \frac{\Gamma(B(R))}{L_R}$ is the RCF Schwarzschild-like burden compression ratio (Section 1.3).

In this regime, the observable sector is only the low-flux shadow of the full relational flow. The interior is not destroyed; its degrees of freedom are operationally hidden by suppression of accessible flux channels.

6.3 Boundary Recovery in the Strong-Sector Regime

6.3.1 The Boundary Record Map

If the interior is hidden by flux suppression, the next physical question is whether the exterior boundary data can reconstruct the interior state. This is the RCF analogue of the holographic principle.

Definition 6.5 (Boundary Record Map).

Let $\mathcal{Z}(R)$ be the set of admissible interior zero-preserving configurations of R .

Let $\mathcal{B}_{\partial R}$ be the space of exterior-accessible boundary data. The

boundary record map is:

$$\mathfrak{B}: \mathcal{Z}(R) \rightarrow \mathcal{B}_{\partial R},$$

which assigns to each interior configuration its boundary-preserved observable content.

6.3.2 Definition — Faithful Boundary Recovery

Definition 6.6 (Faithful Modulo Equivalence).

The boundary map $\mathfrak{B}$ is **faithful modulo equivalence** if:

$\frac{B}{\mathcal{H}_1} = \frac{B}{\mathcal{H}_2} \quad \Longleftrightarrow \quad \mathcal{H}_1 \sim \mathcal{H}_2$

where $\sim$ is the framework's physical equivalence relation on admissible histories (i.e., they differ only by null directions or gauge redundancy).

6.3.3 Theorem — Boundary Recovery

Theorem 6.3 (Boundary Recovery).

Assume the strong black-hole-like regime ($\chi_R \gg 1$, $\forall \epsilon_R \ll 1$). If the boundary record map $\frac{B}{\mathcal{H}}$ is complete and injective on the surviving strong-sector equivalence classes, then the boundary data determine the accessible interior equivalence class up to the constraint quotient.

Proof.

If $\frac{B}{\mathcal{H}}$ is injective on the surviving equivalence classes, then distinct interior classes cannot collapse to the same boundary record. Therefore, given a boundary record $b \in \mathcal{B}_{\partial R}$, there exists at most one interior equivalence class $[\mathcal{H}]_{\text{ext}}$ mapping to b . The interior is recoverable, not by direct access, but by relational reconstruction from the boundary code. $\square$

Interpretation.

This theorem preserves the framework's crucial guardrail: recovery is into the **lowest-burden admissible sector**, not a return of the original matter class. The full 3D relational interior is still there, but the accessible channel is so suppressed that the boundary presentation dominates. The 3D content is preserved in the underlying relational structure, but hidden by projection.

6.4 Entropy-Area Scaling and Lowest-Burden Emission

6.4.1 Boundary Entropy as Record Count

If the interior is hidden behind a projection boundary, the entropy of the black-hole-like region should reflect the number of admissible boundary records.

Theorem 6.4 (Boundary Entropy as Coarse-Grained Record Count).

Let ∂R be the boundary of a strongly burden-loaded region. If the admissible boundary microstates grow exponentially with the effective boundary area, then the boundary entropy satisfies:

$$\boxed{S_{\partial R} \propto \text{Area}(\partial R)} \quad \square$$

Proof Sketch.

Let $N_{\partial R}$ be the number of admissible boundary record states. If $N_{\partial R} \sim e^{\alpha \text{Area}(\partial R)}$, then $S_{\partial R} = \log N_{\partial R} \sim \alpha \text{Area}(\partial R)$. The entropy is governed by boundary area rather than interior volume because the interior degrees of freedom are flux-suppressed and operationally inaccessible. $\square$

6.4.2 Lowest-Burden Emission

As the unreconciled region relaxes, it must emit structure to re-equilibrate with the exterior. What is the nature of this emission?

Theorem 6.5 (Lowest-Burden Emission).

Let R be a strongly burden-loaded region undergoing relaxation. The emitted sector is not the original infallen matter class, but the **lowest-burden admissible output** compatible with the projection-flux boundary.

This output may take either of two forms:

1. **Dimension-independent carriers:** Temperature-linked or energy-like emission that does not carry 3D relational structure.
2. **Boundary-projected 3D relational content:** Three-dimensional relational excitations that are temporarily reduced by projection and then reconstitute after emission.

Proof Sketch.

The projection principle acts by suppressing accessible flux channels. The strongest projection-compatible output is the one carrying the least structural burden. The region relaxes by shedding this simplest admissible carrier sector. $\blacksquare$

> **A black-hole-like region relaxes by emitting the lowest-burden admissible sector, whether dimension-independent or boundary-projected from 3D relational content.**

6.4.3 Hawking-Like Appearance

Under coarse-grained observation, the lowest-burden emission appears thermal.

Theorem 6.6 (Hawking-Like Emission Under Coarse Graining).

Let a black-hole-like region relax by emitting the lowest-burden admissible sector. If the boundary flux channels are strongly suppressed and the emission is observed only at coarse-grained resolution, the outgoing spectrum is approximately thermal.

Proof Sketch.

Thermal spectra appear when transition rates obey approximate detailed balance. In the relational framework, the boundary relaxation process is driven by burden-gradient smoothing. If the outward channel is exponentially suppressed relative to the inward channel, the stationary distribution becomes Gibbs-like, and the emission law is Hawking-like in form. $\blacksquare$

6.5 Guardrails and Summary of Section 6

6.5.1 Guardrails for Section 6

1. **A black hole is not a literal hole.** It is a maximum-constrained, unreconciled relational sector.
2. **A horizon is not a material surface.** It is a boundary of exterior accessibility failure.
3. **A singularity is not a primitive infinite-density point.** It is a projection divergence at maximum internal complexity.
4. **$\alpha_R \rightarrow 0$ is asymptotic suppression, not exact zero.** For finite burden, the clock never literally stops.
5. **Interior causal structure may persist.** Exterior reconstruction failure does not imply internal causal annihilation.
6. **Recovery is not resurrection.** Boundary recovery reconstructs the lowest-burden admissible sector, not the original infallen matter class.
7. **Hawking-like emission is a target.** It follows from coarse-grained boundary relaxation, not a primitive postulate.

6.5.2 Summary of Section 6

Section 6 reframed black holes within the Relational Constraint Framework:

1. **The Unreconciled Region Principle** defined black-hole-like domains as failures of relational structures to reconcile with causal dependency.
2. **Projection as Flux Suppression** formalized how extreme burden attenuates interior visibility.
3. **Dual Horizon Ratios** ($\chi_R \gg 1, \epsilon_R \ll 1$) classified the strong isolation regime, preserving the guardrail that exterior access failure $\not\rightarrow$ internal causal annihilation.
4. **Boundary Recovery** proved that if the boundary record map is injective, the interior equivalence class is recoverable.
5. **Entropy-Area Scaling** was derived as a coarse-grained count of admissible boundary records.
6. **Lowest-Burden Emission** established that relaxation emits the simplest admissible carrier sector, not the original matter.
7. **Hawking-Like Appearance** showed that lowest-burden emission becomes approximately thermal under coarse graining.

The conceptual chain is:

$\boxed{\text{flux suppression} \rightarrow \text{lagging reconciliation} \rightarrow \text{boundary recovery} \rightarrow \text{entropy-area scaling} \rightarrow \text{lowest-burden emission} \rightarrow \text{Hawking-like appearance}}.$

This completes the black-hole layer of the framework.

Section 7 — Records, Classicality, and the Born Rule

7.0 Purpose of Records, Classicality, and Probability

7.0.1 Transition from Quantum Structure to Definite Outcomes

Section 6 reframed black-hole-like domains as unreconciled relational sectors where extreme burden suppresses exterior accessibility. The framework now possesses the full architectural spine: a primitive algebraic foundation, causal dependency, emergent spatial geometry, burden-weighted time, matter modes, gravitational sourcing, and horizon structure.

However, the framework does not yet explain how the definite, classical world of observable facts emerges from the underlying zero-preserving relational algebra. If physical reality is fundamentally a superposition of constraint-compatible relational modes, why do observers experience definite outcomes, stable histories, and predictable frequencies?

The purpose of this section is to reconstruct classicality and probability from the relational constraint structure, without importing external measurement postulates.

> **Classicality emerges when certain zero-preserving records become stable, redundant, and effectively irreversible under constraint-preserving interactions.**

7.0.2 Why Probability Must Emerge from Admissibility

In standard quantum mechanics, the Born rule is postulated as an external probability measure applied to a Hilbert space. In the Relational Constraint Framework, Hilbert space is a derived GNS representation, and the primitive physical condition is the vanishing of the master constraint: $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$.

Therefore, probability cannot be bolted on as a separate axiom. It must emerge as the natural measure of how the zero-constraint is distributed across stable structural alternatives (record sectors).

The central claim of this section is that the Born weights are not an extra probabilistic layer imposed on the theory; they are the record-sector decomposition of the physical state's zero-constraint support.

> **The master constraint remains zero, but the zero is read through stable record weights, so the probability rule is anchored back to admissibility rather than floating free as interpretation.**

7.0.3 What This Section Establishes

This section establishes the following:

- Stable Record Separation**: Decoherence-like burden separation causes distinct record sectors to become approximately orthogonal.
- Redundant Record Robustness**: Classicality arises when records are encoded across many environmental fragments, making them robust against local disturbance.
- Normalized Record Weights**: Positive linear functionals on stable record effects yield mathematically normalized weights ($p_i \geq 0, \sum p_i = 1$).

4. **The Born Rule as Sectorwise Zero-Decomposition**: The master constraint zero condition is shown to distribute over record sectors weighted by squared amplitudes ($|c_i|^2$).
5. **Zero-Envariance (Z-Envariance)**: Symmetries of the zero-preserving state force the branch measure to be proportional to squared amplitude, uniquely selecting the Born form.

7.1 Stable Record Separation

7.1.1 The Problem of Definite Outcomes

The physical state $|\omega\rangle$ evaluates relational expressions. In the GNS representation, this appears as a cyclic vector $|\Omega_\omega\rangle$ in a Hilbert space $\mathcal{H}_\omega$. If the state is a superposition of different macroscopic configurations, standard quantum mechanics asks how one definite outcome is selected.

RCF does not begin with a "measurement postulate" that collapses the state. Instead, it asks: under what conditions do distinct relational configurations become effectively independent, preventing interference?

The answer lies in the **constraint burden**. When different macroscopic configurations interact with their environment, they generate distinct correlation profiles. Maintaining coherence between these distinct profiles requires reconciling vastly different causal dependencies, which imposes a high structural burden. The framework naturally suppresses this high-burden cross-talk.

7.1.2 Definition — Record Sector

Definition 7.1 (Record Sector).

Let the physical state $|\omega\rangle$ generate a GNS representation $(\mathcal{H}_\omega, \pi_\omega, \Omega_\omega)$. A **record sector** $\mathcal{H}_i \subset \mathcal{H}_\omega$ is a closed subspace corresponding to a stable, distinguishable macroscopic relational configuration (a "record").

Let P_i be the orthogonal projector onto $\mathcal{H}_i$. A family of record sectors $\{\mathcal{H}_i\}$ is exhaustive if $\sum_i P_i = I$ on the physical sector.

7.1.3 Theorem — Stable Record Separation

Theorem 7.1 (Stable Record Separation).

Suppose the physical sector decomposes into record-bearing classes such that the constraint-maintenance burden of cross-correlating distinct sectors is sufficiently high. Then, under coarse-graining, the cross-correlation terms between distinct record sectors become negligible.

Specifically, if P_i and P_j project onto distinct record sectors ($i \neq j$), then:

$$\boxed{\langle \Omega_\omega, P_i \rangle_{\pi_\omega(A)} P_j \rangle_{\Omega_\omega} \approx 0 \quad \text{for all } A \in \mathcal{A}_{\{\mathrm{loc}\}}.}$$

Proof Sketch.

Once null directions are removed by the GNS quotient, the remaining structure is correlation geometry. If coarse-graining suppresses the overlap between record classes (due to the high burden of maintaining cross-sector coherence), the induced inner product between distinct branches becomes small. The branches therefore behave like quasi-classical alternatives rather than coherently interfering amplitudes. $\blacksquare$

****Interpretation.****

This is the RCF analogue of decoherence. Interference is not lost by an external "wavefunction collapse"; it is suppressed internally because maintaining cross-sector consistency violates the least-burden reconciliation principle. What remains is a set of independent, stable record sectors.

7.2 Redundant Record Robustness (Classicality)

7.2.1 From Decoherence to Objectivity

Decoherence alone explains why interference vanishes, but it does not fully explain why the world appears classical. For a record to be a classical "fact," it must be stable against local disturbances and redundantly accessible to multiple observers.

RCF formalizes this using the concept of environmental fragments.

7.2.2 Definition — Redundant Encoding

****Definition 7.2 (Environmental Fragments and Redundancy).****

Let R be a zero-preserving record associated with a sector H_i^ω . Let $\{E_k\}$ be a set of environmental fragments (subsystems relationally coupled to R).

The record R is ****redundantly encoded**** if many fragments independently carry recoverable information about R . The ****redundancy degree**** is:

$$\boxed{\mathcal{R}_\eta(R) = \#\{E_k : \mathrm{Corr}_\omega(E_k, R) \geq \eta\},}$$

where η is a recognition threshold.

7.2.3 Definition — m -Robust Record

****Definition 7.3 (m -Robust Record).****

A record R is **** m -robust**** if deleting or disturbing any set of at most m encoding fragments still leaves at least one fragment from which the record value can be recovered.

7.2.4 Theorem — Redundant Record Robustness

****Theorem 7.2 (Redundant Record Robustness).****

Let R be a zero-preserving record with redundancy degree $\mathcal{R}_\eta(R) = N_R$. Assume each of the N_R fragments independently encodes the same record value above the threshold η . If $N_R > m$, then R is m -robust.

Proof.

There are N_R fragments encoding the record. Delete any set D of at most m fragments ($|D| \leq m$). At worst, all deleted fragments are among the N_R record-encoding fragments. The number of surviving record-encoding fragments is at least $N_R - m$. Since $N_R > m$, we have $N_R - m > 0$. Therefore, at least one fragment encoding R remains, and the record value is recoverable. $\square$

****Interpretation.****

Classicality is not a primitive property of macroscopic objects; it is an emergent property of redundant encoding. A classical fact is a stable, redundant, zero-preserving record. Objectivity arises because multiple observers can access the same fact through different fragments without directly disturbing the original record.

7.3 Normalized Record Weights

7.3.1 The Formal Probability Layer

Now that stable, independent record sectors exist, we must assign weights to them. The framework requires a non-negative measure on these sectors compatible with the algebraic state.

****Definition 7.4 (Positive Effect Family).****

Let $\{E_i\}$ be a family of positive operators (effects) in the observable algebra $\mathcal{A}_{\text{obs}}$ that resolve the stable record sector:

$$\boxed{\sum_i E_i = \mathbf{1}.}$$

7.3.2 Definition — Normalized Record Weights

****Definition 7.5 (Record Weights).****

Let ω be a normalised physical state ($\omega(\mathbf{1}) = 1$). The **record weight** assigned to sector i is:

$$\boxed{p_i := \omega(E_i).}$$

7.3.3 Theorem — Normalization and Positivity

****Theorem 7.3 (Normalization and Positivity).****

The record weights p_i satisfy:

$$p_i \geq 0 \quad \text{and} \quad \sum_i p_i = 1.$$

Proof.

Since ω is a positive linear functional and E_i is a positive operator, $p_i = \omega(E_i) \geq 0$.

By linearity of ω and the exhaustivity of the resolution $\sum_i E_i = \mathbf{1}$:
 $\sum_i p_i = \sum_i \omega(E_i) = \omega(\sum_i E_i) = \omega(\mathbf{1}) = 1.$ $\blacksquare$

Interpretation.

This closes the formal probability piece. Positivity makes the weights non-negative, and normalization makes them sum to one. However, this theorem alone does not prove the Born rule ($p_i = |c_i|^2$); it only proves that a consistent probability-like measure exists. The remaining issue is to anchor these weights back to the foundational zero-constraint.

7.4 The Born Rule as Sectorwise Zero-Decomposition

7.4.1 The Core Synthesis

The most critical theoretical move in RCF probability theory is connecting the record weights p_i to the master constraint $\hat{\mathcal{M}}_\omega$.

In standard quantum mechanics, probabilities are imposed after the fact. In RCF, physical admissibility *is* the vanishing of the master constraint. If the universe branches into record sectors, the zero-constraint must be distributed across those sectors.

Theorem 7.4 (Born Weights as Sectorwise Zero-Decomposition).

Let the physical state decompose into decohered record sectors. Let $\hat{C}_i$ be the component of the primitive constraint operator acting on sector i . The sector-weighted master constraint is:

$$\boxed{\mathcal{M}_B = \sum_i p_i \hat{C}_i^\dagger \hat{C}_i, \quad p_i \geq 0, \quad \sum_i p_i = 1.}$$

The physical state condition $\omega(\mathcal{M}_B) = 0$ implies that the zero-constraint is distributed across the stable record sectors according to the weights p_i .

Proof.

Because each $\hat{C}_i^\dagger \hat{C}_i$ is a positive operator and the weights p_i are non-negative, the weighted sum $\mathcal{M}_B$ is positive. The only way for the expectation value $\omega(\mathcal{M}_B)$ to vanish in a physical state is for the constrained sectors to vanish on the physical support.

The master constraint says what must vanish. The Born weights p_i say how the zero is distributed across stable record alternatives. The physical state says the universe occupies the zero-constraint support. Therefore, probability is not added on top of the theory; it is the sectorwise reading of constraint compatibility. $\blacksquare$

> **Born weights do not compete with the master constraint; they refine its zero condition across record sectors.**

7.4.2 Why This Uniquely Selects $|c_i|^2$ (Z-Envariance)

While Theorem 7.4 proves that weights must distribute the zero-constraint, it does not uniquely specify the functional form of p_i . To show that $p_i = |c_i|^2$, the framework employs **Zero-Envariance (Z-Envariance)**, adapting the envariance concept to the RCF constraint structure.

Definition 7.6 (Zero-Envariance).

A local transformation $U_{\mathcal{S}}$ on a system sector is **Z-envariant** if there exists a compensating transformation $U_{\mathcal{E}}$ on the environment sector such that:

1. $(U_{\mathcal{S}} \otimes U_{\mathcal{E}}) |\Psi\rangle = |\Psi\rangle$ (the global physical state is unchanged).
2. The global state remains in the physical kernel: $(U_{\mathcal{S}} \otimes U_{\mathcal{E}}) |\Psi\rangle \in \ker \hat{M}$.

Lemma 7.1 (Phase Invariance from Z-Envariance).

If branch phase rotations are Z-envariant, then the branch measure (probability) must be phase-invariant:

$$p_i(c_i) = p_i(|c_i|).$$

Proof.

A local phase rotation $U_{\mathcal{S}} |s_i\rangle = e^{i\theta} |s_i\rangle$ can be compensated by an inverse environmental phase $U_{\mathcal{E}} |E_i\rangle = e^{-i\theta} |E_i\rangle$. The joint action leaves the entangled state and the master constraint structure unchanged. Since the branch measure is an objective invariant of the physical state, it cannot depend on the phase. $\blacksquare$

Lemma 7.2 (Equal-Magnitude Branch Equality).

If two decohered branches have equal amplitude magnitude ($|c_1| = |c_2|$), and branch swaps are Z-envariant, then their measures are equal:

$$p_1 = p_2.$$

Proof.

By phase invariance, choose phases so $c_1 = c_2$. A swap of the system states combined with a swap of the environment records leaves the global state invariant. The measure cannot distinguish them. $\blacksquare$

7.4.5 Theorem — The Born Rule

Theorem 7.5 (The Born Rule as Normalized Record Weights).

Let $|\Psi\rangle = \sum_i c_i |s_i\rangle |E_i\rangle$ be a decohered physical state in $\ker \hat{M}$. Assume:

1. Branch phase rotations are Z-envariant.
2. Equal-magnitude branch swaps are Z-envariant.
3. Branch measures are objective zero-closure invariants.
4. Branch measures are additive over decohered orthogonal record-sector splits.

Then the unique normalised branch measure is:

$$\boxed{p_i = \frac{|c_i|^2}{\sum_j |c_j|^2}}.$$

If the state is normalised ($\sum_j |c_j|^2 = 1$), then:

$$\boxed{p_i = |c_i|^2.}$$

Proof.

By Lemma 7.1, the measure depends only on $|c_i|$. By Lemma 7.2, equal magnitudes yield equal measures. By additivity over orthogonal splits, the measure function $f(x)$ must satisfy $f(x+y) = f(x) + f(y)$. The unique regular solution is $f(x) = Kx$. Therefore, $p_i \propto |c_i|^2$. Normalization yields the final form. $\blacksquare$

7.5 Guardrails and Summary of Section 7

7.5.1 Guardrails for Section 7

1. **Normalized weights $\neq$ full Born interpretation.** The formal mathematical structure of probability ($p_i \geq 0, \sum p_i = 1$) is established, but the leap to empirical frequencies requires the stability and accessibility of records.
2. **Classicality is emergent.** Records become classical only when they are stable, redundant, and zero-preserving.
3. **The Born rule is not an external postulate.** It is derived as the sectorwise decomposition of the master constraint zero, forced by Z-envariance.
4. **Decoherence is internal.** Suppression of interference arises from the high burden of cross-sector reconciliation, not from an external thermodynamic arrow.

7.5.2 Summary of Section 7

Section 7 reconstructed classicality and probability from the relational constraint framework:

1. **Stable Record Separation** showed that high burden suppresses cross-sector interference, causing record sectors to become orthogonal.
2. **Redundant Record Robustness** proved that classicality arises from m -robust encoding across environmental fragments.
3. **Normalized Record Weights** established the formal probability structure ($p_i \geq 0, \sum p_i = 1$) from positive effects.
4. **The Born Rule as Sectorwise Zero-Decomposition** proved that the master constraint zero distributes over record sectors according to the weights p_i .
5. **Zero-Envariance (Z-Envariance)** forced the branch measure to be proportional to squared amplitude, uniquely selecting $p_i = |c_i|^2$.

The conceptual chain is:

$$\boxed{\text{master constraint} = 0 \quad \rightarrow \quad \text{sector-weighted zero support} \quad \rightarrow \quad \text{Born weights} \quad \rightarrow \quad \text{operational probabilities}.}$$

This completes the probability and classicality layer of the framework.

Section 8 — Cosmology and the Open Universe

8.0 Purpose of Cosmology in RCF

8.0.1 Transition from Local Structure to Global Dynamics

Section 7 established how classicality and probability emerge from the zero-preserving relational substrate. The framework now possesses the full local architecture: physical states, observables, emergent space, emergent time, fields, particles, interactions, gravity, black holes, and records.

However, the universe is not merely a local laboratory. It is a global, historical structure. The framework must explain how the relational substrate unfolds on cosmological scales, why the universe expands, and how large-scale phenomena like cosmic acceleration and clustering emerge without being assumed as primitive background properties.

The purpose of this section is to reconstruct cosmology as the large-scale dynamics of the open relational extension.

> **Cosmic expansion is the large-scale unfolding of causal-relational structure under burden-weighted open extension, stabilized by records and expressed in emergent correlation geometry.**

8.0.2 What This Section Establishes

This section establishes the following:

1. **The Effective Expansion Driver**: Expansion is not driven by a primitive scale factor, but by the coupled action of open extension, burden redistribution, and correlation-distance growth.
2. **Cosmic Expansion as Relational Growth**: The scale factor of the universe is identified with the coarse-grained growth of correlation distance d_{ω} .
3. **Dark-Energy-Like Expansion**: The expansive channel of open extension, dominated by mode burden release, mimics dark-energy-like behaviour.
4. **Dark-Matter-Like Binding**: The binding channel, dominated by relational burden (ρ_{rel}), mimics dark-matter-like halo response.
5. **Friedmann-like Dynamics**: Under assumptions of large-scale homogeneity and isotropy, the effective dynamics reduces to a Friedmann-like expansion law governed by the component-selective burden source.

8.1 The Effective Expansion Driver

8.1.1 Open Extension as the Cosmological Spine

In Section 1, the Open Extension Principle stated that a physical universe is an extendible zero-preserving partial order. The extension label $\$s\$$ was strictly distinguished from physical time.

In the cosmological regime, we must understand *why* the universe extends and *what* drives the pace of that extension.

The **Causal Reconciliation Principle** (Section 1.6) states that constraint satisfaction cannot happen instantaneously. The universe uses admissible relational channels to distribute constraint correction across the causal chain. Open extension exists to preserve the zero constraints under finite propagation speed.

Definition 8.1 (Effective Expansion Driver).

The **effective expansion driver** is the coarse-grained rate at which an open causal chain extends through relationally conditioned structure, redistributes burden, and separates admissible correlation channels across the emergent large-scale geometry. It is a function of: $\boxed{\mathcal{E}_{\mathrm{eff}} = \mathcal{E}(\rho_{\mathrm{mode}}, \rho_{\mathrm{int}}, \rho_{\mathrm{rel}}, D_{\omega}, \mathrm{Ext})}$.

Interpretation.

The causal chain governs open extension (what continuations are allowed). Relational structure conditions how that chain is realized (pace, accessibility, morphology). Expansion is driven by the causal chain, but its pace is shaped by burden redistribution and relational reconciliation.

> **The causal chain governs open extension; relational structure conditions its realization.**

8.1.2 The Role of the Three Burden Sources in Cosmology

The three burden channels introduced in Section 5 play distinct roles in the cosmological engine:

1. **Mode Burden (ρ_{mode})**: Supports local structure and visible matter-like clustering. As the universe extends, mode burden is redistributed, driving the expansive, "unlocking" side of open extension.
2. **Interaction Burden (ρ_{int})**: Supports internal structural complexity and hierarchical binding. It governs how structures form and maintain coherence against dispersal.
3. **Relational Burden (ρ_{rel})**: Supports long-range correlation binding. It acts as a network-level load that preserves coherent structure, generating halo-like responses that counteract pure expansion.

This creates a natural duality within the framework:

- **Expansive Sector**: Dominated by ρ_{mode} and ρ_{int} through burden release and open extension.
- **Binding Sector**: Dominated by ρ_{rel} through correlation preservation.

8.2 Cosmic Expansion as Relational Growth

8.2.1 Redefining the Scale Factor

In standard cosmology, expansion means the stretching of a pre-existing spacetime metric, causing spatial distances between comoving observers to increase.

In RCF, spatial distance is not primitive. It is reconstructed from the correlation kernel K_{ω} . Therefore, cosmic expansion cannot be modelled as the stretching of a background manifold. Instead, it must be understood as the **growth of correlation distance** among admissible event-regions.

Definition 8.2 (Relational Scale Factor).

Let A and B be two coarse-grained relational regions. Let $D_{\omega_s}(A, B)$ be their correlation distance at extension stage s . The **relational scale factor** $a(\tau)$ is defined as the ratio of the large-scale average correlation distance at stage $s+1$ to that at stage s :

$$a(\tau_{s+1}) \sim \frac{\langle D_{\omega_{s+1}}(A, B) \rangle}{\langle D_{\omega_s}(A, B) \rangle}$$

8.2.2 Theorem — Relational Expansion Under Attenuation

Theorem 8.1 (Relational Expansion Under Attenuation).

If persistent relational weights attenuate under open extension such that $W_{s+1}(i, j) = q_{ij} W_s(i, j)$ with $0 < q_{ij} \leq 1$, then the relational distance grows as:

$$D_{\omega_{s+1}}(i, j) = D_{\omega_s}(i, j) + \ell_0 \log\left(\frac{1}{q_{ij}}\right)$$

Consequently, the relational scale factor $a(\tau)$ increases.

Proof.

By definition, $D_{\omega}(i, j) = -\ell_0 \log W(i, j)$. Substituting the attenuated weight:

$$D_{\omega_{s+1}}(i, j) = -\ell_0 \log (q_{ij} W_s(i, j)) = -\ell_0 \log q_{ij} - \ell_0 \log W_s(i, j)$$

Since $0 < q_{ij} \leq 1$, we have $\log q_{ij} \leq 0$, which implies $-\ell_0 \log q_{ij} \geq 0$. Therefore:

$$D_{\omega_{s+1}}(i, j) = D_{\omega_s}(i, j) + \ell_0 \log\left(\frac{1}{q_{ij}}\right)$$

Since $\ell_0 \log(1/q_{ij}) \geq 0$, the distance increases. Averaging over regions yields an increase in the scale factor. $\blacksquare$

Interpretation.

Expansion is not motion through a pre-existing void. It is the stretching of the relational correlation geometry itself. As the universe extends, the correlation weights between distant regions attenuate, causing the emergent spatial distance between them to grow logarithmically.

8.3 Dark-Energy-Like Expansion and Dark-Matter-Like Binding

8.3.1 The Cosmological Duality

The framework's burden decomposition naturally provides a mechanism for the two most puzzling cosmological phenomena: dark energy and dark matter.

> **Dark energy is the expansive, unlocking side of open extension; dark matter is the binding, structure-preserving side that keeps relational systems from dissolving into pure spread.**

8.3.2 Dark-Energy-Like Expansion

If the large-scale sector is dominated by the expansive channel—where open extension permits rapid growth of correlation distance and mode burden is released—the effective dynamics mimic dark-energy-like acceleration.

The cosmological source law (to be formalized in 8.4) will include a term driven by ρ_{mode} and the open extension rate $\mathcal{E}_{\mathrm{eff}}$. When the burden density is dominated by this expansive mode, the scale factor $a(\tau)$ accelerates.

8.3.3 Dark-Matter-Like Binding via Relational Burden

Standard cosmology posits dark matter to explain why galaxies and clusters bind more strongly than visible matter allows. RCF provides a structural candidate for this: **relational burden** (ρ_{rel}).

Suppose matter (mode burden) clusters. As it clusters, the correlation web connecting these structures becomes denser. The framework predicts that $\rho_{\mathrm{rel}} \propto \text{correlation density}$, not merely matter density.

Properties of ρ_{rel} matching dark matter phenomenology:

- Correlation with matter clustering:** ρ_{rel} grows where ρ_{mode} grows.
- Non-luminous:** ρ_{rel} does not belong to individual objects; it belongs to the network.
- Halo extension:** Correlations extend beyond physical boundaries. $R_{\rho_{\mathrm{rel}}} > R_{\rho_{\mathrm{mode}}}$, naturally producing halo-like structures.
- Gravitational response:** ρ_{rel} contributes to the burden tensor $\Theta^{(B)}_{\mu\nu}$, sourcing effective geometry.

> **Relational burden is a candidate effective source for dark-matter-like phenomenology, without introducing exotic particles.**

Guardrail. The framework cannot yet claim $\rho_{\mathrm{rel}} = \rho_{\mathrm{DM}}$ as a theorem. It claims that ρ_{rel} is a candidate effective source that naturally predicts halo formation, non-luminosity, and extra gravitational response. Model-specific matching to galaxy rotation curves remains an open target.

8.4 Friedmann-like Cosmological Emergence

8.4.1 The Conditional Friedmann Theorem

If the emergent geometry is smooth and Lorentz-compatible, and the large-scale sector is homogeneous and isotropic, the effective metric can be cast in a Robertson-Walker-like form. The dynamics of the scale factor $a(\tau)$ must then be governed by the active burden source.

Theorem 8.2 (Friedmann-like Cosmology, Conditional Form).

If the emergent large-scale sector is homogeneous, isotropic, and governed by the covariantly conserved active burden tensor $\Theta^{(B)}_{\mu\nu}$, then the coarse-grained dynamics reduces to a Friedmann-like expansion law:

$$\boxed{H_{\mathrm{eff}}^2 = F(\rho_B, \Pi_B, D_\omega, \dots)}$$

where $H_{\mathrm{eff}} = \frac{\dot{a}}{a}$ is the effective Hubble parameter.

Proof Sketch.

Homogeneity and isotropy eliminate preferred spatial directions in the large-scale limit. This leaves only the scale factor $a(\tau)$ as the surviving geometric degree of freedom.

Covariant conservation of the effective burden source ($\nabla_\mu \Theta^{(B)}_{\mu\nu} = 0$) supplies the evolution equation for $a(\tau)$. The result is an expanding background governed by the effective burden content of the relational state. $\blacksquare$

8.4.2 The Effective Source Law

While a full rigorous derivation of the exact Friedmann equations from the algebraic master constraint remains an open target, the framework posits an effective, conditional source law for cosmological modelling:

$$\nabla^2 \Phi = 4\pi G (\rho_{\mathrm{mode}} + \rho_{\mathrm{int}} + \beta \rho_{\mathrm{rel}})$$

where β is a scaling coefficient relating relational burden to effective gravitational mass. This law encapsulates the idea that visible matter (ρ_{mode}), interaction/binding energy (ρ_{int}), and network correlation load (ρ_{rel}) all contribute to the effective potential driving cosmic structure.

8.5 Guardrails and Summary of Section 8

8.5.1 Guardrails for Section 8

1. **Expansion is not primitive motion.** It is the growth of correlation distance under open extension.

2. **The scale factor is derived.** $a(\tau)$ is reconstructed from the attenuation of relational weights, not assumed as a background function.
3. **Dark-sector claims are phenomenological targets.** ρ_{rel} as dark matter and open extension as dark energy are compelling structural analogues, but require model-specific matching to be elevated to theorems.
4. **Friedmann dynamics is conditional.** The reduction to a Friedmann-like law assumes large-scale homogeneity and isotropy, which are coarse-grained emergent symmetries, not primitive axioms.

8.5.2 Summary of Section 8

Section 8 reconstructed cosmology as the large-scale dynamics of the open relational extension:

1. **The Effective Expansion Driver** was defined as the coupled action of open extension, burden redistribution, and correlation-distance growth.
2. **Cosmic Expansion** was formalized as the growth of correlation distance D_ω under weight attenuation, yielding a relational scale factor $a(\tau)$.
3. **Dark-Energy-Like Expansion** was identified with the expansive channel of open extension and mode burden release.
4. **Dark-Matter-Like Binding** was identified with relational burden ρ_{rel} , which generates halo-like, non-luminous gravitational response.
5. **Friedmann-like Dynamics** was conditionally derived, showing that a homogeneous, isotropic universe governed by a conserved burden tensor reduces to a standard expansion law.

The conceptual chain is:

$\boxed{\text{causal admissibility} \rightarrow \text{relational correlation geometry} \rightarrow \text{burden-weighted clocking} \rightarrow \text{open extension} \rightarrow \text{cosmic expansion}.}$

This completes the cosmological layer of the framework.

Section 9 — Open Targets and Quarantined Phenomenology

9.0 Purpose of the Audit and Consolidation Pass

9.0.1 The Final Synthesis Block

The preceding sections have constructed the complete architectural spine of the Relational Constraint Framework (RCF). We began with a primitive unital $(*)$ -algebra $\mathcal{A}$ and a constraint ideal $\mathcal{I}_C$, derived the physical state space and emergent Hilbert space, reconstructed causal order and spatial geometry, derived burden-weighted time, recovered fields and particles, established the Einstein-like gravitational closure, reframed black holes as unreconciled flux-suppressed sectors, derived the Born rule from zero- ϵ -variance, and reconstructed cosmology as open relational extension.

The framework is now a coherent, mathematically structured research programme. However, the discipline of the framework demands absolute clarity regarding the epistemic status of each claim. A theoretical structure is only as strong as its honesty about its own boundaries.

The purpose of this final section is to perform an audit and consolidation pass. It separates definitions, conditional theorems, open targets, and quarantined speculation. It ensures that the framework says exactly what it proves, and no more.

> **The foundation must say exactly what it proves, and no more.**

9.0.2 The Status Tag System

Throughout the manuscript, claims have been labelled. We now consolidate them into four strict categories:

1. **Definition:** A formal object or rule introduced without need for proof.
2. **Conditional Theorem:** A result that follows logically from definitions under explicitly stated assumptions.
3. **Open Theorem Target:** A precise mathematical statement that the framework aims to prove, but which currently relies on unproven assumptions or missing microscopic derivations.
4. **Quarantined Speculation:** A broad physical interpretation or phenomenological claim that is not yet theorem-ready and must be kept out of the core deductive stack.

This section maps the entire framework into these categories and identifies the precise mathematical locks that remain to be picked.

9.1 The Consolidated Claim-Status Map

The structural hierarchy of RCF can now be audited layer by layer.

Layer 0: The Algebraic Foundation

- * **Status:**  Established (Proven / Definition)
- * **Objects:** $\mathcal{A}$, $\mathcal{I}_C$, ω , $(\mathcal{H}_\omega, \pi_\omega, \Omega_\omega)$, $\hat{\mathcal{M}}_\omega$, $\mathcal{M}(x)$.
- * **Theorems:** Master-zero equivalence, ideal-kernel correspondence, GNS state reconstruction.
- * **Assumptions:** Properness of $\mathcal{I}_C$, existence of positive states.

Layer 1: The Zero-Preserving Causal Foundation

- * **Status:**  Proven Conditionally
- * **Objects:** Zero-preserving observables ($\widetilde{\mathcal{A}}_{\mathrm{obs}}$), Causal dependency ($\prec$), Extension set (Ext), Causal speed limit (c_{RCF}).
- * **Theorems:** Strong observables preserve kernel, causal dependency is a strict partial order, emergent causal speed limit.

* **Assumptions:** Bounded relational step length (ℓ_C), necessity composition for transitivity.

Layer 2: Correlation Geometry and Emergent Space

* **Status:**  Proven Conditionally

* **Objects:** Correlation kernel (K_ω), Correlation distance (d_ω), Emergent points ($\mathcal{X}_\omega$), Emergent direction (T_A^{em}).

* **Theorems:** Pseudometricity under multiplicative consistency, metric quotient, approximate triangle inequality, $D=3$ dimensional closure.

* **Assumptions:** Multiplicative correlation consistency, exact/approximate metricity, independence of existence/position/direction inference channels.

Layer 3: Emergent Time and Burden-Clock Geometry

* **Status:**  Proven Conditionally

* **Objects:** Constraint burden (B_R), Local clock factor (α), Burden-weighted proper time ($\tau[\gamma]$), Burden lapse (α_B).

* **Theorems:** Burden suppresses clock rate, non-uniform burden induces non-uniform proper time, scalar burden-to-lapse bridge.

* **Assumptions:** Rational form of $\alpha(B) = 1/(1+\lambda B)$, definition of reconciliation cost.

Layer 4: Fields, Particles, and Interactions

* **Status:**  Toy-level / Conditional

* **Objects:** Fields (ϕ), Correlation Laplacian (Δ_ω), Particle-like excitations, Interaction functional ($\mathcal{I}$).

* **Theorems:** Particle identity, interaction non-additivity.

* **Assumptions:** Continuum limit of Δ_ω , identification of stable modes with matter.

Layer 5: Gravity as Geometry of Constraint Burden

* **Status:**  Conditional Reduction Theorem

* **Objects:** Burden tensor ($\Theta^{(B)}_{\mu\nu}$), Effective ADM metric (ds_B^2).

* **Theorems:** Active-source conservation, Einstein-like closure (conditional uniqueness).

* **Assumptions:** Smooth 4D Lorentzian continuum limit, Lovelock theorem applicability, constitutive burden-metric dictionary.

Layer 6: Black Holes as Unreconciled Relational Sectors

* **Status:**  Open Theorem Targets

* **Objects:** Unreconciled region, Projection-flux ($\mathcal{P}_{\partial R}$), Dual horizon ratios (χ_R, ε_R).

* **Theorems:** Boundary recovery (conditional), entropy-area scaling (target), Hawking-like emission (target).

* **Assumptions:** Injectivity of boundary record map, exponential scaling of boundary microstates.

Layer 7: Records, Classicality, and the Born Rule

* **Status:**  Open Theorem Targets

- * **Objects:** Record sectors ($\mathscr{H}_i^\omega$), Redundancy degree ($\mathcal{R}_\eta$), Z-Envariance.
- * **Theorems:** Normalized record weights, Born rule as sectorwise zero-decomposition.
- * **Assumptions:** Z-envariance symmetry closure, decoherence from high cross-sector burden.

Layer 8: Cosmology and the Open Universe

- * **Status:** ● Quarantined Phenomenology
- * **Objects:** Effective expansion driver ($\mathcal{E}_{\mathrm{eff}}$), Relational scale factor ($a(\tau)$).
- * **Theorems:** Friedmann-like cosmology (conditional).
- * **Assumptions:** Large-scale homogeneity and isotropy.

9.2 Priority Open Targets

The framework has crossed major theoretical thresholds, but it is not yet a completed theory of physics. It is a structured programme with a short, sharply named list of remaining mathematical locks.

To elevate RCF from a structural framework to a predictive physical theory, the following targets must be rigorously closed:

1. **Microscopic Derivation of Multiplicative Correlation Consistency:**

The emergent spatial metric relies on $K_\omega(A,C) \geq K_\omega(A,B)K_\omega(B,C)$. This must be proven from the dynamics of the constraint algebra $\mathcal{I}_C$ acting on the physical state ω , rather than assumed as a macroscopic regularity.

2. **Smooth Continuum Emergence:**

The framework currently operates on discrete or coarse-grained correlation graphs. A rigorous theorem is needed to show that the emergent metric space $(\mathcal{X}_\omega, D_\omega)$ converges to a smooth manifold in the large-scale limit, justifying the use of differential geometry and the ADM decomposition.

3. **Full Einstein-like Closure from the Master Constraint:**

While Lovelock's theorem guarantees the *form* of the equation ($G_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$), the dynamical derivation of this equation from the primitive master constraint $\hat{\mathfrak{M}}_\omega = 0$ remains open. The effective coupling κ_B must be derived from the algebraic constants ($\epsilon, \ell_0, \lambda$).

4. **Concrete Matter-Burden Identification:**

The burden tensor $\Theta^{(B)}_{\mu\nu}$ is structurally stress-energy-like. A theorem must bridge the gap, proving that the stable field modes (Layer 4) generate a burden response strictly proportional to their physical stress-energy $T_{\mu\nu}$.

5. **Uniqueness of D=3 Closure Defect:**

The dimensional closure theorem (Theorem 2.9) relies on the structural independence of existence, position, and direction inference channels. A formal algebraic proof is required to show that these channels cannot collapse in $D < 3$ and cannot redundantly overlap in $D > 3$ without violating the master constraint.

9.3 Quarantined Phenomenology

The following physical claims are structurally supported by the framework as natural analogues or effective descriptions, but they must remain quarantined from the core theorem stack until the priority open targets are closed and model-specific calculations are performed.

9.3.1 Dark Matter as Relational Burden (ρ_{rel})

The framework predicts that matter clustering increases correlation density, generating a non-local, non-luminous burden (ρ_{rel}) that sources extra geometry. This mimics dark matter halos. However, RCF cannot yet claim $\rho_{\mathrm{rel}} = \rho_{\mathrm{DM}}$ as a theorem. It is a candidate effective source requiring quantitative matching to galaxy rotation curves and cluster lensing.

9.3.2 Dark Energy as Open Extension

The expansive channel of open extension, driven by burden release, mimics dark-energy-like acceleration. While structurally sound, identifying this RCF mechanism with the cosmological constant Λ requires deriving the exact magnitude and equation of state from the microscopic reconciliation rate.

9.3.3 Inflation Replacement (Primordial Common Closure)

RCF suggests that large-scale cosmic uniformity arises because spatially distant regions share common causal-constraint ancestry ($C_0 \prec A, B$), rendering late-time causal contact unnecessary. This is a compelling alternative to cosmic inflation, but it remains a quarantined hypothesis until the framework can derive the exact scale-invariant perturbation spectrum of the CMB from primordial closure fluctuations.

9.3.4 Standard Model Gauge Group Emergence

The framework defines interactions as non-additive constraint burden and gauge structure as relational comparison rules. However, deriving the specific gauge groups ($SU(3) \times SU(2) \times U(1)$), fermionic statistics, and the particle spectrum of the Standard Model from the constraint algebra is strictly quarantined as future work.

9.4 Guardrails for the Final Synthesis

1. **No overclaiming identity.** Structural analogy (e.g., $\Theta^{(B)}_{\mu\nu}$ looking like $T_{\mu\nu}$) is not mathematical identity. The framework must not present analogies as theorems.

2. **No smuggling microscopic assumptions.** The open targets must be closed using only the primitive objects ($\mathcal{A}$, $\mathcal{I}_C$, ω) and their represented dynamics.
3. **Honesty about the continuum limit.** The transition from discrete relational structure to smooth manifold is one of the hardest problems in quantum gravity. RCF must acknowledge this gap rather than assuming it away.

9.5 Summary and Manuscript Closure

The Relational Constraint Framework has been formally established. It began with the claim that physical reality is not a collection of objects in a pre-existing background, but a constrained relational structure whose admissible configurations are those in which inconsistency vanishes.

From this algebraic foundation, the manuscript systematically reconstructed the pillars of physical reality:

- **Space** was reconstructed from correlation profiles.
- **Time** was reconstructed from causal depth weighted by constraint burden.
- **Matter** was reconstructed as stable, localised zero-preserving modes.
- **Gravity** was reconstructed as the macroscopic geometric response to burden distribution.
- **Black Holes** were reconstructed as unreconciled, flux-suppressed relational sectors.
- **Probability** was reconstructed as the sectorwise decomposition of the master constraint zero.
- **Cosmology** was reconstructed as the open extension of the relational network.

The framework is no longer missing its spine or its legs. It is down to the joints where the last bolts of microscopic derivation still need tightening. The remaining work is no longer architectural rescue or conceptual redesign; it is rigorous theorem engineering.

The foundational commitment stands:

$$\boxed{\text{A physical structure is an admissible relational structure whose constraint violations vanish in the physical state.}}$$

This concludes the formal manuscript.